TAMIL NADU AGRICULTURAL UNIVERSITY

B.Sc. (Hons) Agriculture - Third Semester

# AGR 202

# Principles and Practices of Natural Farming

Complete Lecture-Wise Theory Study Material

As per the 2025 Syllabus - All 17 Lectures

**About this study material**

- One numbered chapter per syllabus lecture, in the exact order of the course outline.
- Every sub-topic is expanded with headed sections, plain-English definitions and bullet points.
- Comparison tables, worked numerical examples and text diagrams are placed inside the relevant lecture.
- Each lecture closes with an Exam focus section: key points, short-answer questions with model answers, and long-answer questions with answer outlines.
- Examples are drawn from Indian and Tamil Nadu conditions wherever possible.

**How to use this material**

- Read the lecture notes first, then work through the tables and diagrams - they carry the comparison points that appear in examinations.
- Attempt the short-answer questions without looking at the model answers, then check.
- For long-answer questions, write the outline from memory before expanding it into full paragraphs.
- Use the glossary at the end to revise definitions quickly before the examination.



Mounish Raj · MouRa · Agri AI

# Syllabus Map - Lecture Order

The course is organised into seventeen lectures. Lectures 1 to 8 and 10 to 17 carry the theory of natural farming; Lecture 9 is the mid-semester examination. The table below is the master index of this material.

| Lecture | Topic |
|---|---|
| 1 | Indian Heritage of Ancient Agriculture |
| 2 | History and Evolution of Natural Farming |
| 3 | Concept, Definition and Objectives of Natural Farming |
| 4 | Essential Characteristics and Principles of Natural Farming |
| 5 | Main Pillars of Natural Farming |
| 6 | Scope and Importance of Natural Farming |
| 7 | Methods and Schools of Natural Farming |
| 8 | Characteristics and Design of a Natural Farm |
| 9 | Mid Semester Examination |
| 10 | Ecological Balance, Ecological Engineering and Community Responsibility |
| 11 | Integration of Crops, Trees and Animals in Natural Farming |
| 12 | Biodiversity, Indigenous Seeds and Farm Waste Recycling |
| 13 | Water Conservation, Renewable Energy and Nutrient Management |
| 14 | Insect, Pest, Disease and Weed Management in Natural Farming |
| 15 | Mechanization and Farm Operations in Natural Farming |
| 16 | Processing, Labelling and Economic Viability of Natural Farming |
| 17 | Certification, Standards, Export Potential, Government Initiatives and Entrepreneurship |



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# Lecture 1. Indian Heritage of Ancient Agriculture

**Learning objectives**

- Trace the origins of farming in the Indian subcontinent and the civilisations that shaped it.
- Identify the indigenous wisdom behind traditional soil, water and crop management.
- Explain how ancient practices link to modern sustainable and natural farming.
- Recognise the sources - Vedas, Krishi-Parashara, Kautilya's Arthashastra and regional traditions.

## 1.1 Introduction

India is one of the oldest agricultural civilisations in the world. Farming began here about ten thousand years ago in the fertile plains of the Indus and the Ganga, and the knowledge that grew around it was not merely a means of survival - it was a complete system of living with nature. The farmer of ancient India observed soils, seasons, rainfall, insects, cattle and crops as one connected whole. Natural farming, which is studied in this course, is therefore not a new invention. It is a return to a way of farming that Indian communities practised for centuries before chemical inputs arrived.

The phrase used in the syllabus is 'Indian heritage of ancient agriculture'. Heritage means the knowledge, skills and practices handed down from earlier generations. The task of this lecture is to see what that heritage contained, and why the modern natural farming movement treats it as its root.

## 1.2 Early farming civilisations of India

Indian agriculture evolved in several distinct regions, each with its own crops and methods. Together they form the base of the country's farming tradition.

- **Indus Valley Civilisation (about 3300-1300 BCE):** one of the earliest farming cultures, with evidence of wheat, barley, cotton, peas, sesame and dates. Well-planned granaries, brick-lined storage and canal irrigation show advanced crop management.
- **Vedic period (about 1500-500 BCE):** agriculture became central to settled life. The Rigveda mentions ploughing, irrigation channels (kulya) and the use of cattle manure. Terms such as 'krishi' (agriculture), 'karshaka' (farmer) and 'sita' (furrow) come from this period.
- **Mauryan period (about 320-185 BCE):** Kautilya's Arthashastra records a state department of agriculture, rules on land classification, irrigation, cattle care and crop rotation, and even the protection of forests and wildlife.
- **Sangam and later South Indian period:** Tamil literature describes tank irrigation (eri), land classification into five 'thinai', cattle rearing and rain-fed dryland cropping, all of which shaped Tamil Nadu farming.
- **Medieval period:** Persian and Mughal influence introduced new crops and gardens, but the village-level system of manure, mixed cropping and water harvesting continued unchanged.

## 1.3 Indigenous wisdom in soil management

Ancient Indian farmers understood that the soil is a living body that must be fed with organic matter and protected from harm. Several practices reflect this idea.

### 1.3.1 Feeding the soil, not the crop

The traditional principle was to keep the soil fertile and let the crop draw its nutrition from it. Farmers applied cattle dung, urine, compost, green leaf manure and ash. The Vedic 'gomaya' (cow dung) and 'gomutra' (cow urine) were regarded as purifiers of the soil - the same materials that reappear today in Jivamrit and Bijamrit. The modern scientific explanation is that these materials add organic carbon, humus and beneficial microorganisms.

### 1.3.2 Crop rotation and mixed cropping

Cereals were rotated with pulses so that the nitrogen removed by one crop was replaced by the next. Mixtures of cereals, pulses and oilseeds were grown together - a practice still seen in Tamil Nadu as 'varagu-thuvarai' and similar mixed stands. Rotation broke pest cycles and maintained soil fertility without external inputs.

### 1.3.3 Conservation of soil and moisture

Contour bunding, field bunds, dead furrows, mulching with crop residues and the protection of trees on field boundaries all reduced erosion and conserved moisture. The ancient practice of covering the soil surface with straw or leaves is exactly the 'mulching' pillar of natural farming.

## 1.4 Indigenous wisdom in water management

Water harvesting was a highly developed skill in ancient India, especially in the semi-arid south where rainfall is seasonal.

| Traditional water system | Region | Principle and modern relevance |
|---|---|---|
| Eri (tank) system | Tamil Nadu | Cascading village tanks stored monsoon runoff and recharged wells; the classic example of watershed-based water management. |
| Kudimaramathu | Tamil Nadu | Community labour to desilt and repair tanks and channels before the monsoon; a model of participatory water governance. |
| Kuhl / kulya | Himalayan and northern plains | Gravity channels carried stream water to terraced fields; efficient use of slope and gravity. |
| Stepwells (baoli / vav) | Gujarat and Rajasthan | Deep groundwater access with minimal evaporation loss. |
| Phad system | Rajasthan | Community-managed diversion of river water over a chain of fields. |
| Zabo and bamboo drip | Nagaland and Meghalaya | Hill runoff harvesting and gravity drip irrigation from streams. |

## 1.5 Indigenous wisdom in crop and pest management

Traditional farmers managed pests and diseases through diversity and natural materials rather than poisons.

- **Seed treatment:** seeds were soaked in cow urine, dung slurry, buttermilk or beeja-amrit-like preparations before sowing to improve germination and vigour.
- **Botanical pest control:** neem (Azadirachta indica), nochi (Vitex negundo), erukku (Calotropis), adathoda (Adhatoda vasica), garlic, chilli and tobacco were used as sprays and dusts.
- **Cultural control:** border rows of trap crops, mixed cropping, timely sowing, hand-picking of pests and burning of infested material.
- **Sacred groves and refuges:** patches of vegetation were left undisturbed to shelter natural enemies such as birds, spiders and predatory insects.

- **Astronomical and seasonal calendars:** sowing and harvesting were timed by the position of stars (nakshatra) and the monsoon, a practice encoded in the Tamil 'panchangam' and in the proverbs of 'pattam' and 'paruvam'.

## 1.6 Knowledge sources of ancient Indian agriculture

The heritage survives in a body of texts and in living tradition. The important written sources are listed below.

| Source | Period / author | Contribution |
|---|---|---|
| Rigveda and later Vedas | c. 1500-500 BCE | Earliest references to ploughing, irrigation and cattle-based fertility. |
| Krishi-Parashara | c. 4th-6th century CE (attributed to Parashara) | First systematic Indian text on agriculture: soil types, seed selection, sowing, manuring, rainfall prediction. |
| Kautilya's Arthashastra | c. 4th century BCE | State policy on land, irrigation, cattle, crop protection and forest conservation. |
| Vrikshayurveda | Surapala, c. 10th century CE | Treatise on trees, orchards, plant health and plant nutrition. |
| Agni Purana and Brihat Samhita | c. 6th-9th century CE | Rainfall forecasting, agronomy and plant protection. |
| Sangam Tamil literature | c. 300 BCE-300 CE | Descriptions of dryland farming, tank irrigation and cattle culture in the Tamil country. |
| Village tradition and proverbs | Continuous | Living knowledge of local seeds, soil types, manures and pest control. |

## 1.7 From heritage to natural farming

The table below links each ancient practice to the modern natural farming principle it has become.

| Indigenous practice | Modern scientific basis / natural farming link |
|---|---|
| Cow dung and cow urine application (gomaya, gomutra) | Source of organic carbon, nitrogen, auxins and a rich microbial load; basis of Jivamrit and Bijamrit. |
| Covering soil with crop residues and leaves | Reduces evaporation, moderates soil temperature and suppresses weeds; now called mulching. |
| Cereal-pulse rotation and mixed cropping | Biological nitrogen fixation, break in pest and disease cycles, better use of soil layers. |
| Green leaf manuring (e.g. Glyricidia, neem, Pongamia) | Adds nitrogen and lignin, builds humus, feeds soil organisms. |
| Neem and botanical sprays | Azadirachtin and other compounds act as repellents, antifeedants and growth regulators. |
| Tank and community water harvesting | Watershed recharge, groundwater improvement, drought buffer. |
| Seed selection from the farmer's own harvest | Local adaptation, genetic diversity, seed sovereignty; basis of indigenous seed conservation. |
| Sacred groves and field bunds with trees | Habitat for pollinators and natural enemies; ecological engineering. |

**WORKED EXAMPLE - Heritage practice, modern reading**

- A Tamil Nadu farmer soaks paddy seed in cow urine solution before sowing. Traditionally this is done to 'protect the seed'.

- Modern reading: cow urine is mildly alkaline and carries nitrogen, potassium and microbes. The soak improves germination percentage, hardens the seed and reduces seed-borne fungal infection - an early form of biological seed treatment.

- Link to the course: the same idea is formalised today in the Bijamrit seed treatment of natural farming.

## EXAM FOCUS

**Key points to remember**

- Indian agriculture is about ten thousand years old; the Indus Valley, Vedic and Mauryan periods each contributed practices still in use.
- The heritage rested on feeding the soil with cattle dung, urine, compost and green manure rather than feeding the crop with chemicals.
- Crop rotation, mixed cropping and mulching maintained fertility and broke pest cycles.
- Tank irrigation, the eri system and Kudimaramathu are Tamil Nadu's contribution to community water management.
- Botanical pest control used neem, nochi, erukku and adathoda; cultural control used diversity and timing.
- Krishi-Parashara, the Arthashastra and Vrikshayurveda are the key written sources.
- Every pillar of modern natural farming can be traced to an indigenous practice.

**Short-answer questions with model answers**

Q1. Name any four traditional water-harvesting systems of India.

**Model answer:** Eri (tank) system of Tamil Nadu, kuhl/kulya gravity channels of the hills and northern plains, stepwells (baoli/vav) of Gujarat and Rajasthan, and the phad system of Rajasthan. The zabo system of Nagaland and bamboo drip irrigation of Meghalaya are further examples.

Q2. What is Krishi-Parashara and why is it important?

**Model answer:** It is an ancient Sanskrit treatise on agriculture attributed to the sage Parashara. It is important because it is the first systematic Indian text on farming, covering soil types, seed selection, sowing, manuring and rainfall prediction, and it shows that scientific agronomy existed in India long before modern times.

Q3. How did ancient farmers manage soil fertility without chemical fertilisers?

**Model answer:** They applied cattle dung and urine, compost, ash and green-leaf manure to feed the soil; rotated cereals with nitrogen-fixing pulses; grew mixed crops; and returned crop residues to the field. These practices built organic matter and kept the soil biologically active.

**Long-answer questions with model answer outlines**

Q1. Describe the indigenous wisdom of ancient India in soil, water and crop management, and show how it forms the root of natural farming.

**Outline:**

- Introduction: India as one of the oldest agricultural civilisations; heritage as transmitted knowledge.
- Soil management: cattle dung and urine, compost, green manure, rotation with pulses, mulching with residues.
- Water management: eri tanks, Kudimaramathu, kuhl channels, stepwells, phad, zabo; contour and bunding.
- Crop and pest management: seed treatment, botanical sprays (neem, nochi, erukku), mixed cropping, trap crops, sacred groves.
- Written sources: Krishi-Parashara, Arthashastra, Vrikshayurveda, Sangam literature.
- Linkage: each practice maps to a modern natural farming principle - Jivamrit, mulching, rotation, botanical pest control, water harvesting.
- Conclusion: natural farming is a revival, not an invention.

Q2. Explain the traditional water-harvesting systems of India with special reference to Tamil Nadu and their relevance today.

**Outline:**

- Introduction: seasonal monsoon and the need to store runoff.

- Eri tank system: cascading tanks, catchment, sluice, recharge of wells.
- Kudimaramathu: community desilting and repair as participatory governance.
- Other systems: kuhl, stepwells, phad, zabo, bamboo drip.
- Relevance: groundwater recharge, drought buffer, reduced dependence on borewells, revival under state programmes.
- Conclusion: traditional systems are the template for modern watershed management.

## 1.8 Traditional soil classification and cattle wealth

Ancient Indian farmers classified soils by colour, texture, water-holding capacity and suitability, and matched crops to each. This was practical soil science long before laboratories existed.

| Traditional class | Character | Suitability |
|---|---|---|
| Urvara (fertile) and Anurvara (barren) | General fertility classification | Choice of field for the main crop |
| Krushna (black) | Heavy, moisture-retentive | Cotton and deep-rooted crops |
| Pandu (pale / light) | Light texture, quick to warm | Millets and pulses |
| Lohita (red) | Iron-rich, well drained | Groundnut and pulses |
| Tamil 'thinai' classification | Five ecological zones - kurinji, mullai, marutham, neydal, palai | Matching crops, livestock and livelihood to the landscape |

The Tamil 'thinai' system is a striking early example of ecological land classification, dividing the landscape into hill, forest, agricultural, coastal and dry zones, each with its own crops and practices. Cattle were central to this system: the indigenous cow and bullock provided dung, urine, draught power, milk and leather, and native breeds such as Kangayam, Umbalachery, Pulikulam, Alambadi and Bargur were selected for local conditions. Modern natural farming draws directly on this cattle wealth.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Indian agriculture is about ____ years old.

2 The Indore method of composting was developed by ____.

3 The ancient Tamil water-harvesting system of village tanks is called ____.

4 Krishi-Parashara is attributed to the sage ____.

5 Community labour for desilting village tanks is known as ____.

6 The traditional Tamil land classification of five ecological zones is the ____ system.

### B. State whether true or false

1 The Rigveda contains references to ploughing and irrigation channels. (True)

2 Ancient Indian farmers relied mainly on chemical fertilisers. (False - on cattle dung, urine and compost)

3 Neem was used as a botanical pesticide in ancient India. (True)

4 The Arthashastra has no reference to agriculture. (False - it records a state department of agriculture)

### C. Match the following

| Column A | Column B |
|---|---|
| Eri | Tamil Nadu village tank system |
| Kudimaramathu | Community tank and channel maintenance |
| Vrikshayurveda | Ancient treatise on trees |
| Gomutra | Cow urine used as soil purifier |
| Phad | Community-managed water diversion in Rajasthan |

### D. Answer in one word or one line

1 Name two ancient Indian texts on agriculture.

2 Which traditional practice is the ancestor of modern mulching?

3 Name two botanical pest-control plants used in ancient India.

4 What was the role of cattle in traditional Indian farming?

**Practical application in Tamil Nadu - heritage practices still in use**

- Eri tanks and Kudimaramathu are revived under state watershed and tank restoration programmes.
- Indigenous paddy varieties such as Kallurundai Samba and Mappillai Samba are conserved by farmer groups and seed banks.
- Cow-urine seed soaking and neem sprays are common in natural farming across the state.



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# Lecture 2. History and Evolution of Natural Farming

**Learning objectives**

- Trace the global evolution of natural and organic farming thought.
- Describe the Indian journey from traditional farming to Zero Budget Natural Farming.
- Explain Fukuoka's 'Do-nothing' farming and its influence.
- List the milestones and movements that shaped natural farming in India and Tamil Nadu.

## 2.1 Introduction

Natural farming did not appear suddenly. It grew out of a century of concern about the damage caused by chemical-intensive agriculture - soil erosion, falling fertility, pesticide poisoning, loss of biodiversity and the rising cost of cultivation. This lecture follows that journey across the world and within India, so that the student can see natural farming as the latest stage of a long movement rather than an isolated idea.

## 2.2 The global journey

The modern organic and natural farming movement began in the early twentieth century and passed through several phases.

| Period | Development | Key idea or person |
|---|---|---|
| 1900s-1920s | Biodynamic agriculture in Europe | Rudolf Steiner (1924) - the farm as a self-contained living organism; preparations from manure and herbs. |
| 1930s-1940s | Organic farming concept | Sir Albert Howard (UK, working in India) - 'The Waste Products of Agriculture' and 'An Agricultural Testament'; the Indore method of composting. |
| 1940s | Organic movement formalised | Lady Eve Balfour's Haughley Experiment; Jerome Rodale in the USA popularised 'organic'. |
| 1930s-1970s | Fukuoka's natural farming | Masanobu Fukuoka (Japan) developed 'Do-nothing' or natural farming; wrote 'The One-Straw Revolution' (1975). |
| 1960s-1970s | Green Revolution and its costs | High-yielding varieties, fertilisers and pesticides raised output but caused soil and water degradation. |
| 1980s-1990s | Korean Natural Farming (KNF) | Cho Han-Kyu developed Indigenous Micro-Organisms (IMO) and natural input preparations. |
| 1990s-2000s | ZBNF in India | Subhash Palekar developed Zero Budget Natural Farming after observing natural forests. |
| 2000s onward | Institutional and policy phase | Organic and natural farming missions, certification systems (PGS), and state-scale programmes such as Andhra Pradesh Community Managed Natural Farming. |

## 2.3 Sir Albert Howard and the Indore method

Albert Howard worked as an agricultural adviser in India in the early twentieth century. Studying traditional Indian farmers, he concluded that the health of the soil, the plant, the animal and the human being are one chain. He developed the Indore method of composting, in which crop residues, cattle dung and urine are layered and turned to produce humus. This work became a foundation of the world organic movement and is a direct Indian contribution to it.

## 2.4 Fukuoka and 'Do-nothing' farming

Masanobu Fukuoka was a Japanese plant pathologist who left his job and returned to his family farm. After years of trial he arrived at what he called natural farming or 'Do-nothing' farming. The name does not mean doing nothing at all; it means removing unnecessary human intervention so that nature does the work.

- **No tillage:** the soil is never ploughed; roots, earthworms and microorganisms maintain structure.
- **No chemical fertiliser and no prepared organic fertiliser:** the soil is enriched by returning straw and by green manure crops such as clover.
- **No weeding by tillage or herbicide:** weeds are managed by ground cover, timing and disturbance of the surface layer.
- **No dependence on pesticides:** a balanced ecosystem keeps pests in check.
- His book 'The One-Straw Revolution' (1975) became a global classic and is listed in the reference list of this course.

## 2.5 The Indian journey

India's path to natural farming runs through the traditional practices of Lecture 1, then the Green Revolution, then a corrective phase.

### 2.5.1 Before the Green Revolution

Indian farming up to the 1960s was largely organic by default. Fertility came from cattle dung, compost and green manure; pests were managed with botanicals and cultural methods; and mixed cropping was common. Yields were modest but stable and the system was self-reliant.

### 2.5.2 The Green Revolution

From the mid-1960s India adopted high-yielding varieties of wheat and rice with assured irrigation, chemical fertilisers and pesticides. Food production rose sharply and famine was avoided, which was a genuine achievement. However, the costs appeared later: falling soil organic matter, micronutrient deficiency, groundwater depletion, pesticide residues, pest resurgence and rising input debt for farmers.

### 2.5.3 The corrective phase

From the 1980s awareness grew that chemical-intensive farming was not sustainable. Non-governmental organisations, farmer groups and scientists promoted organic farming. In the 1990s Subhash Palekar developed Zero Budget Natural Farming (ZBNF) in Maharashtra, and the idea spread rapidly across states. In the 2000s and 2010s, state governments, including Andhra Pradesh and Tamil Nadu, launched large programmes, and the central government introduced Paramparagat Krishi Vikas Yojana and the PGS-India certification system.

## 2.6 Subhash Palekar and Zero Budget Natural Farming

Subhash Palekar, a farmer from Maharashtra, observed that natural forests grow luxuriantly without any external input and concluded that the soil has its own capacity to supply nutrition if it is kept alive and covered. He distilled his approach into four pillars (Bijamrit, Jivamrit, Mulching and Waaphasa - see Lecture 5) and called it Zero Budget Natural Farming because the external cash cost of inputs is designed to be close to zero. The word 'budget' here means the cash input cost, not the total cost of cultivation.

| Stage | Approximate period | Character of Indian agriculture |
|---|---|---|
| Traditional / organic by default | up to 1960s | Cattle-based fertility, mixed cropping, botanical pest control, self-reliant. |
| Green Revolution | mid-1960s to 1980s | High-yielding varieties, irrigation, chemical fertiliser and pesticides; rising output and rising cost. |
| Awareness and organic promotion | 1980s to 1990s | NGO and farmer-led organic movements; concern over soil and water degradation. |
| ZBNF and natural farming spread | 1990s to 2010s | Palekar's ZBNF; farmer-to-farmer training; state interest. |
| Policy and programme phase | 2010s onward | Paramparagat Krishi Vikas Yojana, PGS-India, state natural farming missions, AP community-managed natural farming. |

## 2.7 Evolution of natural farming in Tamil Nadu

Tamil Nadu has a long tradition of organic and natural farming rooted in its tank culture, cattle wealth and dryland cropping. The state has supported organic and natural farming through its agriculture department and the Tamil Nadu Organic Certification Department, and through schemes delivered on the Uzhavan platform. Farmer interest is strong in districts such as Erode, Dindigul, Krishnagiri, Salem and the Cauvery delta, where both organic certification and natural farming are practised. Note: specific scheme names, subsidy amounts and years change with each state budget and should be checked in the current government order; this material describes them in general terms.




Fig. 2.1 The evolution of natural farming from tradition to policy.

**WORKED EXAMPLE - Budget: what 'zero budget' really means**

- A natural farmer spends on seed, some labour and land preparation, but not on purchased fertiliser and pesticide.
- Suppose a conventional one-acre crop spends Rs. 9,000 on chemical fertiliser and Rs. 4,000 on pesticide per season.
- A natural farmer replaces these with Jivamrit, Bijamrit, mulching material and botanical sprays prepared on the farm.
- The cash input cost therefore falls close to zero for fertiliser and pesticide, though family labour and organic materials are still used. This is the sense in which ZBNF is 'zero budget'.

## EXAM FOCUS

**Key points to remember**

- Natural farming evolved from a global organic movement and from India's own traditional practices.
- Albert Howard's Indore method of composting is an Indian contribution to world organic farming.
- Fukuoka's 'Do-nothing' farming rests on no tillage, no chemical inputs, no weeding by tillage and no pesticides.
- The Green Revolution raised output but caused soil, water and health problems.
- Subhash Palekar developed ZBNF in the 1990s around four pillars.
- In policy, Paramparagat Krishi Vikas Yojana, PGS-India and state missions carry the movement forward.
- Tamil Nadu's tradition of tanks, cattle and dryland cropping gives natural farming a strong local base.

**Short-answer questions with model answers**

Q1. Who was Masanobu Fukuoka and what is his contribution?

**Model answer:** Fukuoka was a Japanese plant pathologist turned farmer who developed natural or 'Do-nothing' farming. He showed that a farm can be productive without tillage, chemical fertiliser, weeding by tillage or pesticides, and wrote 'The One-Straw Revolution', which inspired the worldwide natural farming movement.

Q2. What is meant by 'zero budget' in ZBNF?

**Model answer:** It means that the external cash cost of inputs is brought close to zero. Fertiliser and pesticide are replaced by on-farm preparations such as Jivamrit and Bijamrit and by mulching. It does not mean there is no cost at all, because land, seed and family labour still involve cost.

Q3. State two achievements and two problems of the Green Revolution.

**Model answer:** Achievements: India became self-sufficient in foodgrain and famine was avoided; yields of wheat and rice rose sharply. Problems: soil organic matter declined and micronutrient deficiencies appeared; groundwater was depleted and pesticide residues and pest resurgence increased, along with farmer indebtedness.

**Long-answer questions with model answer outlines**

Q1. Trace the history and evolution of natural farming from traditional Indian agriculture to the present day.

**Outline:**

- Introduction: define natural farming as a revival of tradition shaped by global thought.
- Global phase: Steiner (biodynamic), Howard (organic, Indore method), Balfour and Rodale, Fukuoka, Korean Natural Farming.
- Indian traditional phase: organic by default with cattle-based fertility and mixed cropping.
- Green Revolution: benefits and costs.
- Corrective phase: organic movements, Palekar's ZBNF and the four pillars.
- Policy phase: PKVY, PGS-India, state missions, AP community-managed natural farming, Tamil Nadu initiatives.
- Conclusion: natural farming today is a mature, policy-supported system.

Q2. Discuss the contributions of Albert Howard, Masanobu Fukuoka and Subhash Palekar to natural farming.

**Outline:**

- Howard: worked in India; Indore method of composting; soil-plant-animal-human health as one chain; foundation of world organic farming.

- Fukuoka: Japan; natural/'Do-nothing' farming; four principles (no tillage, no chemical input, no weeding by tillage, no pesticides); 'The One-Straw Revolution'.
- Palekar: India; observed natural forests; ZBNF; four pillars (Bijamrit, Jivamrit, Mulching, Waaphasa); farmer-to-farmer training.
- Common thread: soil as a living system; reduced external inputs; diversity and natural processes.
- Conclusion: together they converted a local idea into a global and Indian movement.

## 2.8 Milestones of the organic and natural farming movement in India

| Milestone | Significance |
|---|---|
| Albert Howard's work in India | Indore method of composting; foundation of world organic farming. |
| Green Revolution (mid-1960s) | High-yielding varieties and chemical inputs; raised output, later costs appeared. |
| Organic farming movements (1980s-1990s) | NGO and farmer-led promotion of organic methods. |
| ZBNF by Subhash Palekar (1990s) | Four-pillar natural farming; farmer-to-farmer training. |
| National Programme for Organic Production (NPOP) | Set the Indian organic standard and certification framework. |
| PGS-India | Low-cost participatory certification for small farmers. |
| Paramparagat Krishi Vikas Yojana (PKVY) | Cluster-based support for organic and traditional farming. |
| Andhra Pradesh Community Managed Natural Farming | Large state programme; a widely cited model of scale. |
| National Mission on Natural Farming | National push to promote natural farming through demonstration and training. |

Note: the precise year of each milestone should be confirmed from official documents; the sequence and significance above are the important points.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Masanobu Fukuoka wrote the book ____.

2 ZBNF was developed by ____.

3 The Indore method of composting is associated with ____.

4 Korean Natural Farming was developed by ____.

5 The Green Revolution in India began in the mid-____.

6 Fukuoka called his method ____ farming.

### B. State whether true or false

1 The Green Revolution raised yields but had no environmental cost. (False - it caused soil, water and health problems)

2 Albert Howard worked in India and developed a composting method. (True)

3 ZBNF means the total cost of cultivation is zero. (False - it means the external cash cost of inputs is near zero)

4 Fukuoka's method uses chemical fertiliser in small amounts. (False - no chemical fertiliser is used)

### C. Match the following

| Column A | Column B |
|---|---|
| Fukuoka | Do-nothing natural farming, Japan |
| Palekar | ZBNF, India |
| Howard | Indore method of composting |
| Cho Han-Kyu | Korean Natural Farming |
| Rodale | Popularised organic farming in the USA |

### D. Answer in one word or one line

1 Expand ZBNF.

2 Name the four pillars of ZBNF.

3 What is a seed ball?

4 In which decade did the Green Revolution begin in India?

**Practical application in Tamil Nadu - the movement in the state**

- Natural farming and organic farming are promoted through the state agriculture department and its extension network.
- The Tamil Nadu Organic Certification Department supports certification of organic produce.
- Farmer interest is strong in Erode, Dindigul, Krishnagiri, Salem and the Cauvery delta districts.

# Lecture 3. Concept, Definition and Objectives of Natural Farming

**Learning objectives**

- Define natural farming clearly and distinguish it from organic and conventional farming.
- State the ecological and socio-economic objectives of natural farming.
- Explain its importance for climate change, food security and rural sustainability.

## 3.1 What natural farming means

Natural farming is a farming system that works with nature rather than against it. It avoids synthetic chemical fertilisers, pesticides and growth regulators, and it avoids disturbing the soil through deep or repeated tillage. Instead it relies on the natural processes of the soil - the activity of microorganisms, earthworms, organic matter and plant roots - to supply nutrition and to keep pests and diseases in check. Its central idea is simple: if the soil is kept alive, covered and moist, it can feed the crop without purchased inputs.

Several definitions are used. A widely accepted working definition is given below.

**Working definition**

- Natural farming is a chemical-free, low-input, ecologically integrated farming system in which crop nutrition is met from on-farm organic resources and the soil's own biological processes, pest and disease are managed by ecological methods, and the soil is protected from disturbance and exposure, so that the farm becomes self-reliant, sustainable and in harmony with nature.

## 3.2 Natural farming, organic farming and conventional farming

Students often confuse natural farming with organic farming. Both avoid synthetic chemicals, but they differ in their attitude to external inputs and to soil disturbance. The comparison table below is important for the examination.

| Basis | Natural farming | Organic farming | Conventional farming |
|---|---|---|---|
| Core principle | Work with nature; minimum human intervention | Avoid synthetic chemicals; use approved organic inputs | Maximise yield with external inputs |
| Synthetic fertiliser | Not used | Not used | Used heavily |
| Synthetic pesticide | Not used | Not used | Used heavily |
| External organic inputs (compost, oil cakes, rock phosphate) | Minimal; mostly on-farm preparations | Permitted and often purchased | Not the main input |
| Tillage | No tillage or minimum disturbance | Tillage usually allowed | Intensive tillage |
| Mulching | Central practice (crop residue / live mulch) | Practised but not mandatory | Rare |
| Soil cover | Kept covered at all times | Encouraged | Often bare |
| Source of nutrition | Soil biology and on-farm recycling | Organic amendments and manures | Synthetic NPK and micronutrients |
| Input cost | Very low (close to zero cash cost for nutrients) | Moderate to high | High and rising |
| Certification | Often community or self-declared; PGS where applicable | Mandatory certification for premium market (PGS, NPOP) | Not applicable |
| Typical yield in transition | May dip initially, then stabilise | May dip initially, then recover | High but input-dependent |

| Basis | Natural farming | Organic farming | Conventional farming |
|---|---|---|---|
| Main aim | Sustainability, soil health, self-reliance | Chemical-free produce and market | Production and profit |

## 3.3 Ecological objectives

Natural farming has clear ecological aims, which are the reason it is promoted for sustainability.

- **Build soil health:** increase organic carbon, humus and microbial diversity, and improve soil structure and water-holding capacity.
- **Protect water:** reduce chemical runoff and leaching, and improve infiltration and groundwater recharge.
- **Conserve biodiversity:** encourage pollinators, natural enemies and soil organisms through crop diversity and habitat patches.
- **Reduce pollution:** eliminate pesticide residues in soil, water, food and the farmer's body.
- **Mitigate climate change:** sequester carbon in soil, reduce greenhouse gas emissions from fertiliser manufacture and use, and lower fossil energy use.
- **Adapt to climate change:** healthier soils resist drought and heavy rain better, giving resilience to erratic monsoons.

## 3.4 Socio-economic objectives

- **Reduce cost of cultivation:** remove or greatly reduce expenditure on fertiliser and pesticide.
- **Increase and stabilise net income:** lower costs can raise net returns even if gross yield is similar.
- **Free farmers from debt:** reduce dependence on input dealers and moneylenders.
- **Improve food quality:** chemical-free, more nutritious produce for the family and the market.
- **Empower rural communities:** strengthen local knowledge, seed systems and self-reliance.
- **Generate rural employment:** preparation of inputs, mulching, processing and local marketing create work.
- **Improve farmer health:** fewer pesticide-related illnesses.

## 3.5 Importance for climate change, food security and rural sustainability

| Concern | How natural farming helps |
|---|---|
| Climate change mitigation | Soil carbon sequestration through organic matter and no tillage; lower emissions from avoided fertiliser manufacture and use; reduced fossil fuel in tillage. |
| Climate change adaptation | Better soil moisture retention and structure give drought and flood resilience; diverse crops spread risk. |
| Food security | Stable, low-cost production for the household; diversified output (cereals, pulses, vegetables, milk) improves nutrition. |
| Food safety | No pesticide residues; safer produce for consumers and safer work for farmers. |
| Rural sustainability | Lower input cost, local input production and local value addition keep money in the village. |
| Natural resource base | Protects soil, water and biodiversity for future generations. |

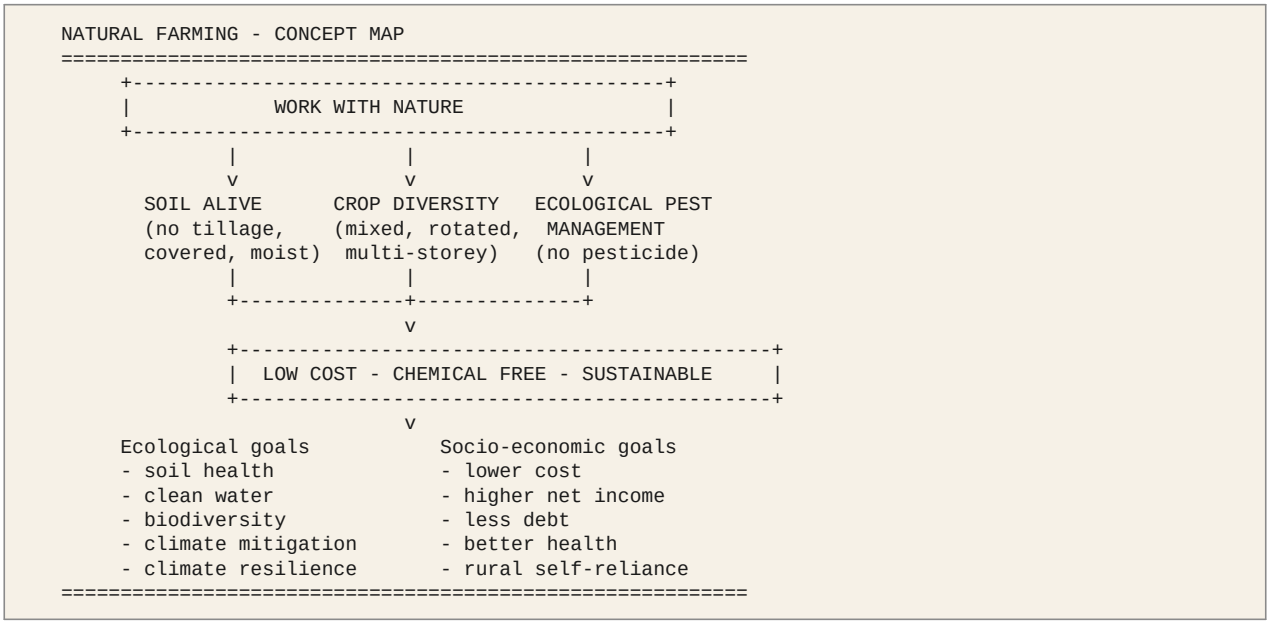



Fig. 3.1 The concept of natural farming and its twin sets of objectives.

## EXAM FOCUS

### Key points to remember

- Natural farming works with nature, avoids synthetic inputs and avoids soil disturbance.
- It is not the same as organic farming: natural farming minimises external inputs and tillage, while organic farming permits approved external organic inputs and tillage.
- Ecological objectives: soil health, clean water, biodiversity, lower pollution, climate mitigation and adaptation.
- Socio-economic objectives: lower cost, higher net income, less debt, better health, rural self-reliance.
- For climate change it both mitigates (carbon sequestration) and helps adaptation (resilience).
- For food security it provides stable, diverse and safe food at low cost.
- For rural sustainability it keeps inputs and value local.

### Short-answer questions with model answers

Q1. Define natural farming.

**Model answer:** Natural farming is a chemical-free, low-input, ecologically integrated farming system in which crop nutrition comes from on-farm organic resources and soil biological processes, pest and disease are managed by ecological methods, and the soil is protected from disturbance and exposure, making the farm self-reliant and sustainable.

Q2. Give any three differences between natural farming and organic farming.

**Model answer:** Natural farming uses no or minimum tillage while organic farming permits tillage. Natural farming relies mainly on on-farm preparations with minimal external inputs while organic farming permits purchased organic inputs such as compost, oil cakes and rock phosphate. Natural farming keeps the soil covered with mulch at all times, whereas mulching is only encouraged and not mandatory in organic farming.

Q3. How does natural farming help in climate change mitigation and adaptation?

**Model answer:** Mitigation: it sequesters carbon in soil through organic matter and no tillage, and it avoids the greenhouse gases and fossil energy associated with fertiliser manufacture and use. Adaptation: healthier soils hold more moisture and resist drought and heavy rain better, and crop diversity spreads risk, so the farm is more resilient to erratic weather.

**Long-answer questions with model answer outlines**

Q1. Define natural farming and explain its ecological and socio-economic objectives.

**Outline:**

- Definition and central idea: work with nature, chemical-free, low input, soil alive and covered.
- Ecological objectives: soil health, water protection, biodiversity, reduced pollution, climate mitigation and adaptation.
- Socio-economic objectives: lower cost, higher net income, less debt, better food quality and health, rural employment and self-reliance.
- Comparison table: natural vs organic vs conventional.
- Conclusion: natural farming meets production and conservation goals together.

Q2. Discuss the importance of natural farming for climate change, food security and rural sustainability.

**Outline:**

- Introduction: the three concerns and why they matter for India.
- Climate change: mitigation (carbon sequestration, lower emissions) and adaptation (resilience, diversity).
- Food security: stable low-cost output, household nutrition through diversification, food safety.
- Rural sustainability: lower input cost, local input production, local value addition, employment, reduced migration.
- Supporting evidence: state programmes and farmer experiences.
- Conclusion: natural farming is a triple-win system for environment, food and rural economy.

## 3.6 Natural farming compared with other sustainable systems

Natural farming is one of several sustainable systems. Students should be able to distinguish them briefly.

| System | Key idea | Difference from natural farming |
|---|---|---|
| Organic farming | Avoid synthetic chemicals; use approved organic inputs | Permits external organic inputs and tillage; requires certification for the premium market |
| Biodynamic farming | Farm as a self-contained organism; special preparations | Uses specific fermented preparations and a cosmic calendar; natural farming does not |
| Permaculture | Design system mimicking natural ecosystems | Emphasis on design of the whole homestead; natural farming focuses on field practice |
| Regenerative agriculture | Restore soil health and sequester carbon | Often permits some external inputs and managed grazing; broader umbrella |
| Conservation agriculture | Minimum tillage with residue retention and rotation | May still use herbicides; natural farming does not |
| Zero Budget Natural Farming | Four pillars; desi cow based; zero external cash cost | A specific school within natural farming |

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Natural farming avoids ____ fertilisers and pesticides.

2 Natural farming keeps the soil ____ at all times.

3 A key socio-economic objective of natural farming is to reduce the ____ of cultivation.

4 Natural farming sequesters carbon in the soil, which helps climate change ____.

5 Natural farming differs from organic farming mainly in minimising ____ inputs and tillage.

6 Food ____ is improved because natural produce is free of pesticide residues.

### B. State whether true or false

1 Natural farming and organic farming are exactly the same. (False - they differ in external inputs and tillage)

2 Natural farming can help both mitigate and adapt to climate change. (True)

3 Natural farming increases the farmer's dependence on input dealers. (False - it reduces dependence)

4 Natural farming has only environmental objectives. (False - it also has socio-economic objectives)

### C. Match the following

| Column A | Column B |
|---|---|
| Ecological objective | Soil, water and biodiversity health |
| Socio-economic objective | Lower cost and higher net income |
| Climate mitigation | Carbon sequestration in soil |
| Climate adaptation | Resilience to drought and heavy rain |
| Food safety | No pesticide residues |

### D. Answer in one word or one line

1 Define natural farming in one sentence.

2 Name two ecological objectives of natural farming.

3 Name two socio-economic objectives.

4 Give one way natural farming helps climate adaptation.

**Practical application in Tamil Nadu - objectives in practice**

- Lower input cost is the strongest motivation for small farmers adopting natural farming.
- Chemical-free produce is sold through organic stores, farmers' markets and online platforms at a premium.
- Improved soil moisture retention matters greatly in the state's dryland districts.

# Lecture 4. Essential Characteristics and Principles of Natural Farming

**Learning objectives**

- State the essential characteristics that define a natural farm.
- Explain the principles of no tillage, no chemical inputs and minimum disturbance.
- Describe reliance on natural processes for fertility and pest control.

## 4.1 Essential characteristics

The characteristics below distinguish a natural farm from any other farming system. A farm need not satisfy every point perfectly at the start, but these are the direction of travel.

- **No synthetic chemical input:** no urea, DAP, complex fertiliser, synthetic pesticide, herbicide or growth regulator.
- **No tillage or minimum tillage:** the soil is not inverted or pulverised; structure built by roots and soil life is preserved.
- **Soil always covered:** crop residues, straw or live mulch keep the surface shaded and moist.
- **Soil always moist (Waaphasa):** a balance of air and moisture is maintained in the soil pores.
- **Reliance on soil biology:** microorganisms, earthworms and mycorrhiza supply and mobilise nutrients.
- **Crop diversity:** mixed cropping, rotation and multi-storey systems instead of monoculture.
- **On-farm input production:** inputs such as Jivamrit and botanical sprays are prepared from farm and local materials.
- **Integration of crops, trees and livestock:** the farm recycles nutrients within its own boundary.
- **Ecological pest management:** prevention through diversity and natural enemies, with botanicals as a last resort.
- **Seed from the farm and indigenous varieties:** local seed adapted to the soil and climate.
- **Low external dependence and self-reliance:** minimal purchase, minimal debt.
- **Long-term soil and ecosystem health as the primary goal.**

## 4.2 Principle of no tillage

Tillage is the mechanical disturbance of soil. Conventional farming ploughs and pulverises the soil to prepare a seedbed and control weeds. Natural farming rejects this because tillage destroys soil structure, breaks fungal networks, oxidises organic matter, exposes soil to erosion and kills earthworms and other soil life.

- Soil is a living structure of aggregates, pores and channels built over years by roots and organisms.
- Tillage collapses aggregates, seals pores and reduces infiltration.
- Tillage accelerates the loss of organic carbon to the atmosphere.
- No tillage keeps root channels and earthworm burrows intact, improving water movement and aeration.
- Seeds are sown by dibbling, direct seeding or transplanting into a mulch layer without ploughing.

## 4.3 Principle of no chemical inputs

Synthetic fertilisers and pesticides give a quick response but harm the soil ecosystem and create dependence.

- Chemical fertilisers supply only a few nutrients and do not build organic matter.
- Excess nitrogen acidifies soil and leaches into groundwater as nitrate.
- Pesticides kill beneficial insects, pollinators and natural enemies, causing pest resurgence.
- Residues enter the food chain and harm farmer and consumer health.
- Natural farming substitutes Jivamrit, compost, green manure and botanical preparations.

## 4.4 Principle of minimum disturbance and ecological balance

Besides tillage, other disturbances are also minimised - unnecessary weeding, burning of residues, removal of all biomass and excessive grazing. The aim is to keep the farm's living communities intact so that they regulate themselves. A balanced ecosystem has predators that control pests, decomposers that recycle nutrients and pollinators that set seed and fruit.

## 4.5 Reliance on natural processes for fertility

In natural farming, fertility is a product of biological activity rather than of applied chemicals. The main natural processes are given below.

| Natural process | How it supplies nutrition |
|---|---|
| Organic matter decomposition | Soil microbes break down mulch and residues into humus, releasing nitrogen, phosphorus, potassium and micronutrients slowly. |
| Biological nitrogen fixation | Rhizobia in legume root nodules and free-living bacteria fix atmospheric nitrogen. |
| Mycorrhizal association | Fungal networks extend root reach and help absorb phosphorus, zinc and water. |
| Nutrient mobilisation | Phosphate-solubilising and potassium-mobilising bacteria release locked nutrients from soil minerals. |
| Earthworm activity | Earthworms mix organic matter with soil and produce nutrient-rich casts. |
| Green manuring | Leguminous green manure crops (sunhemp, daincha) add nitrogen and organic matter. |
| Mulching | Residue mulch feeds soil life and conserves moisture, driving the whole process. |

## 4.6 Reliance on natural processes for pest control

| Natural process | How it controls pests |
|---|---|
| Natural enemies | Predators and parasitoids (ladybird beetles, spiders, lacewings, Trichogramma) keep pest populations low. |
| Crop diversity | Mixed cropping confuses pests, hosts alternative prey for predators and slows spread. |
| Botanical preparations | Neem, nochi, erukku and adathoda act as repellents, antifeedants and growth regulators. |
| Healthy soil | Balanced plant nutrition from organic matter makes plants less attractive to pests and more tolerant of damage. |
| Cultural methods | Timely sowing, trap crops, border rows, sanitation and hand-picking. |
| Habitat for enemies | Bund vegetation, flowering plants and trees shelter beneficial insects and birds. |

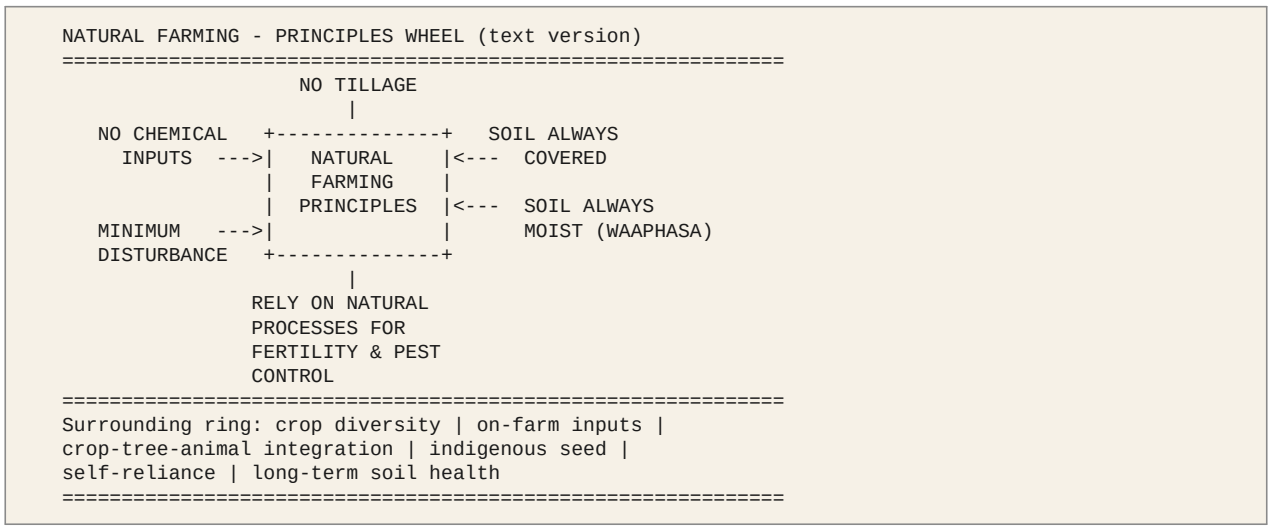



Fig. 4.1 The principles wheel - five core principles with supporting ideas around them.

**WORKED EXAMPLE - Applying the principles on one acre**

- No tillage: sow with a dibble or seed drill directly into the previous crop's mulch; do not plough.
- No chemical input: replace the usual urea top-dress with Jivamrit; replace the usual pesticide with a neem-based spray only if the pest crosses the threshold.
- Soil covered: spread 2-3 tonnes of straw or crop residue per acre as mulch.
- Soil moist: keep the mulch layer intact and irrigate only when the soil below the mulch begins to dry.
- Diversity: sow a cereal with a pulse and a border row of marigold or castor to shelter natural enemies.

## EXAM FOCUS

**Key points to remember**

- The defining characteristics are no chemical input, no or minimum tillage, permanent soil cover, moist soil (Waaphasa), reliance on soil biology, diversity and self-reliance.
- No tillage protects soil structure, organic matter and soil life.
- No chemical input avoids soil acidification, pollution and pest resurgence.
- Minimum disturbance extends to weeding, residue burning and grazing.
- Fertility comes from decomposition, nitrogen fixation, mycorrhiza, nutrient mobilisation, earthworms, green manuring and mulch.
- Pest control comes from natural enemies, diversity, botanicals, healthy plants and cultural methods.
- The five principles together make the farm self-regulating.

**Short-answer questions with model answers**

Q1. Why does natural farming avoid tillage?

**Model answer:** Tillage destroys soil structure, breaks fungal networks, oxidises organic carbon, seals pores, increases erosion and kills earthworms and soil microorganisms. No tillage preserves root channels and burrows, keeps organic matter in place and maintains the biological activity on which natural farming depends for nutrition.

Q2. Name four natural processes that supply nutrition in a natural farm.

**Model answer:** Decomposition of organic matter and mulch into humus; biological nitrogen fixation by legumes and free-living bacteria; mycorrhizal association helping uptake of phosphorus, zinc and water; and nutrient mobilisation by phosphate-solubilising and potassium-mobilising bacteria. Earthworm activity and green manuring are further examples.

Q3. What is minimum disturbance in natural farming?

**Model answer:** It means reducing all unnecessary interference with the farm ecosystem beyond tillage - avoiding excessive weeding, not burning crop residues, leaving biomass on the field and controlling grazing - so that the living communities of soil, plants and beneficial insects remain intact and regulate themselves.

**Long-answer questions with model answer outlines**

Q1. State the essential characteristics and principles of natural farming and explain any four of them.

**Outline:**

- List the characteristics: no chemical input, no tillage, permanent cover, Waaphasa, soil biology, diversity, on-farm inputs, integration, ecological pest management, indigenous seed, self-reliance.
- Explain no tillage: soil structure, organic carbon, soil life.
- Explain no chemical input: pollution, resurgence, health.
- Explain reliance on soil biology for fertility: decomposition, fixation, mycorrhiza.
- Explain ecological pest management: enemies, diversity, botanicals.
- Conclusion: together they make the farm self-regulating and sustainable.

Q2. Explain how natural farming relies on natural processes for fertility and pest control instead of external inputs.

**Outline:**

- Fertility: organic matter decomposition, nitrogen fixation, mycorrhiza, nutrient mobilisation, earthworms, green manuring, mulching.
- Pest control: natural enemies, crop diversity, botanical preparations, healthy plants, cultural methods, habitat for enemies.
- Why it works: a balanced ecosystem regulates itself; diversity and soil health are the foundation.
- Contrast with conventional reliance on fertiliser and pesticide.
- Conclusion: natural processes replace purchased inputs.

## 4.7 Common mistakes in applying the principles

Knowing the mistakes helps the student apply the principles correctly.

- **Stopping all inputs at once:** an abrupt change can cause a severe yield drop; transition gradually, especially reducing fertiliser over a season or two.
- **Neglecting mulch:** without mulch, Waaphasa cannot be maintained and the soil dries out; mulch is not optional.
- **Over-irrigation:** flooding destroys Waaphasa and drives out air from the soil.
- **Poorly prepared Jivamrit:** wrong proportions or too long a fermentation reduce its microbial value.
- **Using non-desi cattle products:** the microbial richness of desi cow dung and urine is considered essential in ZBNF.
- **Expecting immediate results:** soil biology takes time to rebuild; the first year or two may be a learning period.
- **Ignoring diversity:** monoculture under natural farming is fragile; diversity is part of the system.
- **Spraying botanicals routinely:** even botanicals can harm beneficial insects; spray only above the action threshold and in the evening.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 The five core principles are no tillage, no chemical input, minimum disturbance, permanent ____ and ____.

2 Tillage accelerates the loss of soil organic ____.

3 Biological nitrogen fixation is carried out by ____ in legume root nodules.

4 Fungal networks that help roots absorb phosphorus are called ____.

5 Earthworms produce nutrient-rich ____.

6 A balanced ecosystem controls pests through natural ____.

### B. State whether true or false

1 Natural farming uses deep ploughing to prepare the seedbed. (False - no tillage)

2 Tillage improves soil structure in the long run. (False - it destroys structure)

3 Mycorrhiza help plants absorb phosphorus and water. (True)

4 Natural farming sprays pesticide on a fixed calendar. (False - only above the action threshold)

### C. Match the following

| Column A | Column B |
|---|---|
| No tillage | Preserves soil structure and soil life |
| No chemical input | Avoids pollution and pest resurgence |
| Waaphasa | Balance of air and moisture in soil pores |
| Rhizobium | Nitrogen fixation in legumes |
| Green manuring | Adds nitrogen and organic matter |

### D. Answer in one word or one line

1 Name the five core principles of natural farming.

2 Why is tillage harmful to soil?

3 What is mycorrhiza?

4 Name two natural processes that supply nutrition.

**Practical application in Tamil Nadu - principles on the ground**

- No-till sowing with a dibbler or no-till drill is used for rain-fed crops such as millets and pulses.
- Mulching with crop residues is common in coconut and banana gardens to conserve moisture.
- Botanical sprays made from neem and nochi are widely used instead of synthetic pesticides.

# Lecture 5. Main Pillars of Natural Farming

**Learning objectives**

- Name the four pillars of natural farming and explain each.
- Give the composition and function of Bijamrit and Jivamrit.
- Explain mulching and Waaphasa and their scientific basis.
- Describe supporting practices such as Whapasa management and botanical inputs.

## 5.1 The four pillars

Subhash Palekar's ZBNF rests on four pillars: Bijamrit (seed treatment), Jivamrit (soil inoculum), Mulching (soil cover) and Waaphasa (soil moisture-air balance). Together they keep the soil alive, covered, moist and biologically active, so that the crop is nourished without external inputs.

| Pillar | Meaning | Main composition | Function |
|---|---|---|---|
| Bijamrit | Seed treatment / seed inoculum | Cow dung (desi cow), cow urine, lime (calcium), a little soil from the farm bund, water; often jaggery | Coats the seed with beneficial microbes; protects against seed-borne disease; improves germination and early vigour. |
| Jivamrit | Living microbial culture for the soil | Desi cow dung, desi cow urine, jaggery or jaggery powder, pulse flour (besan), and soil from the farm bund, mixed in water | Inoculates the soil with a large population of beneficial microorganisms; accelerates decomposition of mulch; mobilises nutrients. |
| Mulching | Covering the soil surface | Crop residues, straw, dry leaves, weeds, green biomass, or a live mulch crop | Conserves moisture, moderates temperature, suppresses weeds, feeds soil life and adds organic matter. |
| Waaphasa | Soil moisture-air balance in the pores | Achieved through mulching and judicious irrigation, not a material | Maintains the ideal proportion of air and water in soil pores so that aerobic microbes and roots thrive. |

## 5.2 Bijamrit - seed treatment

Bijamrit means 'nectar of the seed'. It is a coating applied to seeds before sowing. The purpose is to surround the seed with beneficial microorganisms and a mild protective layer so that it germinates strongly and resists early disease.

### 5.2.1 Preparation

A common preparation uses the following materials. Quantities are for treating seed for about one acre; adjust in proportion.

| Ingredient | Quantity (typical) | Role |
|---|---|---|
| Desi cow dung (fresh) | 5 kg | Carries a large microbial population; provides organic matter. |
| Desi cow urine | 5 litre | Mildly alkaline; nitrogen, potassium and microbes; disinfecting action. |
| Lime (calcium) or chuna | 50-100 g | Raises pH slightly, provides calcium, helps coat the seed. |
| Farm soil from the bund | a handful (about 100 g) | Introduces locally adapted microbes. |
| Water | sufficient to make a slurry | Medium for mixing and coating. |
| Jaggery (optional) | 50-100 g | Food for microbes in the slurry. |

**Method:** Mix the cow dung and cow urine in water to form a thin slurry. Add lime, farm soil and jaggery and stir well. Soak the seed in the slurry for a short time, then either sow immediately or dry the seed in shade and sow the same day. The seed should be evenly coated but not left wet for long.

## 5.3 Jivamrit - the living soil inoculum

Jivamrit means 'nectar of life'. It is a microbial culture that is applied to the soil or through irrigation to multiply beneficial microorganisms. These organisms decompose mulch, release nutrients and improve soil structure.

### 5.3.1 Preparation

| Ingredient | Quantity (for one acre) | Role |
|---|---|---|
| Desi cow dung (fresh) | 10 kg | Base microbial source and organic matter. |
| Desi cow urine | 10 litre | Nitrogen, potassium, microbes, mild disinfectant. |
| Jaggery or jaggery powder | 2 kg | Energy source for rapid microbial multiplication. |
| Pulse flour (besan or any pulse) | 2 kg | Protein and nutrient source for microbes. |
| Farm soil from the bund | a handful (about 100 g) | Locally adapted microbial inoculum. |
| Water | 200 litre (in a drum) | Medium for the culture. |

**Method:** Dissolve jaggery in about 200 litres of water in a drum. Add cow dung, cow urine, pulse flour and farm soil. Stir well and allow to ferment for about two to three days in shade, stirring twice a day. The culture is ready when a mild fermenting smell develops. Apply through irrigation water or as a soil drench. Note: the exact ratio and fermentation time vary with climate and source material; the quantities above are the standard ZBNF range.

## 5.4 Mulching

Mulching is the covering of the soil surface with organic material. In natural farming it is not optional - it is a pillar, because it drives the other three.

- **Types:** straw mulch, crop residue mulch, dry leaf mulch, green mulch and live mulch (a cover crop grown in place).
- **Moisture conservation:** mulch cuts evaporation from the soil surface, so the same irrigation goes further.
- **Temperature moderation:** it keeps the soil cooler in summer and warmer in winter, protecting roots and soil life.
- **Weed suppression:** a thick mulch layer blocks light and reduces weed germination.
- **Soil life:** as mulch decomposes it feeds earthworms and microorganisms and builds humus.
- **Erosion control:** mulch absorbs the impact of raindrops and reduces runoff.

## 5.5 Waaphasa - the soil moisture-air balance

Waaphasa (also written Whapasa) is the state in which soil pores hold both air and moisture in the right proportion for plant roots and aerobic microorganisms. It is not merely 'wet soil'. Saturated soil has no air and anaerobic conditions develop; dry soil has no moisture. Waaphasa is the middle, balanced condition.

- Achieved mainly through mulching, which slows moisture loss and prevents surface crusting.
- Judicious, light irrigation rather than flooding helps maintain the balance.
- In a well-aggregated soil with good organic matter, pores of different sizes hold air and water together.
- Aerobic microbes need oxygen; Waaphasa keeps them active and prevents putrefaction.

- Over-irrigation destroys Waaphasa by filling all pores with water and driving out air.

## 5.6 Supporting practices

Beyond the four pillars, natural farming uses several supporting practices.

| Practice | Purpose |
|---|---|
| Achhadana / straw mulching of paths and bunds | Keep the whole farm covered, not only the cropped area. |
| Beejamrit and Jivamrit through drip | Deliver microbes efficiently with irrigation water. |
| Ghana Jivamrit (solid form) | A concentrated solid Jivamrit for dryland and plantation crops. |
| Botanical sprays (neem, nochi, erukku, adathoda) | Ecological pest and disease management. |
| Dravajivamrit / liquid organic preparations | Quick nutrition and microbial boost as a foliar or soil drench. |
| Green manuring and cover cropping | Add nitrogen and organic matter and provide live mulch. |
| Crop-tree-animal integration | Internal nutrient recycling and risk reduction. |
| Indigenous seed conservation | Local adaptation and seed sovereignty. |

**WORKED EXAMPLE - Preparing Bijamrit and Jivamrit for 1 acre**

- Bijamrit: 5 kg desi cow dung + 5 L cow urine + 50-100 g lime + a handful of farm soil, mixed in water to a slurry; coat the seed before sowing.

- Jivamrit: 10 kg desi cow dung + 10 L cow urine + 2 kg jaggery + 2 kg pulse flour + a handful of farm soil in 200 L water; ferment 2-3 days; apply to the soil.

- For 2 acres, double every quantity: Bijamrit 10 kg dung + 10 L urine; Jivamrit 20 kg dung + 20 L urine + 4 kg jaggery + 4 kg pulse flour in about 400 L water.

- Rule of thumb: the ratio of dung, urine, jaggery and pulse flour is kept in a fixed proportion and the water is adjusted to the area.

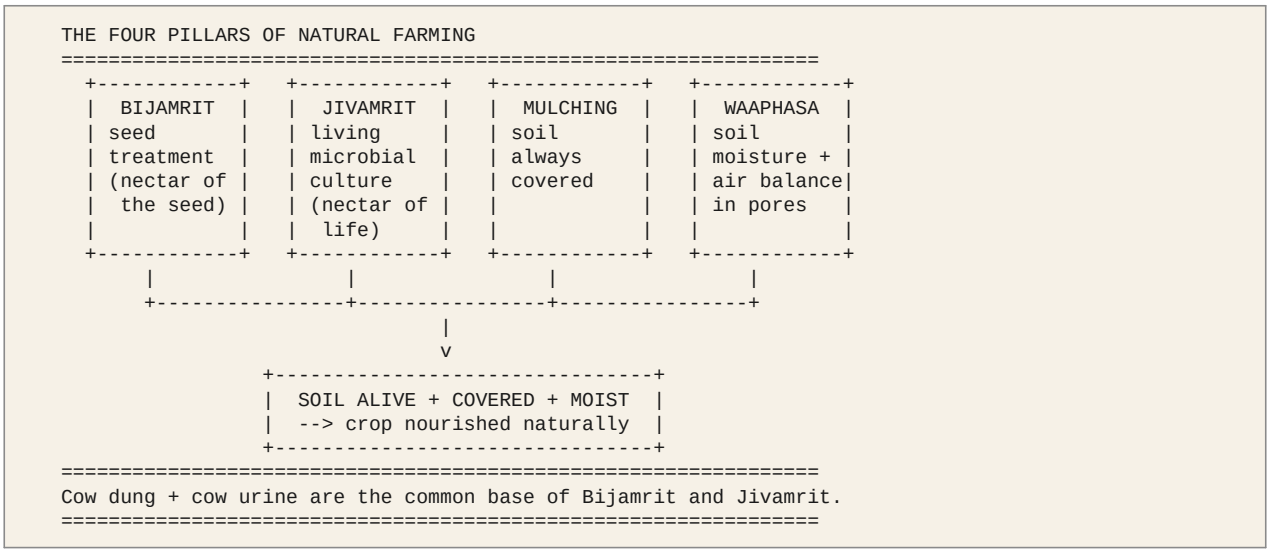



Fig. 5.1 The four pillars and how they combine to nourish the crop.

## EXAM FOCUS

**Key points to remember**

- The four pillars are Bijamrit (seed treatment), Jivamrit (soil microbial culture), Mulching (soil cover) and Waaphasa (moisture-air balance).
- Bijamrit coats the seed with beneficial microbes; typical base is desi cow dung, cow urine and lime with farm soil.
- Jivamrit is a fermented microbial culture of cow dung, cow urine, jaggery, pulse flour and farm soil in water.

- Mulching conserves moisture, moderates temperature, suppresses weeds and feeds soil life.
- Waaphasa is the balanced state of air and water in soil pores; over-irrigation destroys it.
- Cow dung and cow urine are the common base of the preparations.
- Supporting practices include Ghana Jivamrit, botanical sprays, green manuring and drip application.

**Short-answer questions with model answers**

Q1. Name the four pillars of natural farming.

**Model answer:** Bijamrit (seed treatment), Jivamrit (living microbial culture applied to the soil), Mulching (permanent soil cover) and Waaphasa (the balance of moisture and air in soil pores).

Q2. What is Jivamrit and how is it prepared?

**Model answer:** Jivamrit is a living microbial culture applied to the soil. It is prepared by dissolving jaggery in about 200 litres of water and adding desi cow dung, desi cow urine, pulse flour and a handful of farm soil, then fermenting for two to three days in shade with stirring. It multiplies beneficial microorganisms that decompose mulch and mobilise nutrients.

Q3. Define Waaphasa and state why it is important.

**Model answer:** Waaphasa is the state in which soil pores hold air and moisture in the correct proportion for roots and aerobic microbes. It is important because saturated soil becomes anaerobic and dry soil starves the plant; the balanced state keeps aerobic decomposition and root activity at their best. Mulching and light irrigation maintain Waaphasa.

**Long-answer questions with model answer outlines**

Q1. Explain the four pillars of natural farming with their composition and function.

**Outline:**

- Bijamrit: meaning, composition (cow dung, cow urine, lime, farm soil), function (seed protection, germination).
- Jivamrit: meaning, composition (cow dung, cow urine, jaggery, pulse flour, farm soil, water), function (microbial multiplication, decomposition, nutrient mobilisation).
- Mulching: types and functions (moisture, temperature, weeds, soil life, erosion).
- Waaphasa: definition and importance, role of mulching and irrigation.
- How the four work together.
- Conclusion: they keep the soil alive, covered and moist, replacing external inputs.

Q2. Describe the preparation of Bijamrit and Jivamrit for one acre and explain their role in soil fertility.

**Outline:**

- Bijamrit method: mix cow dung and urine with water, add lime, farm soil and jaggery, coat the seed.
- Jivamrit method: dissolve jaggery in 200 L water, add dung, urine, pulse flour, farm soil, ferment 2-3 days, apply.
- Role in fertility: microbial inoculation, decomposition of mulch, nitrogen fixation and nutrient mobilisation, improved soil structure.
- Scaling to larger areas by proportion.
- Precautions: use desi cow products, shade fermentation, apply fresh.
- Conclusion: these preparations convert on-farm materials into a substitute for purchased fertiliser.

## 5.7 Precautions and quality checks for preparations

- **Use fresh desi cow dung and urine:** stale material loses microbial activity.

- **Ferment in shade:** direct sunlight can kill microorganisms and dry the culture.
- **Stir twice daily:** stirring aerates the culture and promotes aerobic microbes.
- **Do not over-ferment:** excessive fermentation produces a foul smell and reduces quality.
- **Use clean water:** chlorinated water can harm the microbes; allow it to stand before use.
- **Apply fresh:** use Jivamrit within a short period of preparation; its value declines with storage.
- **Filter before spraying:** particles clog sprayer nozzles.
- **Scale by proportion:** keep the ratio of dung, urine, jaggery and pulse flour fixed and adjust the water to the area.

A simple quality check: a well-prepared Jivamrit has a mild sweet-fermenting smell, a slightly frothy surface and no foul odour. If it smells bad, discard and prepare again.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 The four pillars are Bijamrit, Jivamrit, Mulching and ____.

2 Jivamrit contains cow dung, cow urine, jaggery and ____ flour.

3 Bijamrit is a treatment applied to the ____.

4 Mulching reduces ____ from the soil surface.

5 Waaphasa is the balance of air and ____ in soil pores.

6 Over-irrigation destroys ____ by filling all pores with water.

### B. State whether true or false

1 Jivamrit is a chemical fertiliser. (False - it is a microbial culture)

2 Mulching suppresses weeds by blocking light. (True)

3 Bijamrit is applied to the soil, not the seed. (False - it is a seed treatment)

4 Waaphasa means fully saturated soil. (False - it is a balance of air and moisture)

### C. Match the following

| Column A | Column B |
|---|---|
| Bijamrit | Seed treatment |
| Jivamrit | Living microbial soil culture |
| Mulching | Permanent soil cover |
| Waaphasa | Moisture-air balance |
| Ghana Jivamrit | Solid form of Jivamrit |

### D. Answer in one word or one line

1 Name the four pillars of natural farming.

2 Name four ingredients of Jivamrit.

3 What does Bijamrit mean?

4 Give two benefits of mulching.

**Practical application in Tamil Nadu - preparing inputs locally**
- Jivamrit drums are commonly set up near the cattle shed in natural farms across the state.
- Desi cow breeds such as Kangayam, Umbalachery and Pulikulam are valued for their dung and urine.
- Rice straw from the delta is used as mulch and in Jivamrit preparation.



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# Lecture 6. Scope and Importance of Natural Farming

**Learning objectives**

- Explain the economic, environmental and social significance of natural farming.
- Assess its scope in Indian agriculture and its contribution to the Sustainable Development Goals.
- Identify the conditions under which natural farming is most suitable.

## 6.1 Introduction

Scope means the range of situations in which a practice can be applied and the benefits it can deliver. Importance means why it matters. Natural farming has a wide scope in India because most Indian farms are small, rain-fed or partially irrigated, and run by families who cannot absorb the rising cost of purchased inputs. It also matters globally because agriculture is both a victim of and a contributor to environmental degradation.

## 6.2 Economic significance

- **Lower cost of cultivation:** fertiliser and pesticide, usually the largest cash cost, are replaced by on-farm preparations.
- **Higher net income:** even where gross yield is similar, the reduction in cost raises net return per acre.
- **Price premium:** chemical-free and certified natural produce often fetches a higher price in niche and export markets.
- **Reduced risk:** the farmer is less exposed to input price shocks and to crop failure from input misuse.
- **Less debt:** reduced borrowing for inputs and reduced dependence on moneylenders.
- **Diversified income:** integration of crops, livestock, trees, honey, mushrooms and value-added products gives several income streams.
- **Employment:** input preparation, mulching, processing and local marketing create rural jobs.
- **Foreign exchange:** certified organic and natural produce has export potential.

## 6.3 Environmental significance

- **Soil health:** rising organic carbon, better structure, higher microbial and earthworm activity.
- **Water:** less chemical contamination of groundwater and surface water; better infiltration and recharge.
- **Biodiversity:** more pollinators, natural enemies and soil organisms; conservation of indigenous varieties.
- **Climate mitigation:** carbon sequestration in soil and lower greenhouse gas emissions.
- **Climate resilience:** soils with more organic matter withstand drought and heavy rain better.
- **Reduced pollution:** no pesticide drift, no nitrate leaching, no residue in food.

## 6.4 Social significance

- **Farmer health:** fewer pesticide-poisoning incidents and lower exposure to agrochemicals.
- **Consumer health:** safe, residue-free, often more nutritious food.

- **Food sovereignty:** farmers control their own seed and inputs instead of depending on the market.
- **Knowledge revival:** traditional and local knowledge becomes valuable again.
- **Women's participation:** input preparation, seed conservation and home gardens involve and empower women.
- **Community strength:** group farming, community seed banks and participatory certification build social capital.

## 6.5 Scope in India and Tamil Nadu

| Situation | Suitability of natural farming |
|---|---|
| Small and marginal holdings | High - low cash cost suits limited capital. |
| Rain-fed and dryland areas | High - mulching and moisture conservation reduce drought risk. |
| Tribal and hilly regions | High - long tradition of organic practices and rich local seed. |
| Intensive irrigated paddy and sugarcane | Moderate - possible, but needs careful water and mulch management in the transition. |
| Plantation crops (coconut, mango, cashew) | High - mulching, cover crops and intercropping work well. |
| Horticulture and vegetables | High - premium market and quick returns from value addition. |
| Peri-urban and market-linked farms | High - direct and niche markets reward chemical-free produce. |

## 6.6 Natural farming and the Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) are seventeen global goals for 2030. Natural farming contributes directly to several of them.

| SDG | How natural farming contributes |
|---|---|
| SDG 1 - No poverty | Lower input cost and diversified income raise net farm income. |
| SDG 2 - Zero hunger | Stable, diverse, low-cost food production and better household nutrition. |
| SDG 3 - Good health | Residue-free food and reduced farmer exposure to pesticides. |
| SDG 6 - Clean water | Less chemical runoff and leaching; better infiltration and recharge. |
| SDG 8 - Decent work | Rural employment in input preparation, processing and marketing. |
| SDG 12 - Responsible consumption | Sustainable production and reduced chemical use. |
| SDG 13 - Climate action | Carbon sequestration and lower emissions; climate resilience. |
| SDG 15 - Life on land | Soil health, biodiversity and conservation of indigenous species. |

## 6.7 Limitations and the honest picture

A balanced answer also notes the difficulties, which examiners appreciate.

- Yield may fall in the first one to three years of transition while the soil rebuilds its biology.
- The system is knowledge-intensive; wrong preparation or timing of inputs can reduce results.
- Mulch material may be scarce where crop residues are used for cattle feed or fuel.
- Certification and market linkage are still developing for natural farming produce.
- Large-scale evidence on long-term yield is still being built; claims should be treated with care.

**WORKED EXAMPLE - Economics of scope: net return per acre**

- Conventional paddy, one acre, one season: gross value of produce Rs. 45,000; input cost (fertiliser, pesticide, seed, labour) about Rs. 26,000; net return about Rs. 19,000.
- Natural farming paddy, one acre: gross value of produce Rs. 43,000 (slightly lower yield, small price premium); input cost (seed, some labour, on-farm inputs) about Rs. 11,000; net return about Rs. 32,000.
- Difference: net return is higher by about Rs. 13,000 per acre per season despite a lower gross value, because the input cost is much lower.
- Note: figures are illustrative for a teaching example; actual values depend on season, variety, price and location.

## EXAM FOCUS

**Key points to remember**

- Scope means where natural farming can be applied; importance means why it matters.
- Economic: lower cost, higher net income, premium price, less debt, diversified income.
- Environmental: soil health, clean water, biodiversity, climate mitigation and resilience.
- Social: farmer and consumer health, food sovereignty, knowledge revival, women's participation.
- It fits small, rain-fed, tribal, plantation and horticulture farms best.
- It contributes to SDGs 1, 2, 3, 6, 8, 12, 13 and 15.
- Limitations include a transition yield dip, knowledge intensity, mulch scarcity and weak market linkage.

**Short-answer questions with model answers**

Q1. List any five Sustainable Development Goals supported by natural farming.

**Model answer:** SDG 1 No poverty, SDG 2 Zero hunger, SDG 3 Good health, SDG 6 Clean water, SDG 13 Climate action. SDG 8, 12 and 15 are also supported. Natural farming raises net income, produces safe food, protects water, and sequesters carbon in soil.

Q2. Why is natural farming especially suited to small and marginal farmers?

**Model answer:** Because its cash input cost is very low, which suits farmers with limited capital; it does not require borrowing for fertiliser and pesticide; it can use family labour and local materials; and it gives diversified output for home consumption and local sale, reducing risk.

Q3. State any three limitations of natural farming.

**Model answer:** Yield may decline for one to three years during the transition while soil biology is rebuilt; the system is knowledge-intensive and errors in preparing inputs reduce results; and mulch material and market or certification linkage may be limited. Long-term large-scale yield evidence is still being built.

**Long-answer questions with model answer outlines**

Q1. Discuss the economic, environmental and social significance of natural farming.

**Outline:**

- Introduction: define scope and importance.
- Economic significance: cost, net income, premium, debt, diversification, employment.
- Environmental significance: soil, water, biodiversity, climate mitigation and resilience.
- Social significance: health, food sovereignty, knowledge, women, community.
- Scope in India and Tamil Nadu with examples of suitable situations.
- Limitations for balance.
- Conclusion: natural farming is significant on all three fronts.

Q2. Explain the contribution of natural farming to the Sustainable Development Goals and its scope in India.

**Outline:**

- Introduction: the SDGs and agriculture's role.
- Goal-wise contribution: 1, 2, 3, 6, 8, 12, 13, 15 with one line each.
- Scope: small holdings, rain-fed areas, tribal regions, plantations, horticulture, peri-urban farms.
- Conditions needed: knowledge, mulch availability, market linkage.
- Conclusion: natural farming aligns production with the SDG agenda.

## 6.8 Natural farming and the Indian farmer's economy

India has a very large number of small and marginal holdings, and a large share of farm households carry debt. The economics of natural farming must be judged against this reality.

- Small and marginal holdings dominate Indian agriculture, so low-cost methods have wide relevance.
- Input cost, especially fertiliser and pesticide, is a large share of the cash cost for many crops.
- Removing this cost reduces the need to borrow at the start of the season.
- Diversified output (grain, milk, vegetables, fruit) improves household food security.
- A price premium, where available, further improves net return.
- The main constraint is the knowledge and labour needed, and the market linkage for the produce.

Note: precise national figures on indebtedness and holding size change with each agricultural census; the general pattern above is what matters for the examination.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Natural farming reduces the ____ of cultivation.

2 Chemical-free produce often earns a price ____.

3 Natural farming contributes to SDG ____ (zero hunger).

4 Natural farming is well suited to ____ and marginal holdings.

5 A limitation of natural farming is a possible ____ dip during transition.

6 Natural farming protects ____ by reducing chemical runoff.

### B. State whether true or false

1 Natural farming always gives higher yield than conventional farming in the first year. (False - yield may dip initially)

2 Natural farming can earn a premium price in niche markets. (True)

3 Natural farming has no social benefits. (False - it improves health and food sovereignty)

4 Natural farming reduces the farmer's debt burden. (True)

### C. Match the following

| Column A | Column B |
|---|---|
| SDG 2 | Zero hunger |
| SDG 6 | Clean water |
| SDG 13 | Climate action |
| SDG 15 | Life on land |
| SDG 3 | Good health |

### D. Answer in one word or one line

1 Name three economic benefits of natural farming.

2 Name two environmental benefits.

3 Which farms are best suited to natural farming?

4 State one limitation of natural farming.

**Practical application in Tamil Nadu - scope in the state**

- Rain-fed millet and pulse areas of the southern districts suit natural farming well.
- Hill and tribal areas of the Western and Eastern Ghats have a long organic tradition.
- Plantation crops such as coconut, mango and cashew benefit from mulching and intercropping.



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# Lecture 7. Methods and Schools of Natural Farming

**Learning objectives**

- Describe the major schools of natural farming and their unique principles.
- Compare Fukuoka natural farming, ZBNF, Rishi Krishi and Korean Natural Farming.
- Recognise the practices shared by all schools and those that differ.

## 7.1 Introduction

Several schools of natural farming have developed in different countries. They share the core idea of working with nature and avoiding synthetic chemicals, but they differ in their specific techniques, their attitude to external organic inputs and their philosophical base. This lecture compares the four main schools named in the syllabus.

## 7.2 Fukuoka natural farming (Japan)

Developed by Masanobu Fukuoka on his farm in Japan, this is the most radical of the schools. It is also called 'Do-nothing' farming or the natural way of farming.

- **No tillage:** the soil is never ploughed.
- **No chemical fertiliser and no prepared organic fertiliser:** fertility comes from returning straw and from a green manure crop such as clover grown with the main crop.
- **No weeding by tillage or herbicide:** weeds are managed by timing, a clover ground cover and temporary flooding in paddy.
- **No pesticides:** the balanced ecosystem keeps pests in check.
- **Seed balls:** seeds are enclosed in clay balls and broadcast, so no nursery or transplanting is needed.
- **Philosophy:** the farm should be as close to nature as possible; human effort is reduced to observation and minimal timely intervention.

## 7.3 Subhash Palekar's Zero Budget Natural Farming (India)

ZBNF is the most widely adopted school in India. It grew from Palekar's observation that natural forests are fertile without external inputs.

- **Four pillars:** Bijamrit (seed treatment), Jivamrit (microbial soil inoculum), Mulching and Waaphasa (soil moisture-air balance).
- **Desi cow based:** the dung and urine of the indigenous cow are considered essential, because of their rich microbial load.
- **Zero external cash input:** the aim is to eliminate purchased fertiliser and pesticide.
- **No tillage and permanent mulch:** soil is kept covered and undisturbed.
- **Crop diversity and multi-storey systems:** mixed and layered cropping is encouraged.
- **Farmer-to-farmer training:** knowledge spreads through farmer training camps rather than formal institutions.

## 7.4 Rishi Krishi (India)

Rishi Krishi, or 'farming of the sages', is a school based on ancient Indian texts and the traditional practices of rishis. It emphasises the sacred and holistic dimension of farming.

- **Rooted in Vedic and Puranic texts:** practices are drawn from Krishi-Parashara, Vrikshayurveda and traditional wisdom.
- **Amritpani and Amritjal:** liquid preparations of cow dung, cow urine, jaggery and gram flour used as soil and foliar inputs.
- **Cow-centred:** the indigenous cow is central; panchagavya and other preparations are used.
- **Spiritual and ritual element:** the farm is treated as sacred; timings may follow traditional calendars.
- **No chemical inputs and no deep tillage;** emphasis on soil health and the farmer's ethical relationship with the land.
- **Relevance:** it is closest to India's heritage and is often grouped with natural farming under traditional practices.

## 7.5 Korean Natural Farming (KNF)

Korean Natural Farming was developed by Cho Han-Kyu in South Korea. It is a highly technical system built around cultivating and using local microorganisms.

- **Indigenous Micro-Organisms (IMO):** collected from a forest floor, multiplied in cooked rice and then in a larger substrate, and applied to the soil.
- **Oriented to local organisms:** the aim is to strengthen the microbes already adapted to the local environment.
- **Fermented plant and fruit extracts:** preparations such as FPJ (Fermented Plant Juice) and FFJ (Fermented Fruit Juice) supply nutrition.
- **Mineral and fish preparations:** e.g. Oriental Herbal Nutrient (OHN), water-soluble calcium and phosphorus preparations.
- **Lactic acid bacteria and yeast:** used in preparations for decomposition and soil conditioning.
- **Systematic and measurable:** it uses specific recipes and stages, unlike the more philosophical Fukuoka approach.
- **No synthetic chemicals:** the entire input set is made from local natural materials.



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## 7.6 Comparison of the schools

| Basis | Fukuoka natural | ZBNF (Palekar) | Rishi Krishi | Korean Natural Farming |
|---|---|---|---|---|
| Origin | Japan | India (Maharashtra) | India (traditional/Vedic) | South Korea |
| Founder | Masanobu Fukuoka | Subhash Palekar | Based on ancient texts and gurus | Cho Han-Kyu |
| Central idea | Do-nothing; let nature work | Four pillars; desi cow based | Farming of the sages; sacred cow | Cultivate local microorganisms |
| Tillage | None | None | None to minimal | Minimal |
| Main inputs | Straw, clover, seed balls | Bijamrit, Jivamrit, mulch, Waaphasa | Amritpani, Amritjal, panchagavya | IMO, FPJ, FFJ, OHN, LAB |
| External organic input | Discouraged | Minimised | Minimised | Minimised; local materials only |
| Cow dependency | Not central | Essential (desi cow) | Essential (desi cow) | Not central; uses local resources |
| Signature practice | Seed balls, clover ground cover | Jivamrit and mulching | Amritpani and ritual calendar | IMO collection from forest |
| Philosophical base | Very strong; minimal intervention | Practical; farmer training | Spiritual and Vedic | Technical and systematic |

## 7.7 Common ground among the schools

Despite their differences, all four schools agree on the essentials, and these are the points to write in an examination.

- No synthetic chemical fertiliser and no synthetic pesticide.
- Soil life is the foundation of fertility; keep it alive.
- Soil should be covered and protected, not left bare.
- Crop diversity is preferred over monoculture.
- Inputs should be made from local, on-farm materials.
- The farmer should be self-reliant rather than dependent on the market.
- Long-term soil and ecosystem health is the goal.

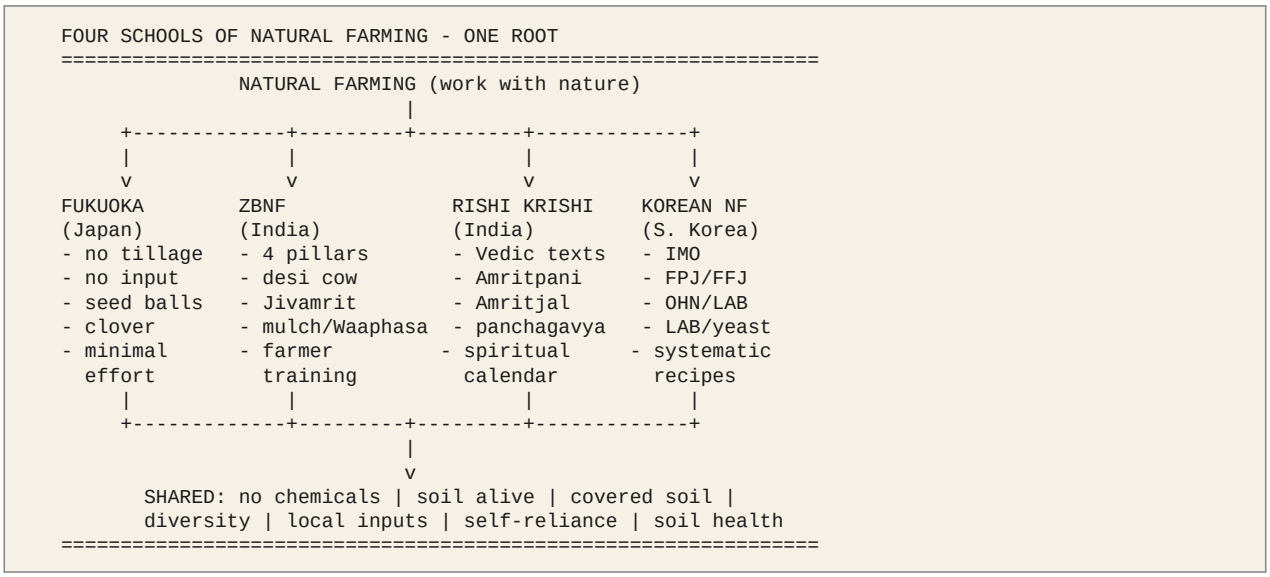



Fig. 7.1 The four schools and their shared principles.

**WORKED EXAMPLE - Choosing a school for a Tamil Nadu farm**

- A dryland farmer in Virudhunagar with a few desi cows and limited capital: ZBNF suits best - Jivamrit and mulching use the cattle and crop residues available on the farm.
- A traditional paddy grower near Kumbakonam interested in the least possible labour: Fukuoka-style no-till paddy with clover and straw return is an option, though it needs careful water management.
- A farmer with a strong cultural and religious orientation and access to panchagavya: Rishi Krishi fits the farm's ethos.
- A horticulture grower willing to learn precise recipes: Korean Natural Farming's IMO and FPJ preparations give a structured input programme.
- Note: most Indian farmers mix methods; the four pillars of ZBNF are the most common practical base.

## EXAM FOCUS

**Key points to remember**

- Four main schools: Fukuoka natural farming, ZBNF, Rishi Krishi and Korean Natural Farming.
- Fukuoka: no tillage, no input, no weeding by tillage, no pesticides, seed balls, clover cover, minimal intervention.
- ZBNF: four pillars, desi cow based, zero external cash input, farmer-to-farmer training.
- Rishi Krishi: Vedic base, Amritpani and Amritjal, panchagavya, sacred-cow-centred, spiritual calendar.
- Korean Natural Farming: Indigenous Micro-Organisms, FPJ, FFJ, OHN, LAB, systematic recipes.
- All schools share no chemicals, live soil, covered soil, diversity, local inputs and self-reliance.
- Indian farmers often combine methods, usually around the ZBNF pillars.

**Short-answer questions with model answers**

Q1. What is Korean Natural Farming and what is its signature technique?

**Model answer:** Korean Natural Farming, developed by Cho Han-Kyu in South Korea, is a technical system built on cultivating and using local microorganisms. Its signature technique is the collection and multiplication of Indigenous Micro-Organisms (IMO) from a forest floor, along with fermented plant and fruit juice preparations, Oriental Herbal Nutrient and lactic acid bacteria.

Q2. State the four principles of Fukuoka's natural farming.

**Model answer:** No tillage; no chemical fertiliser and no prepared organic fertiliser; no weeding by tillage or herbicide; and no dependence on pesticides. Fertility comes from returning straw and a clover ground cover, and weeds are managed by timing and ground cover rather than by cultivation.

Q3. Name two preparations used in Rishi Krishi.

**Model answer:** Amritpani and Amritjal - liquid preparations of cow dung, cow urine, jaggery and gram flour used as soil and foliar inputs. Rishi Krishi also uses panchagavya, a preparation of five cow products.

**Long-answer questions with model answer outlines**

Q1. Describe the four main schools of natural farming and compare their principles.

**Outline:**

- Fukuoka natural farming: origin, four principles, seed balls, clover, philosophy.
- ZBNF: origin, four pillars, desi cow base, zero external cash input, farmer training.
- Rishi Krishi: Vedic base, Amritpani/Amritjal/panchagavya, sacred cow, spiritual calendar.
- Korean Natural Farming: IMO, FPJ, FFJ, OHN, LAB, systematic recipes.
- Comparison table on origin, founder, tillage, inputs, cow dependency, signature practice.

- Common ground and conclusion.

Q2. Explain ZBNF and Korean Natural Farming in detail and state how they differ.

**Outline:**

- ZBNF: Palekar, four pillars, desi cow, mulching, Waaphasa, no tillage, zero budget.
- KNF: Cho Han-Kyu, IMO collection, FPJ/FFJ, OHN, LAB, structured recipes.
- Differences: cow dependency, philosophical vs technical, input recipes, method of microbial cultivation.
- Similarities: no chemicals, local inputs, soil biology, self-reliance.
- Conclusion: both are natural farming schools with different techniques.

## 7.8 Convergence of schools and hybrid practice

In practice, farmers rarely follow one school strictly. They combine methods to suit their resources, climate and market. This convergence is a strength.

- A farmer may use ZBNF's Jivamrit and mulching with Korean Natural Farming's IMO for a horticulture crop.
- Panchagavya from the Rishi Krishi tradition is used widely even by ZBNF farmers.
- Fukuoka's no-till and cover-crop ideas combine well with ZBNF's four pillars.
- Botanical sprays from traditional practice are used across all schools.
- The common thread - no chemicals, live soil, diversity and local inputs - remains the same.

For the examination, describe the schools separately but note that farmers often combine them, which shows a practical understanding of the subject.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Fukuoka's method is also called ____ farming.

2 ZBNF stands for ____.

3 Rishi Krishi uses preparations called Amritpani and ____.

4 Korean Natural Farming is built on ____ Micro-Organisms.

5 FPJ stands for Fermented ____ Juice.

6 Panchagavya is a preparation of ____ cow products.

### B. State whether true or false

1 Korean Natural Farming was developed in Japan. (False - in South Korea, by Cho Han-Kyu)

2 Rishi Krishi is based on ancient Indian texts. (True)

3 Fukuoka's method uses a clover ground cover. (True)

4 ZBNF does not use cow dung. (False - desi cow dung is essential)

### C. Match the following

| Column A | Column B |
|---|---|
| Fukuoka | Seed balls and clover cover |
| ZBNF | Four pillars |
| Rishi Krishi | Amritpani and panchagavya |
| Korean Natural Farming | Indigenous Micro-Organisms |
| Palekar | Founder of ZBNF |

### D. Answer in one word or one line

1 Name the four schools of natural farming.

2 What is a seed ball?

3 Name two preparations used in Rishi Krishi.

4 What is IMO in Korean Natural Farming?

**Practical application in Tamil Nadu - schools in practice**

- ZBNF is the most widely practised school in the state because of its simple four-pillar method.
- Panchagavya is widely used by organic farmers in Tamil Nadu.
- Korean Natural Farming methods are adopted by some horticulture growers for precise input programmes.

# Lecture 8. Characteristics and Design of a Natural Farm

**Learning objectives**

- Describe how a natural farm is structured and laid out.
- Explain crop diversity, livestock integration and natural resource use.
- Design a simple natural farm layout for a small holding.

## 8.1 Introduction

A natural farm is not simply a field where chemicals are avoided. It is designed as an ecosystem, with diversity of plants and animals, a permanent soil cover, water conservation structures and internal nutrient recycling. Good design reduces the work needed to manage the farm and increases its stability.

## 8.2 Characteristics of a natural farm

- **Diversity of crops:** cereals, pulses, oilseeds, vegetables, fodder and trees grown together or in rotation.
- **Integration of livestock:** cattle, goats, poultry or fish provide dung, urine, draught power and income.
- **Permanent soil cover:** mulch or live cover on the cropped area and on bunds.
- **No chemical input:** all nutrition and protection from on-farm and local materials.
- **Water conservation structures:** farm ponds, bunds, trenches, recharge pits and drip or sprinkler where feasible.
- **Tree component:** fruit trees, timber trees, fodder trees and nitrogen-fixing trees on boundaries and as intercrops.
- **Seed and input self-reliance:** own seed and own preparations.
- **Habitat for biodiversity:** flowering plants, bund vegetation and undisturbed patches for pollinators and natural enemies.
- **Internal nutrient cycling:** residues, dung and crop by-products returned to the soil.
- **Low external dependence and family-labour based operation.**

## 8.3 Design principles

| Design principle | What it means in practice |
|---|---|
| Mimic nature | Use the structure of a natural forest: ground cover, shrubs, trees and animals together. |
| Maximise diversity | Grow many species and varieties to spread risk and support beneficial organisms. |
| Close the nutrient loop | Return all crop residues, dung and urine to the soil; minimise export of nutrients. |
| Use vertical space | Multi-storey cropping with tall trees, medium shrubs and ground crops. |
| Conserve water | Harvest rainfall, mulch heavily and use efficient irrigation. |
| Use local resources | Prefer local seed, local cow breeds and locally available organic material. |
| Plan for resilience | Spread crops and income streams so that one failure does not sink the farm. |
| Integrate components | Crop, livestock and tree components should support each other, not compete. |

## 8.4 Crop diversity and cropping systems

Diversity is the backbone of a natural farm. It is achieved through several cropping systems used at the same time.

| Cropping system | Description | Benefit in a natural farm |
|---|---|---|
| Mixed cropping | Two or more crops grown together in the same field. | Spreads risk, uses different soil layers, confuses pests. |
| Intercropping | A main crop with one or more intercrops in rows. | Extra income and nitrogen fixation if a legume is included. |
| Crop rotation | Different crops in sequence over seasons. | Breaks pest and disease cycles; balances nutrients. |
| Multi-storey cropping | Layers of trees, shrubs and ground crops. | Uses vertical space; mimics a forest; high total biomass. |
| Strip cropping | Alternate strips of different crops. | Reduces erosion and pest spread. |
| Cover cropping / green manure | Legumes grown to cover soil and then incorporated or mulched. | Adds nitrogen and organic matter; live mulch. |

## 8.5 Livestock integration

Livestock convert crop residues and fodder into dung, urine, milk and draught power, and they close the nutrient loop on the farm.

- **Desi cattle:** source of dung and urine for Bijamrit and Jivamrit, draught power and farmyard manure; their dung is preferred in natural farming for its microbial richness.
- **Goats and sheep:** convert shrubs and crop residues into manure and provide income.
- **Poultry:** eat insects and weed seeds, provide manure and eggs, and reduce pest pressure.
- **Fish and ducks:** in farm ponds, they use farm waste and add nutrients to pond silt.
- **Bee keeping:** pollination service and honey income; strongly supported by crop diversity.

## 8.6 Layout of a natural farm

A well-designed natural farm arranges its components so that they support one another. The text diagram below shows a typical layout for a small holding.

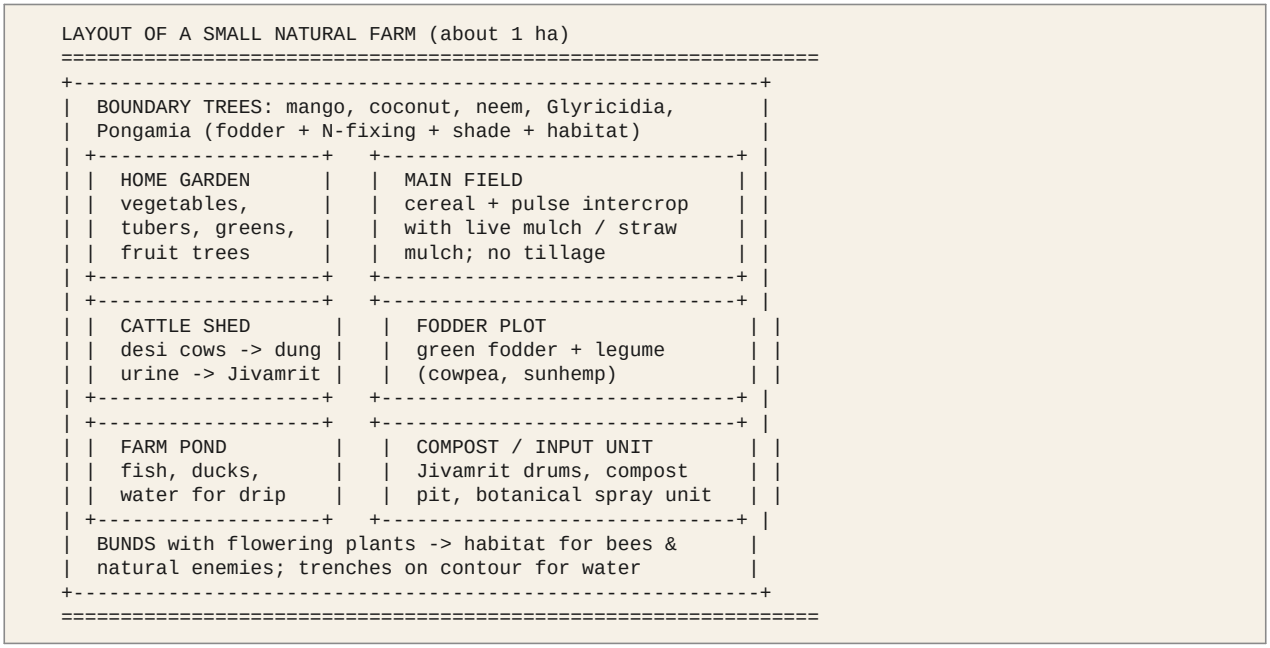


Fig. 8.1 A designed natural farm: fields, home garden, cattle, fodder, pond, input unit and boundary trees.

## 8.7 Natural resource use on a natural farm

| Resource | How a natural farm uses it |
|---|---|
| Soil | Kept covered and undisturbed; fed with mulch and Jivamrit; protected from erosion. |
| Water | Rainfall harvested in ponds and trenches; mulch reduces evaporation; drip or sprinkler used sparingly. |
| Crop residues | Used as mulch, compost and cattle feed - not burnt. |
| Cattle dung and urine | Converted into Jivamrit, Bijamrit and compost. |
| Sunlight | Captured by multi-storey cropping and used for drying and solar equipment. |
| Trees | Provide fodder, fuel, fruit, shade, nitrogen fixation and habitat. |
| Local biodiversity | Natural enemies, pollinators and soil organisms are treated as free farm labour. |

**WORKED EXAMPLE - Designing a one-hectare natural farm**

- Divide the holding: about 0.6 ha main field, 0.15 ha home garden, 0.1 ha fodder plot, 0.05 ha farm pond, 0.1 ha input and cattle area and boundary trees.
- Main field: cereal + legume intercrop with straw mulch; rotate with a different family each season.
- Cattle: keep two or three desi cows; their dung and urine supply all Bijamrit and Jivamrit needs.
- Water: dig a farm pond to harvest runoff; use it to irrigate the home garden and fodder plot by drip.
- Inputs: set up three or four Jivamrit drums and a compost pit near the cattle shed.
- Result: the farm produces grain, pulses, vegetables, fodder, milk and fruit, and recycles nearly all its nutrients internally.

## EXAM FOCUS

**Key points to remember**

- A natural farm is designed as an ecosystem, not just a chemical-free field.
- Key characteristics: diversity, livestock integration, permanent cover, no chemicals, water structures, trees, seed self-reliance and biodiversity habitat.
- Design principles: mimic nature, maximise diversity, close the nutrient loop, use vertical space, conserve water, use local resources, plan for resilience.
- Cropping systems used together: mixed, inter, rotation, multi-storey, strip and cover cropping.
- Livestock - especially desi cattle - supply dung and urine for inputs and close the nutrient loop.
- A good layout arranges field, garden, cattle, fodder, pond, input unit and boundary trees to support one another.
- All natural resources are used in a way that recycles and conserves.

**Short-answer questions with model answers**

Q1. What are the main characteristics of a natural farm?

**Model answer:** Crop diversity, integration of livestock, permanent soil cover with mulch, no chemical input, water conservation structures, a tree component, seed and input self-reliance, habitat for biodiversity and internal nutrient cycling, all operated with low external dependence.

Q2. Why is livestock integration important in a natural farm?

**Model answer:** Livestock, especially desi cattle, supply the dung and urine needed for Bijamrit and Jivamrit, provide draught power and farmyard manure, convert crop residues and fodder into useful products, and close the nutrient loop by returning nutrients to the soil. They also add income and reduce risk.

Q3. Name any four cropping systems used in a natural farm.

**Model answer:** Mixed cropping, intercropping, crop rotation and multi-storey cropping. Strip cropping and cover cropping or green manuring are further systems used.

**Long-answer questions with model answer outlines**

Q1. Describe the characteristics and design of a natural farm with a suitable layout.

**Outline:**

- Introduction: a natural farm is a designed ecosystem.
- Characteristics: diversity, livestock, cover, no chemicals, water structures, trees, self-reliance, habitat, recycling.
- Design principles: mimic nature, diversity, closed nutrient loop, vertical space, water conservation, local resources, resilience.
- Layout: field, home garden, cattle shed, fodder plot, farm pond, input unit, boundary trees and bund vegetation.
- Natural resource use.
- Conclusion: good design reduces work and increases stability.

Q2. Explain crop diversity and livestock integration in the design of a natural farm.

**Outline:**

- Crop diversity: cropping systems (mixed, inter, rotation, multi-storey, strip, cover), benefits.
- Livestock integration: desi cattle, goats, poultry, fish, ducks, bees; roles and products.
- How they support each other: residues to cattle, dung to soil, fodder from farm, pest control by poultry.
- Nutrient loop closure and risk reduction.
- Conclusion: diversity plus livestock is the core of natural farm design.

## 8.8 Common mistakes in farm design

- **Too many animals for the available fodder:** the farm then has to buy feed, defeating self-reliance.
- **Monoculture despite the design:** a single crop ignores the diversity principle.
- **No water storage:** without a pond or recharge structure, the farm cannot buffer a dry spell.
- **Leaving the soil bare:** mulch must cover the soil, including bunds and paths.
- **Placing the cattle shed far from the field:** increases the labour of carrying dung and urine.
- **Ignoring boundary trees:** trees provide fodder, fuel, shade, nitrogen and habitat at little cost.
- **No input unit:** without drums and a compost pit, on-farm input production is difficult.
- **Planning for the short term only:** trees and soil health need a multi-year view.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 A natural farm integrates crops, ____ and animals.

2 Multi-storey cropping mimics the structure of a ____.

3 Desi cattle supply dung and ____ for natural inputs.

4 A farm ____ harvests runoff for supplementary irrigation.

5 Nitrogen-fixing trees such as ____ provide fodder and fix nitrogen.

6 Poultry help control ____ and weeds on the farm.

### B. State whether true or false

1 A natural farm should be a monoculture for efficiency. (False - diversity is the backbone)

2 Boundary trees provide fodder, fuel and habitat. (True)

3 Livestock integration helps close the nutrient loop. (True)

4 Water conservation structures are not needed in natural farms. (False - they are important)

### C. Match the following

| Column A | Column B |
|---|---|
| Home garden | Vegetables, tubers and fruit trees |
| Cattle shed | Source of dung and urine |
| Farm pond | Harvests runoff and supports fish |
| Input unit | Jivamrit drums and compost pit |
| Boundary trees | Fodder, fuel, shade and habitat |

### D. Answer in one word or one line

1 Name three components of a natural farm layout.

2 Why is crop diversity important?

3 Name two nitrogen-fixing trees used in natural farms.

4 What is the role of a farm pond?

**Practical application in Tamil Nadu - designing a natural farm**

- Homesteads in the state commonly follow a multi-storey pattern with coconut, banana, spices and vegetables.
- Farm ponds are promoted under watershed development and irrigation schemes.
- Boundary planting with Glyricidia and Pongamia is common for fodder and nitrogen.

# Lecture 9. Mid Semester Examination

This lecture is the mid-semester examination of the course. It is placed here in the syllabus to mark the point at which the first half of the course - Lectures 1 to 8 - is assessed. The examination covers the heritage, history, concept, principles, pillars, scope, schools and design of natural farming. The purpose of this chapter is to consolidate revision for that examination.

## 9.1 Coverage of the mid-semester examination

The mid-semester examination tests the following units.

| Unit | Lecture | Weightage to prepare |
|---|---|---|
| Indian heritage of ancient agriculture | 1 | Definitions, examples, sources |
| History and evolution of natural farming | 2 | Timeline, key persons, movements |
| Concept, definition and objectives | 3 | Definition, comparison, objectives |
| Characteristics and principles | 4 | Principles, natural processes |
| Main pillars | 5 | Composition and function of four pillars |
| Scope and importance | 6 | Economic, environmental, social, SDGs |
| Methods and schools | 7 | Four schools and comparison |
| Design of a natural farm | 8 | Characteristics, layout, integration |

## 9.2 Revision summary - the first half in one page

- **Heritage:** Indian agriculture is about ten thousand years old; cattle-based fertility, mixed cropping, tank irrigation and botanical pest control are its core practices; sources include Krishi-Parashara, the Arthashastra and Vrikshayurveda.
- **History:** global organic movement (Howard, Steiner, Balfour, Rodale), Fukuoka's natural farming, Korean Natural Farming, then Palekar's ZBNF; the Green Revolution raised output but degraded resources.
- **Concept:** natural farming works with nature, avoids synthetic inputs and soil disturbance, and relies on soil biology; it differs from organic farming in minimising external inputs and tillage.
- **Objectives:** ecological (soil, water, biodiversity, climate) and socio-economic (cost, income, health, self-reliance).
- **Principles:** no tillage, no chemical input, minimum disturbance, permanent cover, Waaphasa, reliance on natural processes, diversity, self-reliance.
- **Pillars:** Bijamrit (seed treatment), Jivamrit (microbial soil inoculum), Mulching (cover) and Waaphasa (moisture-air balance).
- **Scope:** best suited to small, rain-fed, tribal, plantation and horticulture farms; supports several SDGs.
- **Schools:** Fukuoka, ZBNF, Rishi Krishi and Korean Natural Farming; all avoid chemicals and rely on soil biology.
- **Design:** diversity, livestock integration, permanent cover, water structures, trees and internal nutrient recycling.

## 9.3 Model question paper

The following model paper follows the usual TNAU pattern for a mid-semester examination in a theory course. Marks and duration are indicative.

**Model question paper (indicative)**

- Time: 1.5 hours. Maximum marks: 50. Answer all questions in Part A, any two in Part B and any one in Part C.

- PART A - Objective and very short answer (10 x 1 = 10 marks):

- 1. Name the four pillars of natural farming. 2. Who wrote 'The One-Straw Revolution'? 3. Expand ZBNF. 4. What is Waaphasa? 5. Name two traditional water-harvesting systems of Tamil Nadu. 6. What is Bijamrit? 7. Name one nitrogen-fixing tree used in natural farms. 8. What is IMO in Korean Natural Farming? 9. Define mulching. 10. Name one SDG supported by natural farming.

- PART B - Short answer (2 x 5 = 10 marks):

- 11. Differentiate natural farming from organic farming. 12. Explain the role of cow dung and cow urine in natural farming. 13. Describe any two cropping systems used in a natural farm. 14. Write short notes on Rishi Krishi.

- PART C - Long answer (1 x 15 = 15 marks):

- 15. Explain the four pillars of natural farming with their composition and function. 16. Describe the history and evolution of natural farming in India. 17. Discuss the design of a natural farm with a suitable layout.

- PART D - Application (1 x 15 = 15 marks):

- 18. Prepare a plan for converting a one-hectare chemically farmed holding in your district into a natural farm.

## 9.4 Examination technique

- Read the whole paper first and budget time in proportion to marks.
- For comparison questions, always draw a table - it earns marks quickly and clearly.
- For 'explain with composition' questions, give ingredients, quantities and function.
- Use Indian and Tamil Nadu examples wherever possible; they show application.
- Draw a simple text diagram (pillars, principles wheel, farm layout) for long answers on those topics.
- Define the key term in the first line of every answer.
- Leave time to re-read; correct spelling of technical terms such as Jivamrit, Bijamrit and Waaphasa.

## EXAM FOCUS

**Key points to remember**

- The mid-semester examination covers Lectures 1 to 8.
- Expect a table-based comparison of natural, organic and conventional farming.
- The four pillars and their composition are almost certain to appear.
- Be ready to name key persons: Fukuoka, Palekar, Howard, Cho Han-Kyu.
- Know the ZBNF preparations and their ingredients.
- Be able to sketch a natural farm layout and a principles wheel.
- Use Indian and Tamil Nadu examples in every long answer.

**Short-answer questions with model answers**

Q1. How should a comparison question be answered?

**Model answer:** By drawing a table with a 'basis' column and one column for each system being compared, then filling rows such as chemical input, tillage, mulching, source of nutrition, input cost and certification. A table earns marks quickly and shows clarity.

Q2. Which topics are most likely to appear as long answers in the mid-semester examination?

**Model answer:** The four pillars of natural farming with composition and function; the history and evolution of natural farming; the characteristics and design of a natural farm with a layout; and the comparison of natural, organic and conventional farming.

Q3. Why should Indian examples be used in answers?

**Model answer:** Because they demonstrate application of the theory to real conditions, which examiners reward. Examples such as Palekar's ZBNF, Andhra Pradesh community-managed natural farming, Tamil Nadu's eri tanks and PGS-India certification show that the student can connect the concept to practice.

**Long-answer questions with model answer outlines**

Q1. Write a complete answer on the main pillars of natural farming.

**Outline:**

- Introduction: name the four pillars and their origin in ZBNF.
- Bijamrit: definition, composition, method, function.
- Jivamrit: definition, composition, method, function.
- Mulching: types and functions.
- Waaphasa: definition, importance, how maintained.
- How the pillars work together; supporting practices.
- Conclusion.

Q2. Describe how you would convert a chemically farmed one-hectare holding into a natural farm.

**Outline:**

- Baseline survey: soil test, water source, cattle, crops, residue availability.
- Plan components: main field, home garden, fodder, pond, cattle, input unit, boundary trees.
- Year 1: introduce mulching, Jivamrit and Bijamrit; stop pesticide first, then reduce fertiliser gradually.
- Year 2-3: build diversity, add trees and livestock, strengthen water structures.
- Inputs: prepare Jivamrit, Bijamrit, botanical sprays; conserve own seed.
- Monitoring: soil organic carbon, yield, cost, net income.
- Conclusion: transition is gradual and the farm becomes self-reliant.

# Lecture 10. Ecological Balance, Ecological Engineering and Community Responsibility

**Learning objectives**

- Explain ecological balance and how natural farming maintains it.
- Define ecological engineering and describe its tools in a natural farm.
- Describe the collective community role in sustainability and biodiversity conservation.

## 10.1 Ecological balance

Ecological balance is the state in which the living and non-living parts of an ecosystem are in stable relationship, so that no single species or process runs out of control. In a farm, balance means that pests do not multiply without limit because predators keep them down, that nutrients are recycled rather than lost, and that the soil, plants, animals and microorganisms support one another. Natural farming seeks this balance deliberately; conventional farming, by killing pests with broad-spectrum poisons and by adding nutrients artificially, tends to disturb it.

## 10.2 How natural farming maintains ecological balance

| Mechanism | How balance is maintained |
|---|---|
| Predator-prey relationship | Natural enemies (ladybird beetles, spiders, lacewings, predatory bugs, birds) keep pest numbers low. |
| Parasitism and pathogens | Parasitoids such as Trichogramma and entomopathogens such as Beauveria and Metarhizium control pests. |
| Competition | Diverse plant cover competes with weeds and reduces their dominance. |
| Nutrient cycling | Decomposers convert residues into plant nutrients, preventing loss and buildup. |
| Pollination | Bees, butterflies and other pollinators sustain seed and fruit set. |
| Soil food web | Bacteria, fungi, protozoa, nematodes and earthworms regulate each other and release nutrients. |
| Habitat and refuge | Undisturbed patches, bund vegetation and trees shelter beneficial organisms. |

## 10.3 Ecological engineering

Ecological engineering means designing the farm so that the environment itself supports natural pest control and pollination. Instead of relying on a pesticide, the farmer manipulates the habitat so that natural enemies thrive. This is the modern, scientific form of the traditional practice of leaving refuge vegetation.

### 10.3.1 Tools of ecological engineering

- **Habitat manipulation:** growing flowering plants on bunds and borders to provide nectar and pollen to adult parasitoids and predators.
- **Banker plants:** plants that carry alternative prey or hosts, on which natural enemies build up before the pest arrives.
- **Trap crops:** attractive plants that draw pests away from the main crop and are then destroyed.
- **Border and refuge vegetation:** strips of grass, shrubs and trees that shelter enemies and pollinators.
- **Beetle banks:** raised grassy strips that shelter ground beetles and spiders.
- **Bird perches and nest boxes:** encourage insect-eating birds.

- **Sacred groves and undisturbed patches:** preserve the local gene pool of beneficial organisms.
- **Water bodies:** farm ponds support dragonflies, frogs and other predators.

### 10.3.2 Examples of plants used

| Plant | Role in ecological engineering |
|---|---|
| Marigold (Tagetes) | Trap crop for nematodes; nectar for beneficial insects. |
| Sunflower and cosmos | Nectar and pollen for parasitoids and predators. |
| Castor | Trap crop for Spodoptera; host for some parasitoids. |
| Cowpea and other legumes | Nitrogen fixation and cover; support predatory mites. |
| Glyricidia and Pongamia | Nitrogen fixation, fodder, shade, perching sites for birds. |
| Tulsi, nochi and other herbs | Botanical source and aromatic pest repellent. |
| Sunhemp (Crotalaria) | Green manure and trap crop for nematodes. |

## 10.4 Community responsibility

Ecological balance cannot be maintained by one farmer alone, because pests, water and biodiversity cross farm boundaries. Community action is therefore essential.

- **Group farming and clusters:** neighbouring farmers adopting natural farming together prevent re-infestation from chemically farmed fields and enable collective marketing.
- **Community seed banks:** conserve indigenous varieties and ensure seed availability for all members.
- **Community water management:** the traditional Kudimaramathu model of joint tank and channel maintenance.
- **Participatory certification:** the PGS-India system is built on peer inspection by farmer groups.
- **Joint pest and disease monitoring:** early warning and coordinated action across a village.
- **Biodiversity conservation:** protecting sacred groves, village trees and common lands as reservoirs of beneficial species.
- **Knowledge sharing:** farmer-to-farmer training, field days and demonstration plots.
- **Collective bargaining:** groups get better prices and can meet the volume demanded by buyers.

## 10.5 Biodiversity conservation as a community duty

Biodiversity is the variety of living things - crops, livestock, trees, insects, microbes and wildlife. Natural farming depends on it and also protects it.

| Level of biodiversity | Examples on a natural farm | Community role |
|---|---|---|
| Genetic diversity | Land races and indigenous varieties of paddy, millets, pulses, vegetables | Seed banks, seed exchanges, conservation of land races |
| Species diversity | Many crops, trees, livestock breeds, bees, natural enemies | Avoid monoculture belts; maintain common land |
| Ecosystem diversity | Fields, ponds, bunds, groves, homesteads | Protect tanks, groves and village forests |

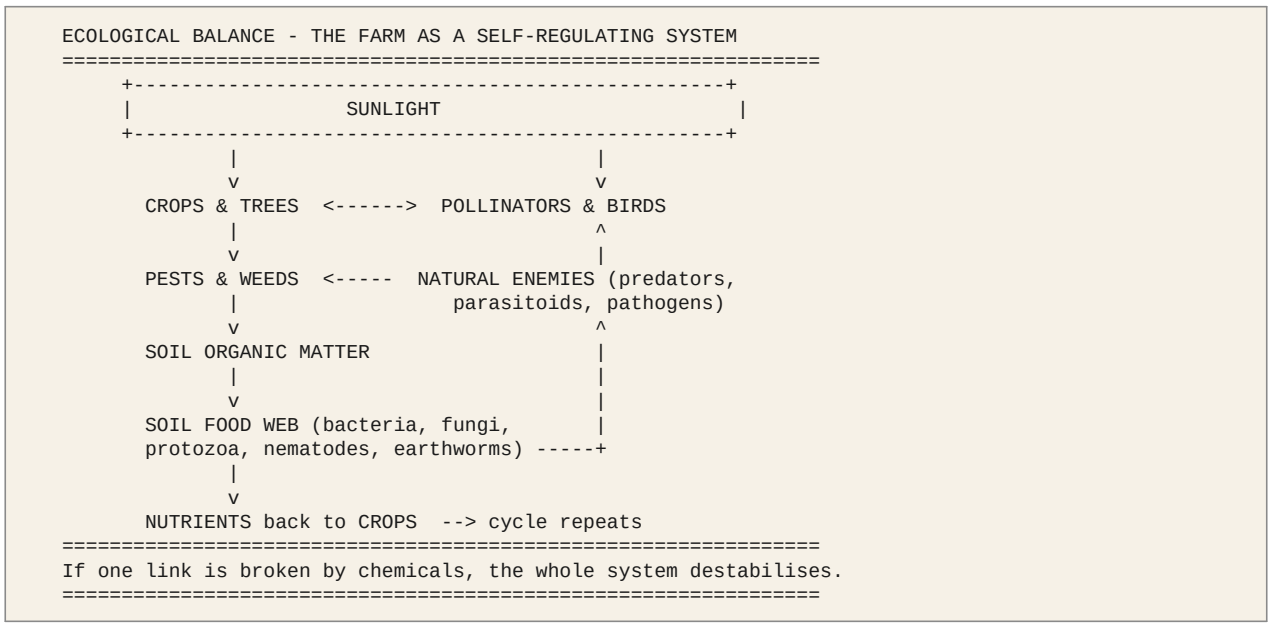



Fig. 10.1 Ecological balance as a closed, self-regulating cycle on the farm.

**WORKED EXAMPLE - Ecological engineering on one acre of cotton**

- Border: two rows of castor as a trap crop for Spodoptera, and marigold as a trap for nematodes and nectar source.
- Bunds: flowering plants such as sunflower and cosmos to feed adult parasitoids; grassy strips as beetle banks.
- Intercrop: cowpea as a cover and nitrogen source, supporting predatory mites.
- Bird perches: 10 perches per acre to attract insectivorous birds.
- Result: pest pressure is reduced so that a botanical spray is needed only if the pest crosses the action threshold, not on a fixed calendar.

## EXAM FOCUS

**Key points to remember**

- Ecological balance is the stable relationship among the living and non-living parts of the farm ecosystem.
- Natural farming maintains balance through predators, parasitoids, competition, nutrient cycling, pollination and the soil food web.
- Ecological engineering designs the habitat so that natural enemies and pollinators thrive.
- Tools include habitat manipulation, banker plants, trap crops, border vegetation, beetle banks, bird perches, ponds and undisturbed patches.
- Community responsibility is essential because pests, water and biodiversity cross farm boundaries.
- Community tools: group farming, seed banks, joint water management, participatory certification and knowledge sharing.
- Biodiversity operates at genetic, species and ecosystem levels, and all three need community protection.

**Short-answer questions with model answers**

Q1. Define ecological engineering and give two examples.

**Model answer:** Ecological engineering is the design of the farm environment so that natural pest control and pollination are supported by the habitat itself. Examples: growing flowering plants such as sunflower and marigold on bunds to provide nectar and pollen to parasitoids and predators, and growing a trap crop such as castor to draw Spodoptera away from the main crop.



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Q2. Why is community responsibility important in natural farming?

**Model answer:** Because pests, water, pollinators and biodiversity move across farm boundaries, so one farmer acting alone cannot maintain ecological balance. Group adoption prevents re-infestation from neighbouring fields, community seed banks secure indigenous seed, joint water management maintains tanks and channels, and participatory certification and collective marketing strengthen the whole village.

Q3. Name three tools of ecological engineering.

**Model answer:** Habitat manipulation with flowering plants on bunds; trap crops such as marigold and castor; and banker plants that carry alternative prey for natural enemies. Beetle banks, bird perches and undisturbed refuge patches are further tools.

**Long-answer questions with model answer outlines**

Q1. Explain ecological balance and ecological engineering, and describe how a natural farm maintains both.

**Outline:**

- Ecological balance: definition and why it matters.
- Mechanisms: predators, parasitoids, competition, nutrient cycling, pollination, soil food web, habitat.
- Ecological engineering: definition and purpose.
- Tools: habitat manipulation, banker plants, trap crops, border vegetation, beetle banks, perches, ponds, groves.
- Examples of plants and their roles.
- Conclusion: design replaces chemicals.

Q2. Discuss the role of the community in maintaining ecological balance and conserving biodiversity in natural farming.

**Outline:**

- Introduction: ecological processes cross farm boundaries.
- Group farming and clusters to prevent re-infestation and enable marketing.
- Community seed banks and conservation of land races.
- Community water management (Kudimaramathu model).
- Participatory certification and joint pest monitoring.
- Protection of groves, tanks and common lands; knowledge sharing.
- Conclusion: natural farming is a community enterprise as much as a farm practice.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Ecological balance is maintained by natural ____ such as predators and parasitoids.

2 Designing the habitat to support natural enemies is called ecological ____.

3 A ____ crop draws pests away from the main crop.

4 A ____ plant hosts alternative prey for natural enemies.

5 Community labour for tank maintenance is called ____.

6 Biodiversity operates at genetic, species and ____ levels.

### B. State whether true or false

1 Ecological engineering replaces pesticides with habitat design. (True)

2 One farmer alone can maintain ecological balance regardless of neighbours. (False - processes cross farm boundaries)

3 Trap crops protect the main crop. (True)

4 Participatory certification is part of the PGS-India system. (True)

### C. Match the following

| Column A | Column B |
|---|---|
| Predators | Keep pest numbers low |
| Trap crop | Draws pests away from main crop |
| Banker plant | Hosts alternative prey |
| Seed bank | Conserves indigenous varieties |
| PGS-India | Peer-based certification |

### D. Answer in one word or one line

1 Define ecological balance.

2 Define ecological engineering.

3 Name two tools of ecological engineering.

4 Why is community action important?

**Practical application in Tamil Nadu - community initiatives**

- Farmer groups and clusters adopt natural farming together to prevent re-infestation from neighbouring fields.
- Community seed banks conserve traditional paddy and millet varieties.
- Joint tank maintenance follows the traditional Kudimaramathu model.

# Lecture 11. Integration of Crops, Trees and Animals in Natural Farming

**Learning objectives**

- Explain integrated farming systems and their components.
- Describe how crops, trees and animals recycle resources on a natural farm.
- Show how integration builds resilience and farm sustainability.

## 11.1 Integrated farming system

An integrated farming system (IFS) combines two or more enterprises - crops, livestock, trees, poultry, fish, bees or mushrooms - on the same farm so that the waste of one becomes the input of another. On a natural farm, integration is not just an add-on; it is how the nutrient loop is closed and how risk is spread.

## 11.2 Components and their interactions

| Component | Provides | Receives |
|---|---|---|
| Crops | Grain, fodder, crop residues, mulch | Dung and urine based inputs, compost |
| Trees | Fodder, fuel, fruit, timber, shade, nitrogen | Compost, mulch, water |
| Cattle and buffaloes | Dung, urine, milk, draught power | Fodder, crop residues, water |
| Goats and sheep | Manure, meat, income | Fodder, shrubs, crop residues |
| Poultry | Manure, eggs, meat, insect control | Grain, kitchen waste, insects |
| Fish and ducks | Protein, pond silt as fertiliser | Farm waste, pond water |
| Bees | Honey, pollination | Nectar and pollen from diverse crops and trees |
| Mushrooms | Protein, income | Crop residues and straw |

## 11.3 The crop-tree-animal nutrient cycle

The heart of integration is a cycle in which nutrients move from soil to plant to animal and back to soil, losing as little as possible.

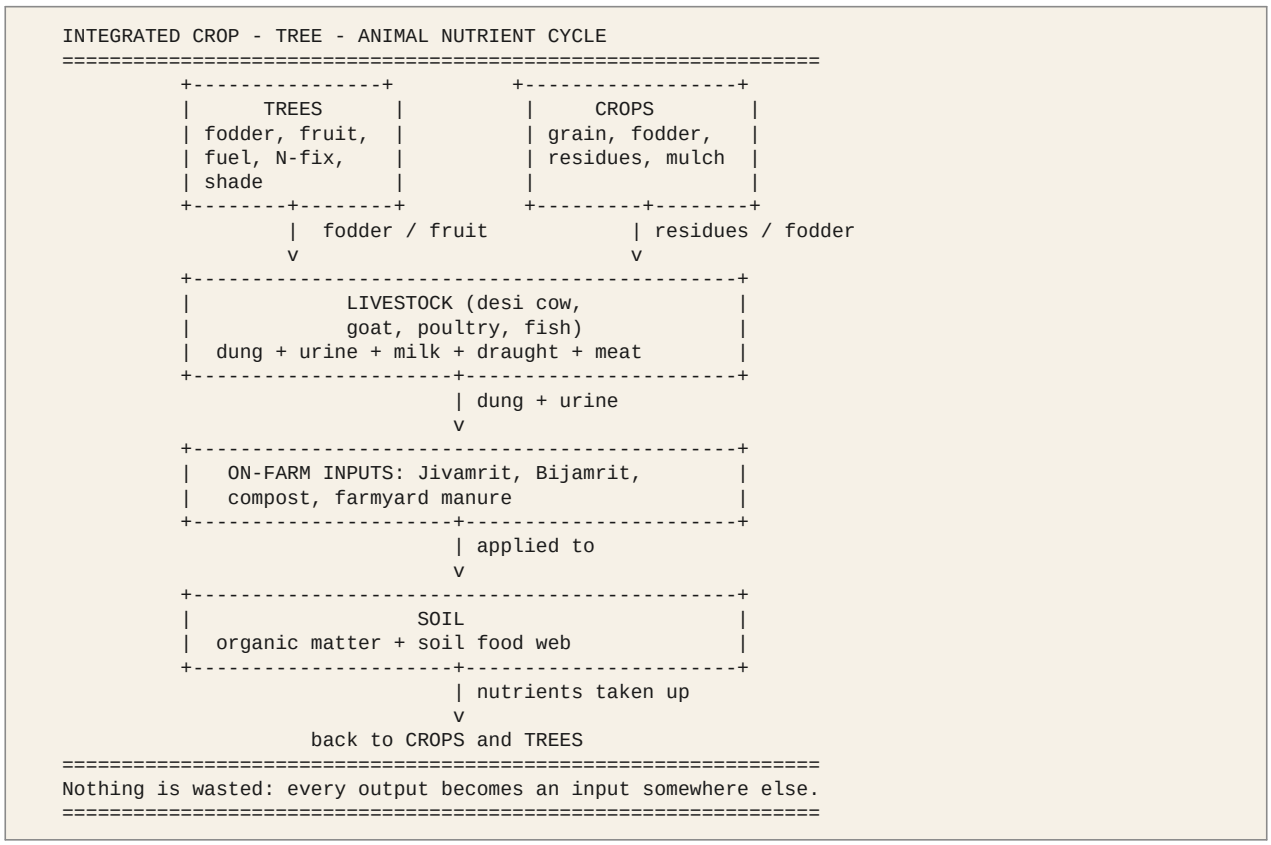



Fig. 11.1 The integrated crop-tree-animal nutrient cycle.

## 11.4 Common integrated models

| Model | Components | Suitability |
|---|---|---|
| Crop + dairy | Field crops, fodder, two to five desi cows | Most small holdings; supplies dung and urine for inputs. |
| Crop + poultry + vegetables | Cereal, poultry, home garden | Homesteads; poultry controls insects and weeds. |
| Crop + goat + tree | Field crops, goats, fodder and fruit trees | Dryland and hilly areas. |
| Crop + fish + duck | Paddy or pond, fish, ducks | Low-lying and tank-fed areas; ducks control pests and add nutrients. |
| Agri-horti-forestry | Field crops with fruit and timber trees | Rain-fed and degraded lands; long-term income with short-term crops. |
| Crop + bees | Diverse crops and trees with bee boxes | Any diverse farm; pollination plus honey income. |
| Multi-storey homestead | Coconut, banana, spices, vegetables, tubers, livestock | Tamil Nadu homesteads; high output from small area. |

## 11.5 Benefits of integration

- **Resource recycling:** residues, dung and urine are used on the farm instead of being wasted or bought outside.
- **Reduced cost:** the farm supplies its own inputs, fodder and much of its food.
- **Diversified income:** grain, milk, eggs, meat, fruit, honey and timber give several income streams.
- **Risk reduction:** if one enterprise fails, others support the household.
- **Improved soil:** continuous addition of manure and residues builds organic matter.



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- **Employment:** livestock and tree components need and provide year-round labour.
- **Nutrient balance:** livestock convert low-value residues into high-value manure.
- **Climate resilience:** trees provide shade and reduce heat stress; diversity buffers weather shocks.

## 11.6 Designing integration on a small farm

Integration should be planned so that components match the farm's resources and the family's needs. The steps below are a practical guide.

1. Assess resources: land, water, labour, capital, existing livestock and trees.
2. Identify the family's needs: food, fodder, fuel, cash and risk tolerance.
3. Choose a main enterprise (usually the crop) and supporting enterprises (dairy, poultry, trees).
4. Match component sizes: the number of animals should match the fodder and residue available, so that the farm does not need to buy feed.
5. Locate components for efficiency: cattle shed near the field and input unit; fodder plot close to the shed; pond downhill to harvest runoff.
6. Plan the nutrient flow: dung and urine to the input unit; residues to mulch or feed; pond silt to the field.
7. Add trees on boundaries and as intercrops for fodder, nitrogen and shade.
8. Monitor and adjust each season based on yields, costs and income.

**WORKED EXAMPLE - Nutrient budget of an integrated one-hectare farm**

- Cattle: 3 desi cows producing about 8-10 kg dung and 8-10 L urine per cow per day.
- Dung per year: about 3 cows x 9 kg x 365 = about 9,850 kg (say 10 tonnes) of fresh dung.
- Urine per year: about 3 x 9 L x 365 = about 9,850 L of urine - ample for the season's Jivamrit and Bijamrit needs.
- If half the dung is composted and applied, and urine is used in preparations, the farm supplies most of its nitrogen and potassium requirement from internal sources.
- Crop residues from 0.6 ha of cereals and pulses: about 2-3 tonnes of straw, used as mulch and cattle feed, which returns carbon and nutrients to the soil.
- Result: the farm's nutrient exports (grain, milk) are substantially offset by internal recycling, reducing the need for any external input.
- Note: these are indicative teaching figures; actual values depend on animal size, feed quality and crop yields.

## EXAM FOCUS

**Key points to remember**

- An integrated farming system combines crops, livestock, trees and other enterprises so that the waste of one becomes the input of another.
- Components interact: crops give residues and fodder; animals give dung, urine, milk and power; trees give fodder, fuel, fruit, nitrogen and shade.
- The crop-tree-animal nutrient cycle closes the loop from soil to plant to animal and back to soil.
- Common models: crop+dairy, crop+poultry+vegetables, crop+goat+tree, crop+fish+duck, agri-horti-forestry, crop+bees, multi-storey homestead.
- Benefits: resource recycling, lower cost, diversified income, risk reduction, better soil, employment and climate resilience.

- Design should match component sizes to available fodder and residues.
- Integration is the practical way to achieve self-reliance in natural farming.

**Short-answer questions with model answers**

Q1. What is an integrated farming system?

**Model answer:** An integrated farming system combines two or more enterprises, such as crops, livestock, trees, poultry, fish or bees, on the same farm so that the by-product or waste of one enterprise becomes the input of another. This closes the nutrient loop, reduces cost, diversifies income and spreads risk.

Q2. Name any four integrated farming models suitable for Tamil Nadu.

**Model answer:** Crop plus dairy, crop plus poultry plus vegetables, crop plus fish plus duck, and multi-storey homestead with coconut, banana, spices, vegetables and livestock. Agri-horti-forestry and crop plus bees are further models.

Q3. How does livestock integration close the nutrient loop?

**Model answer:** Livestock consume crop residues and fodder grown on the farm and convert them into dung and urine, which are used to prepare Jivamrit, Bijamrit and compost. These are applied back to the soil, so nutrients that would otherwise be exported or wasted return to the crop. Nothing leaves the farm as waste.

**Long-answer questions with model answer outlines**

Q1. Explain the integration of crops, trees and animals in natural farming with a nutrient-cycle diagram.

**Outline:**

- Introduction: definition and purpose of integration.
- Components and what each provides and receives.
- The crop-tree-animal nutrient cycle drawn as a diagram.
- Common integrated models with suitability.
- Benefits: recycling, cost, income, risk, soil, employment, resilience.
- Conclusion: integration is how a natural farm becomes self-reliant.

Q2. Describe how you would design an integrated natural farm for a one-hectare holding and estimate its nutrient budget.

**Outline:**

- Resource assessment and family needs.
- Choice of main and supporting enterprises.
- Matching animal numbers to fodder and residues.
- Layout: shed, fodder plot, pond, input unit, boundary trees.
- Nutrient budget: dung, urine, residues and their use.
- Monitoring and adjustment.
- Conclusion: internal recycling reduces external inputs.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 An integrated farming system combines crops, livestock and ____.

2 Livestock convert crop residues into ____ and urine.

3 Trees provide fodder, fuel, fruit and ____ fixation.

4 In crop plus fish plus duck systems, ducks control ____ and add nutrients.

5 Bee keeping provides pollination and ____ income.

6 The crop-tree-animal cycle closes the ____ loop.

### B. State whether true or false

1 Integrated farming increases risk by concentrating on one enterprise. (False - it spreads risk)

2 Matching animal numbers to available fodder reduces the need to buy feed. (True)

3 Trees compete with crops and are never useful. (False - they provide many benefits)

4 Integration reduces the cost of inputs. (True)

### C. Match the following

| Column A | Column B |
|---|---|
| Crop + dairy | Dung and urine for inputs |
| Crop + poultry | Insect and weed control |
| Crop + fish + duck | Low-lying and tank-fed areas |
| Agri-horti-forestry | Short and long-term income together |
| Multi-storey homestead | High output from a small area |

### D. Answer in one word or one line

1 Define integrated farming system.

2 Name three components of an integrated farm.

3 Give two benefits of integration.

4 How does livestock close the nutrient loop?

**Practical application in Tamil Nadu - integrated models**

- Crop plus dairy is the most common integrated model on smallholdings in the state.
- Coconut-based multi-storey homesteads combine coconut, banana, spices, vegetables and livestock.
- Crop plus fish plus duck suits the tank-fed low-lying areas of the Cauvery delta.

# Lecture 12. Biodiversity, Indigenous Seeds and Farm Waste Recycling

**Learning objectives**

- Explain the importance of biodiversity and the conservation of indigenous species.
- Describe indigenous seed conservation and seed treatment practices.
- Explain how farm wastes are converted into inputs for soil enrichment.

## 12.1 Biodiversity on the farm

Biodiversity is the variety of life - genes, species and ecosystems. On a farm it includes crops, livestock breeds, trees, insects, birds, microbes and the ecosystems of field, pond and grove. High biodiversity makes a farm stable: more species means more ways for the system to function if one is disturbed.

- **Genetic diversity:** different varieties and land races within a crop, and different breeds of livestock.
- **Species diversity:** many kinds of plants and animals on the farm.
- **Ecosystem diversity:** fields, bunds, ponds, groves, homesteads and wetlands.
- **Functional biodiversity:** the organisms that do useful work - pollinators, natural enemies, decomposers, nitrogen fixers.

## 12.2 Indigenous seeds

Indigenous seeds are the varieties developed and maintained by farmers over generations. They are also called land races or traditional varieties. They are adapted to local soil, climate and pests, and they carry genetic diversity that modern varieties often lack.

### 12.2.1 Why indigenous seeds matter

- Adapted to local conditions, so they perform well with low inputs and in variable rainfall.
- More tolerant of pests, diseases and drought than many high-yielding varieties.
- Better taste, cooking quality and nutrition in many cases.
- Genetic diversity is the raw material for future crop improvement.
- They give the farmer independence from purchased seed and protect seed sovereignty.
- Many are suited to mixed cropping and to the farmer's own cropping calendar.

### 12.2.2 Tamil Nadu examples

| Crop | Indigenous varieties / land races (examples) |
|---|---|
| Paddy | Traditional varieties such as Kallurundai Samba, Karuppu Kavuni (black rice), Mappillai Samba, Kuzhiyadichan, Samba and Seeraga Samba. |
| Millets | Kambu (pearl millet), Kelvaragu (finger millet), Varagu (kodo), Samai (little millet), Thinai (foxtail) and Panivaragu (proso) land races. |
| Pulses | Traditional cowpea, blackgram, greengram and redgram land races. |
| Oilseeds | Traditional groundnut, sesame (gingelly) and castor varieties. |
| Vegetables | Local brinjal, chilli, okra, gourds and country tomato land races. |

Note: the variety names above are widely known Tamil Nadu land races; the exact list held by any seed bank varies, and students should treat these as examples rather than a complete catalogue.

### 12.2.3 Seed conservation and treatment

- **Community seed banks:** farmer groups store and exchange seed of indigenous varieties.
- **On-farm conservation:** growing the variety every year keeps it alive and adapted.
- **Seed selection:** selecting the best plants and grain for seed maintains vigour.
- **Seed treatment:** Bijamrit, cow urine, dung slurry, buttermilk and botanical extracts protect the seed and improve germination.
- **Storage:** dry seed stored with neem leaves, ash or in airtight containers to prevent insect damage.
- **Seed exchange and seed festivals:** events that distribute and celebrate indigenous seed.

## 12.3 Farm waste recycling

A natural farm treats waste as a resource. Crop residues, animal dung, kitchen waste and weeds are all converted into inputs, so that nothing is burnt or thrown away and no nutrient is lost.

### 12.3.1 Types of farm waste and their use

| Farm waste | Recycling method | Product and use |
|---|---|---|
| Crop residues and straw | Mulching, composting, cattle feed, mushroom substrate | Mulch, compost, manure, mushrooms |
| Cattle dung | Composting, Jivamrit, Bijamrit, vermicompost | Organic manure and microbial inoculum |
| Cattle urine | Jivamrit, Bijamrit, botanical sprays, pest repellent | Liquid fertiliser and repellent |
| Green weeds and prunings | Green manure, compost, liquid manure | Nitrogen and organic matter |
| Kitchen and market waste | Composting, vermicomposting | Nutrient-rich compost |
| Poultry litter | Composting, direct application | Nitrogen-rich manure |
| Pond silt | Applied to fields | Nutrient-rich silt |
| Banana pseudostem and other crop trash | Composting, mulch, bio-enzyme | Mulch and liquid fertiliser |

### 12.3.2 Composting methods

- **Indore method:** layered composting of residues with dung and urine, turned periodically - the method of Albert Howard.
- **Bangalore method:** pit composting suited to drier conditions.
- **Vermicomposting:** earthworms convert organic waste into nutrient-rich vermicompost.
- **NADEP composting:** a low-cost method using biomass, dung and soil in an aerated pit.
- **Waste decomposer / microbial consortia:** cultures that speed up decomposition of residues.
- **Liquid manures:** Jivamrit, panchagavya and dung-urine extracts as quick nutrition.

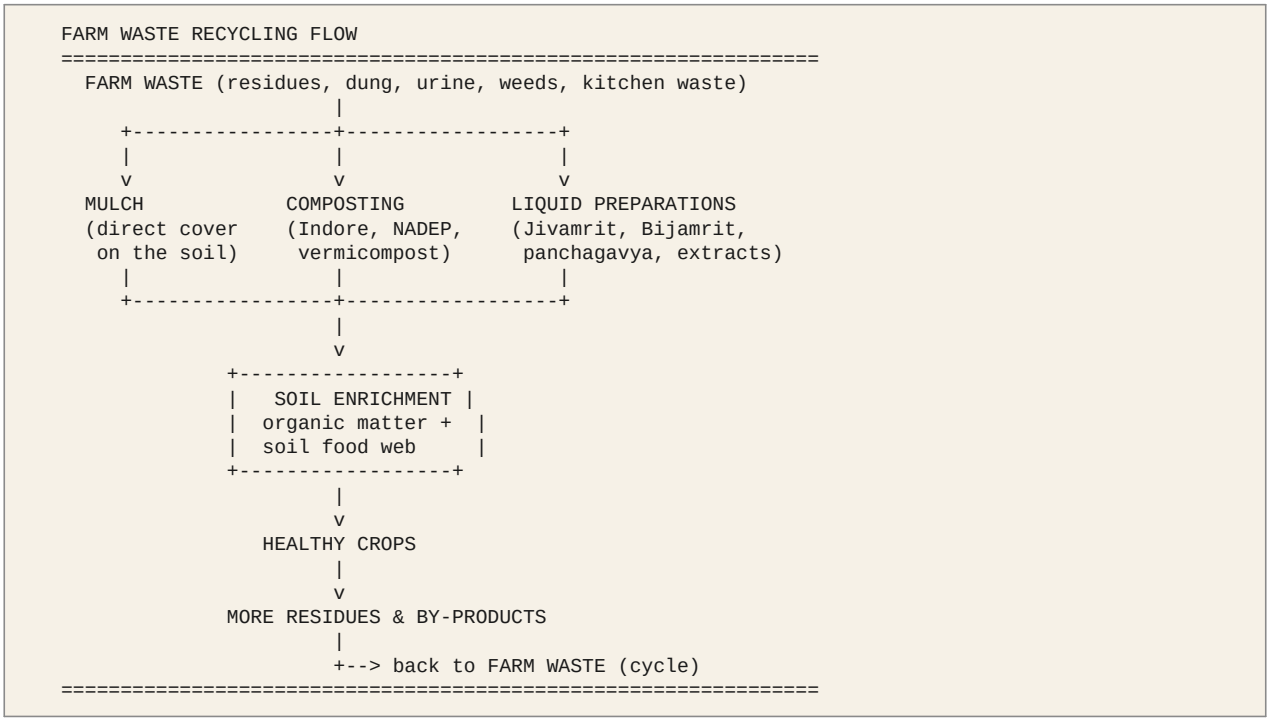



Fig. 12.1 Farm waste recycling - waste becomes input, input becomes crop, crop becomes waste again.

## 12.4 Nutrient value of recycled materials

| Material | Approximate nutrient content (per cent, on fresh/dry basis as noted) | Note |
|---|---|---|
| Farmyard manure (well rotted) | N 0.5, P 0.2, K 0.5 (dry basis, indicative) | Values vary widely with source and storage. |
| Compost | N 0.5-1.5, P 0.3-0.8, K 0.5-1.5 | Range depends on feedstock and maturity. |
| Vermicompost | N 1.0-2.0, P 0.5-1.0, K 0.8-1.5 | Also rich in microbes and humus. |
| Cattle urine (fresh) | N 0.5-1.0, K 0.5-1.5 | Used diluted; source of nitrogen and potassium. |
| Poultry manure | N 1.5-3.0, P 1.5-2.5, K 1.0-1.5 | Rich but should be composted before use. |
| Green manure (sunhemp) | Adds about 40-60 kg N per ha when grown well | Depends on biomass produced. |

Note: nutrient percentages of organic materials vary greatly with source, age and handling. The ranges above are the commonly cited values; exact figures should be obtained from a laboratory test where precision is needed.

**WORKED EXAMPLE - Converting one tonne of crop residue**

- One tonne of dry crop residue contains roughly 5-10 kg nitrogen, 1-2 kg phosphorus and 10-15 kg potassium, depending on the crop.
- If burnt, almost all of this is lost and carbon is released as carbon dioxide.
- If composted, most of the nutrients are retained and returned to the soil, and about 0.2-0.3 tonnes of stable organic matter is added.
- If used as mulch, the nutrients are released slowly over the season, moisture is conserved and soil life is fed.
- Conclusion: recycling one tonne of residue is worth several hundred rupees of fertiliser and, more importantly, builds long-term soil health.



Mounish Raj · MouRa · Agri AI

## EXAM FOCUS

**Key points to remember**

- Biodiversity operates at genetic, species and ecosystem levels, plus functional biodiversity of useful organisms.
- Indigenous seeds are farmer-maintained land races adapted to local conditions; they give low-input performance, tolerance and seed sovereignty.
- Tamil Nadu has rich land races of paddy, millets, pulses, oilseeds and vegetables.
- Seed conservation uses community seed banks, on-farm growing, selection, Bijamrit treatment and careful storage.
- Farm waste is a resource: residues, dung, urine, weeds and kitchen waste are recycled.
- Recycling methods: mulching, composting (Indore, NADEP, vermicompost) and liquid preparations.
- Recycled materials supply nutrients and, more importantly, organic matter and soil life.

**Short-answer questions with model answers**

Q1. What are indigenous seeds and why are they important?

**Model answer:** Indigenous seeds are traditional varieties or land races developed and maintained by farmers over generations. They are important because they are adapted to local soil, climate and pests, perform well with low inputs, tolerate drought and pests better than many modern varieties, often have better taste and nutrition, carry genetic diversity for future breeding, and give the farmer independence from purchased seed.

Q2. Name any four methods of composting farm waste.

**Model answer:** The Indore method, the Bangalore method, vermicomposting and NADEP composting. Microbial waste decomposer cultures and liquid preparations such as Jivamrit are further ways of recycling farm waste.

Q3. Give two Tamil Nadu indigenous paddy varieties and state one benefit of using them.

**Model answer:** Kallurundai Samba and Karuppu Kavuni (black rice) are examples, along with Mappillai Samba and Seeraga Samba. A benefit is that they are adapted to local conditions and perform well under low-input natural farming, while also fetching a premium in niche markets for traditional rice.

**Long-answer questions with model answer outlines**

Q1. Explain the importance of biodiversity and indigenous seed conservation in natural farming.

**Outline:**

- Biodiversity: definition and levels (genetic, species, ecosystem, functional).
- Role of biodiversity in stability, pest control and pollination.
- Indigenous seeds: definition and advantages.
- Tamil Nadu examples of land races.
- Conservation methods: community seed banks, on-farm conservation, selection, treatment and storage.
- Conclusion: biodiversity and indigenous seed are the genetic base of natural farming.

Q2. Describe how farm wastes are recycled into inputs for soil enrichment on a natural farm.

**Outline:**

- Introduction: waste as a resource.
- Types of waste and their recycling methods (table).
- Composting methods: Indore, Bangalore, vermicompost, NADEP.
- Liquid preparations: Jivamrit, panchagavya.

- Nutrient value of recycled materials.
- Waste recycling flow diagram.
- Conclusion: recycling builds organic matter and closes the nutrient loop.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Biodiversity operates at genetic, species and ____ levels.

2 Traditional varieties maintained by farmers are called land ____.

3 A farmer-managed store of indigenous seed is a community ____ bank.

4 The Indore method is a method of ____.

5 Earthworms produce nutrient-rich ____.

6 Green manure crops such as sunhemp add ____ to the soil.

### B. State whether true or false

1 Indigenous seeds are less tolerant of local pests than modern varieties. (False - often more tolerant)

2 Farm waste should be burnt to keep the field clean. (False - it should be recycled)

3 Vermicomposting uses earthworms. (True)

4 Community seed banks conserve indigenous varieties. (True)

### C. Match the following

| Column A | Column B |
|---|---|
| Karuppu Kavuni | Black rice land race |
| Sunhemp | Green manure crop |
| Vermicompost | Earthworm-produced manure |
| NADEP | Low-cost composting method |
| Mulching | Direct use of residue as cover |

### D. Answer in one word or one line

1 Name two Tamil Nadu indigenous paddy varieties.

2 Name three methods of composting.

3 What is a community seed bank?

4 Name two benefits of using indigenous seed.

**Practical application in Tamil Nadu - biodiversity and seed**
- Traditional paddy varieties such as Mappillai Samba and Karuppu Kavuni fetch premium prices.
- Millet land races of the southern districts are conserved by farmer groups and seed savers.
- Vermicompost units are widely promoted for recycling farm and kitchen waste.

# Lecture 13. Water Conservation, Renewable Energy and Nutrient Management

**Learning objectives**

- Explain water conservation practices suited to natural farming.
- Describe renewable energy applications on the farm.
- Explain natural nutrient sources and nutrient management in natural farming.

## 13.1 Water conservation

Water is a limiting resource in most of Tamil Nadu, where rainfall is seasonal and often erratic. Natural farming conserves water both by storing it and by reducing losses, and by improving the soil's capacity to hold moisture.

### 13.1.1 Soil-based water conservation

- **Organic matter:** every one per cent increase in soil organic carbon raises the water-holding capacity of the soil substantially; mulch and manure build this.
- **Mulching:** a mulch layer cuts evaporation from the soil surface by a large fraction, so the same rainfall or irrigation lasts longer.
- **No tillage:** undisturbed soil has continuous pores and root channels that improve infiltration.
- **Cover crops:** live mulch protects the surface and reduces runoff.

### 13.1.2 Farm-level water conservation structures

| Structure | Purpose |
|---|---|
| Farm pond | Stores runoff for supplementary irrigation and for fish and ducks. |
| Contour bund and trench | Slows runoff, increases infiltration on sloping land. |
| Water absorption trench | Recharges groundwater along the contour. |
| Percolation pond | Recharges groundwater and improves well yield. |
| Check dam | Slows stream flow and recharges the channel bed. |
| Field bund | Retains rainwater in the field and prevents soil loss. |
| Roof water harvesting | Collects rainwater for drinking and homestead use. |
| Recharge pit | Directs runoff into the ground near the source of use. |

### 13.1.3 Efficient irrigation methods

- **Drip irrigation:** delivers water directly to the root zone, saving a large fraction of water compared with flooding.
- **Micro-sprinklers:** suited to close-grown crops and to sandy soils.
- **Alternate wetting and drying:** in paddy, allowing the field to dry partially between irrigations saves water without much yield loss.
- **Irrigation scheduling:** irrigate based on soil moisture and crop stage, not on a fixed calendar.
- **Mulch plus drip:** the most water-efficient combination, because evaporation loss is minimised.

## 13.2 Renewable energy on the farm

Natural farming reduces dependence on purchased energy and makes use of the sun, wind, animals and farm waste.

| Renewable energy source | Application on the farm |
|---|---|
| Solar photovoltaic | Water pumps, lighting, fencing, drying and charging batteries. |
| Solar thermal | Water heating and solar drying of produce. |
| Biogas from dung | Cooking gas and lighting; the digested slurry is a rich manure. |
| Wind (small) | Water pumping in windy locations. |
| Draught animal power | Ploughing, carting, water lifting and threshing. |
| Bullock / animal power | Reduces fossil fuel dependence for field operations. |
| Bio-energy from residues | Fuel and compost; no residue is wasted. |
| Human power | Hand tools and cycle-based pumps for small operations. |

A biogas plant is particularly suited to a natural farm because it uses cattle dung, which is already the base of the input system, and it returns a nutrient-rich slurry that can be applied to crops.

## 13.3 Nutrient management in natural farming

Nutrient management means supplying the crop with the nutrients it needs, in the right balance and at the right time, from natural sources. In natural farming the approach is to feed the soil, not the crop, and to let soil biology deliver the nutrients.

### 13.3.1 Natural nutrient sources

| Source | Main nutrients | How it is used |
|---|---|---|
| Farmyard manure / compost | N, P, K and micronutrients, plus organic matter | Broadcast or placed at sowing; base dressing. |
| Vermicompost | N, P, K, humus and microbes | Applied to nursery and field crops. |
| Jivamrit | Microbes, N, K | Soil drench or through irrigation. |
| Cattle urine | N, K | Diluted as a spray or in preparations. |
| Green manure (sunhemp, daincha) | N | Grown and incorporated or mulched before flowering. |
| Legume rotation | N | Fixes atmospheric nitrogen in root nodules. |
| Oil cakes (neem, groundnut, castor) | N, P and pest repellent action | Composted or applied as slurry. |
| Wood ash | K, Ca, P | Applied sparingly to potassium-demanding crops. |
| Rock phosphate and bone meal | P, Ca | Used in organic systems; sparingly in natural farming. |
| Panchagavya and botanical extracts | Growth promoters and micronutrients | Foliar spray at critical stages. |

### 13.3.2 Principles of nutrient management

- **Feed the soil:** build organic matter and soil life; the crop is nourished indirectly.
- **Balance:** supply a mix of sources so that nitrogen, phosphorus, potassium and micronutrients are all available.
- **Timeliness:** apply at sowing, at critical growth stages and after stress.
- **Slow release:** organic sources release nutrients slowly, matching crop demand and reducing loss.
- **Recycling:** return all residues and manures to the soil to reduce nutrient export.
- **Monitor:** use soil tests and crop observation to adjust inputs.

**WORKED EXAMPLE - Nutrient budget from on-farm inputs for one acre**

- Compost: 2 tonnes per acre of good compost at roughly 0.5-1.5 per cent N gives about 10-30 kg nitrogen, plus phosphorus and potassium.
- Jivamrit: two or three applications per season supply a microbial boost and some readily available nitrogen and potassium.
- Cattle urine: 200 L diluted and sprayed supplies readily available nitrogen and potassium and acts as a mild pest repellent.
- Legume intercrop or green manure: a good sunhemp crop can add the equivalent of 40-60 kg nitrogen per hectare (about 16-24 kg per acre).
- Mulch: crop residue returning to the soil recycles several kilograms each of N, P and K per tonne.
- Total: on-farm sources can meet a large part of the crop's nitrogen and potassium needs; phosphorus is the nutrient most likely to need supplementation in the early years.
- Note: figures are indicative; a soil test is the reliable guide.

## EXAM FOCUS

**Key points to remember**

- Water is conserved by building soil organic matter, mulching, no tillage and cover crops.
- Farm-level structures include farm ponds, contour bunds and trenches, percolation ponds, check dams and recharge pits.
- Efficient irrigation uses drip, micro-sprinklers, alternate wetting and drying, and moisture-based scheduling.
- Renewable energy on the farm includes solar pumps and dryers, biogas from dung, wind, draught animals and hand tools.
- Biogas is especially suited to natural farms because it uses dung and returns a rich slurry.
- Natural nutrient sources include compost, vermicompost, Jivamrit, cattle urine, green manure, legumes, oil cakes and ash.
- Nutrient management principles: feed the soil, balance the sources, apply on time, use slow-release organics and recycle residues.

**Short-answer questions with model answers**

Q1. Name any five farm-level water conservation structures.

**Model answer:** Farm pond, contour bund and trench, percolation pond, check dam and recharge pit. Field bunds and roof water harvesting are further structures.

Q2. How is biogas suited to a natural farm?

**Model answer:** Biogas plants run on cattle dung, which is already the base of the natural farming input system. The plant provides cooking gas and lighting, reducing fuel cost, and the digested slurry that comes out is a nutrient-rich manure that can be applied to crops. Thus a waste product becomes both energy and fertiliser.

Q3. List four natural sources of nutrients in natural farming.

**Model answer:** Farmyard manure or compost, vermicompost, Jivamrit and green manure such as sunhemp or daincha. Cattle urine, oil cakes, wood ash, legume rotation and panchagavya are further sources.

**Long-answer questions with model answer outlines**

Q1. Explain water conservation and renewable energy practices suitable for a natural farm.

**Outline:**

- Water conservation: soil organic matter, mulching, no tillage, cover crops.
- Farm structures: farm pond, contour bunds, trenches, percolation ponds, check dams, recharge pits.

- Efficient irrigation: drip, micro-sprinkler, alternate wetting and drying, scheduling.
- Renewable energy: solar pump and dryer, biogas, wind, draught animals, hand tools.
- Biogas integration with dung and slurry.
- Conclusion: conservation of water and energy reduces cost and increases resilience.

Q2. Describe nutrient management in natural farming with reference to natural nutrient sources.

**Outline:**

- Definition and approach: feed the soil, not the crop.
- Natural nutrient sources with nutrients supplied (table).
- Principles: balance, timeliness, slow release, recycling, monitoring.
- Worked nutrient budget from on-farm inputs.
- Role of Jivamrit and mulch in nutrient release.
- Conclusion: soil biology is the delivery system for nutrients.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Mulching reduces ____ from the soil surface.

2 A ____ pond stores runoff for supplementary irrigation.

3 Drip irrigation delivers water to the ____ zone.

4 A ____ plant uses cattle dung to produce cooking gas and rich slurry.

5 Green manure crops such as ____ add nitrogen to the soil.

6 Every one per cent increase in soil organic carbon raises water-holding ____.

### B. State whether true or false

1 Tillage improves the soil's water-holding capacity. (False - organic matter does)

2 Biogas slurry is a nutrient-rich manure. (True)

3 Alternate wetting and drying saves water in paddy. (True)

4 Organic nutrient sources release nutrients very quickly. (False - they release them slowly)

### C. Match the following

| Column A | Column B |
|---|---|
| Farm pond | Stores runoff |
| Percolation pond | Recharges groundwater |
| Biogas | Energy from cattle dung |
| Vermicompost | Nutrient-rich manure |
| Jivamrit | Microbial soil inoculum |

### D. Answer in one word or one line

1 Name three farm-level water conservation structures.

2 Name two renewable energy applications on a farm.

3 Name three natural sources of nutrients.

4 What is alternate wetting and drying?

**Practical application in Tamil Nadu - water and energy**

- Drip and micro-sprinkler irrigation are promoted with subsidy under state schemes.
- Farm ponds and recharge structures are built under watershed development programmes.
- Solar pumps are widely used to reduce electricity and diesel cost.

# Lecture 14. Insect, Pest, Disease and Weed Management in Natural Farming

**Learning objectives**

- Explain the principles of ecological pest management.
- Describe botanical formulations used in natural farming.
- Explain biodiversity-based prevention and non-chemical weed management.

## 14.1 The natural farming attitude to pests

In natural farming a pest is not an enemy to be destroyed but a symptom of imbalance. A healthy, diverse, well-nourished crop in a living soil rarely suffers serious pest damage, because natural enemies keep pests in check and because the plant's own defences are strong. The aim is therefore prevention through a healthy ecosystem, and only then control by ecological means. A spray is a last resort, not a routine.

## 14.2 Principles of ecological pest management

- **Prevention first:** a healthy crop in a diverse system is the best protection.
- **Tolerate the harmless:** a small pest population feeds the natural enemies and does not reduce yield; do not spray at the first sight of an insect.
- **Use the action threshold:** act only when pest numbers cross the level at which economic damage begins.
- **Conserve natural enemies:** avoid broad-spectrum poisons and provide habitat for predators and parasitoids.
- **Use diversity:** mixed cropping, rotation, trap crops and border rows reduce pest build-up.
- **Prefer the least disruptive method:** cultural and mechanical methods before botanicals.
- **Treat the cause:** correct soil, nutrition and water problems that make plants vulnerable.

## 14.3 Categories of pest management methods

| Category | Methods | Examples in natural farming |
|---|---|---|
| Cultural | Practices that prevent pest build-up | Timely sowing, crop rotation, mixed cropping, sanitation, removal of infested parts, summer ploughing where permitted, flooding in paddy. |
| Mechanical and physical | Direct removal or exclusion | Hand-picking of larvae and egg masses, light traps, yellow and blue sticky traps, pheromone traps, bird perches, netting. |
| Biological | Use of living organisms | Conservation and release of predators and parasitoids (ladybird beetles, Chrysoperla, Trichogramma), entomopathogens (Beauveria, Metarhizium, Bacillus thuringiensis, NPV). |
| Botanical | Plant-based preparations | Neem, nochi, erukku, adathoda, garlic, chilli, tobacco, tulsi, pungam. |
| Habitat / ecological engineering | Designing the environment | Flowering plants on bunds, banker plants, trap crops, beetle banks, refuge patches. |
| Biological control of disease | Antagonistic microbes | Trichoderma, Pseudomonas fluorescens, Bacillus subtilis for soil-borne disease. |

## 14.4 Botanical formulations

Botanical preparations are made from locally available plants. They are less harmful than synthetic pesticides but must still be used carefully, because even plant extracts can harm beneficial insects if applied indiscriminately.

| Plant | Active principle / action | Typical use |
|---|---|---|
| Neem (Azadirachta indica) | Azadirachtin - antifeedant, repellent, growth regulator | Seed kernel extract or neem oil spray for sucking pests and larvae. |
| Nochi (Vitex negundo) | Repellent and insecticidal compounds | Leaf extract spray for caterpillars and sucking pests. |
| Eruku / Erukku (Calotropis gigantea) | Latex compounds, repellent | Leaf extract in combination sprays. |
| Adathoda (Adhatoda vasica) | Alkaloids with insecticidal action | Leaf extract spray for larvae. |
| Garlic and onion | Sulphur compounds, repellent | Bulb extract spray against sucking pests and as a fungicide. |
| Chilli | Capsaicin, irritant | Combined with garlic and soap for a repellent spray. |
| Tobacco | Nicotine - insecticidal | Decoction, used sparingly; harmful to humans and bees, so use with care. |
| Tulsi (Ocimum) | Aromatic repellent | Leaf extract spray. |
| Pongam (Pongamia pinnata) | Karanjin - insecticidal | Seed oil spray. |

A common combination spray is 'neem-garlic-chilli-soap': an extract of neem kernel, garlic, chilli and a little soap solution to help the spray stick. It is used against sucking pests and small caterpillars.

## 14.5 Disease management

- **Healthy soil:** a biologically active soil suppresses soil-borne pathogens; this is the foundation of disease control.
- **Antagonists:** Trichoderma and Pseudomonas fluorescens applied to seed and soil compete with and suppress pathogens.
- **Resistant varieties:** indigenous and locally adapted varieties tolerate disease better.
- **Cultural practice:** crop rotation, sanitation, spacing for air movement and avoiding excess moisture reduce disease.
- **Botanical and mineral sprays:** neem, garlic and dilute milk or buttermilk sprays for some fungal diseases; lime and ash for certain problems.
- **Seed treatment:** Bijamrit and warm-water treatment reduce seed-borne disease.

## 14.6 Weed management without herbicides

Weeds are managed by making the crop more competitive and by covering the soil, rather than by killing plants with chemicals.

| Method | How it works |
|---|---|
| Mulching | A thick mulch layer blocks light and prevents weed germination. |
| Live mulch and cover crops | Smother weeds and occupy the space. |
| Competitive crops and close spacing | The crop shades out weeds. |
| Crop rotation | Changes the weed flora and prevents a single weed from dominating. |

| Method | How it works |
|---|---|
| Timely hand weeding | Remove weeds young, before they set seed. |
| Stale seedbed and solarisation | Allow weeds to germinate, then destroy them before sowing. |
| Green manuring with smother crops | Dense crops such as sunhemp suppress weeds. |
| Animals | Ducks and poultry graze young weeds and insects in some systems. |

In Fukuoka's system, weeds are managed by a clover ground cover, by timing and by brief flooding in paddy, rather than by tillage or herbicide.

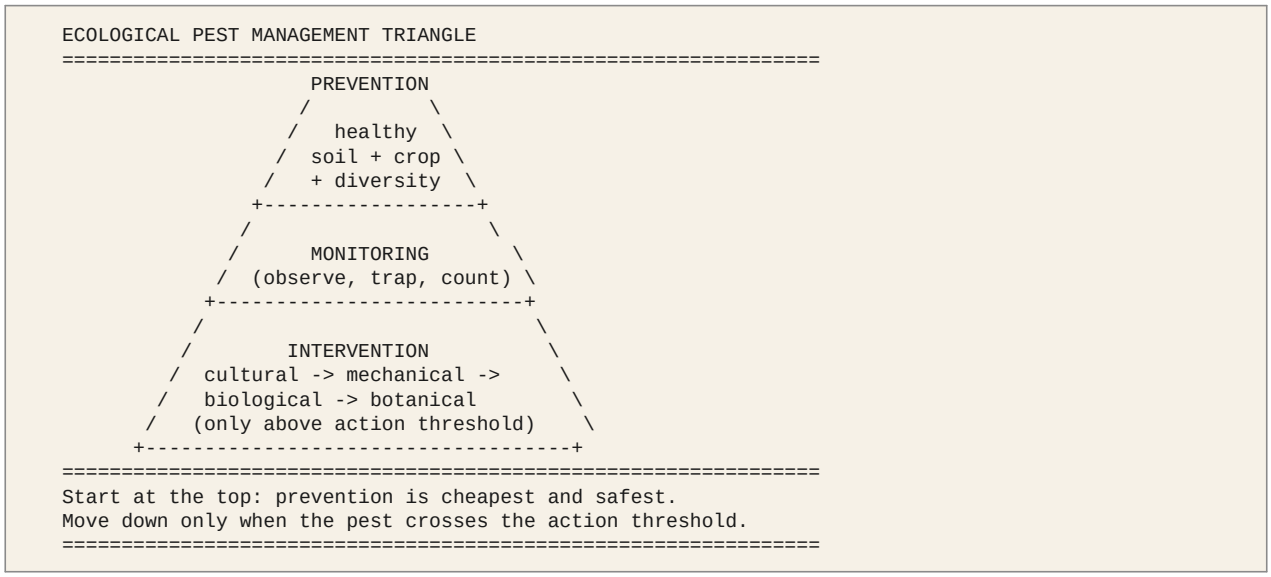



Fig. 14.1 The ecological pest management triangle - prevention first, intervention last.

**WORKED EXAMPLE - Managing fruit borer in tomato by ecological means**

- Prevention: grow tomato with a border of marigold (trap for nematodes and nectar for enemies), rotate away from the previous solanaceous crop, and keep the soil covered with mulch.
- Monitoring: set pheromone traps and yellow sticky traps; count moths and adults twice a week.
- Cultural: remove and destroy infested fruits and fallen fruit; keep the field clean.
- Biological: conserve ladybird beetles and spiders; spray Bacillus thuringiensis only against early-instar larvae.
- Botanical: if the count crosses the action threshold, spray neem kernel extract at the recommended dilution, in the evening to protect bees.
- Result: no synthetic pesticide is used, and the pest is kept below the economic injury level.

## EXAM FOCUS

### Key points to remember

- In natural farming a pest is a symptom of imbalance, so prevention comes first and spraying is a last resort.
- Principles: prevention, tolerance, action threshold, conserving natural enemies, diversity and treating the cause.
- Method categories: cultural, mechanical and physical, biological, botanical, habitat design and biological disease control.
- Botanical plants: neem, nochi, erukku, adathoda, garlic, chilli, tobacco, tulsi and pongam.
- Disease control rests on healthy soil, antagonists such as Trichoderma and Pseudomonas, resistant varieties and cultural practice.
- Weeds are managed by mulching, live mulch, competition, rotation, timely hand weeding and smother crops.



Mounish Raj · MouRa · Agri AI

- The pest management triangle: prevention, then monitoring, then intervention only above threshold.

**Short-answer questions with model answers**

Q1. State the four principles of ecological pest management.

**Model answer:** Prevention through a healthy, diverse crop in a living soil; tolerance of harmless pest levels; action only when the pest crosses the action threshold; and conservation of natural enemies by avoiding broad-spectrum poisons and providing habitat. Diversity and correction of the underlying cause are part of the same approach.

Q2. Name any four plants used for botanical pest control and one action of each.

**Model answer:** Neem (azadirachtin - antifeedant and growth regulator), nochi (repellent and insecticidal against caterpillars), garlic (sulphur compounds - repellent and fungicidal) and pongam (karanjin - insecticidal). Eruku, adathoda, chilli and tobacco are further examples.

Q3. How are weeds managed in natural farming without herbicides?

**Model answer:** By mulching to block light, using live mulch and cover crops to smother weeds, growing competitive crops and close spacing to shade them out, rotating crops to change the weed flora, removing weeds young by hand, using stale seedbed and solarisation, growing smother crops such as sunhemp, and in some systems using ducks or poultry to graze young weeds.

**Long-answer questions with model answer outlines**

Q1. Explain ecological pest management in natural farming with a suitable diagram.

**Outline:**

- Attitude: pest as a symptom of imbalance; prevention first.
- Principles of ecological pest management.
- Method categories with examples (cultural, mechanical, biological, botanical, habitat).
- The pest management triangle diagram.
- Botanical formulations and their use.
- Conclusion: a healthy ecosystem is the best pesticide.

Q2. Describe botanical formulations, disease management and weed management in natural farming.

**Outline:**

- Botanical plants, active principles and uses (table).
- Combination spray example and precautions for bees.
- Disease management: healthy soil, antagonists, resistant varieties, cultural practice, seed treatment.
- Weed management: mulching, live mulch, competition, rotation, hand weeding, smother crops.
- Fukuoka's approach to weeds.
- Conclusion: all three rely on ecology rather than chemicals.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 In natural farming a pest is treated as a symptom of ____.

2 Control is taken only when the pest crosses the action ____.

3 Azadirachtin is the active compound in ____.

4 Trichoderma is used for biological control of ____.

5 A thick ____ layer suppresses weeds by blocking light.

6 Broad-spectrum poisons kill natural ____.

### B. State whether true or false

1 Natural farming sprays pesticide on a fixed calendar. (False - only above the action threshold)

2 Botanical sprays are completely harmless to bees. (False - they can harm bees, so spray in the evening)

3 Neem acts as an antifeedant and growth regulator. (True)

4 Diversity helps reduce pest build-up. (True)

### C. Match the following

| Column A | Column B |
|---|---|
| Neem | Azadirachtin |
| Nochi | Repellent and insecticidal |
| Trichoderma | Disease control |
| Bacillus thuringiensis | Bacterial control of larvae |
| Marigold | Trap crop for nematodes |

### D. Answer in one word or one line

1 Name three principles of ecological pest management.

2 Name four botanical pest-control plants.

3 How is disease controlled in natural farming?

4 Name two non-chemical methods of weed control.

**Practical application in Tamil Nadu - pest management**
- Neem-based sprays and panchagavya are widely used by organic farmers.
- Pheromone and sticky traps are used for monitoring in vegetable and fruit crops.
- Border rows of marigold and castor are common in cotton and vegetable fields.

# Lecture 15. Mechanization and Farm Operations in Natural Farming

**Learning objectives**

- Explain the principles that guide mechanisation in natural farming.
- Describe suitable tools, machines and low-cost technologies.
- Show how farm operations are carried out with minimum soil and ecosystem disturbance.

## 15.1 The place of mechanisation in natural farming

Mechanisation means using tools and machines to carry out farm operations. In natural farming, mechanisation is welcome when it reduces drudgery and cost without harming the soil or the ecosystem. It is avoided when it disturbs the soil, compacts it or destroys soil life. The guiding rule is: use the machine that does the job with the least disturbance, and prefer the smallest machine that will do it.

## 15.2 Principles of mechanisation in natural farming

- **Minimum soil disturbance:** avoid the mouldboard plough and heavy discing; prefer no-till seed drills, dibblers and surface tools.
- **Minimum compaction:** use light machines, low tyre pressure and controlled traffic, and avoid working wet soil.
- **Low cost:** prefer small, locally made and repairable equipment over large, expensive machines.
- **Appropriate scale:** match the machine to the size of the holding and the availability of labour.
- **Energy efficiency:** prefer human, animal, solar and small-engine power where suitable.
- **Ecosystem protection:** avoid machines and practices that kill soil life, harm beneficial insects or damage habitat.
- **Timeliness:** mechanisation should allow operations at the right time, which is often more important than speed.

## 15.3 Tools and implements by operation

| Operation | Suitable tools and machines | Note on natural farming |
|---|---|---|
| Sowing without tillage | No-till seed drill, zero-till drill, dibbler, jab planter, seed balls | Places seed in the soil with minimal disturbance; seed can be sown into mulch. |
| Nursery and transplanting | Tray nursery, mat nursery, manual and semi-automatic transplanters | Reduces labour and root damage. |
| Weeding | Hand hoe, wheel hoe, cono-weeder, brush cutter (surface only) | Shallow surface weeding only; never deep cultivation. |
| Mulch management | Straw spreader, hand tools, bale chopper | Spreads mulch evenly without disturbing soil. |
| Spraying botanicals | Knapsack sprayer, battery sprayer, boom sprayer | Use low-drift nozzles; spray in the evening to protect bees. |
| Irrigation | Drip and micro-sprinkler systems, solar pump, hose reel | Water-efficient and root-zone targeted. |
| Harvesting | Sickles, reapers, small combine for large farms | Keep stubble as mulch; avoid burning. |
| Threshing and winnowing | Pedal thresher, small power thresher, winnower | Retains straw for mulch and fodder. |
| Transport and haulage | Bullock cart, small tractor trailer, cycle cart | Low-cost, low-compaction options. |
| Post-harvest processing | Cleaner, grader, solar dryer, mill, oil expeller | Adds value on the farm. |

## 15.4 Low-cost and appropriate technologies

- **Seed ball making:** a simple hand or drum method to coat seed in clay and compost for broadcasting.
- **Dibbler and jab planter:** hand tools that place seed and fertiliser without ploughing.
- **Cono-weeder:** a light, hand-pulled weeder that works on the surface between rows.
- **Battery knapsack sprayer:** cheaper and cleaner than engine sprayers for small farms.
- **Solar pump:** reduces diesel and electricity cost for irrigation.
- **Bullock-drawn implements:** low-cost, low-compaction tools for small and medium farms.
- **Cycle-based pumps and carts:** low-cost human-powered equipment.
- **Farm-made compost and Jivamrit units:** drums, pits and simple stirrers for input preparation.
- **Solar dryer and improved storage:** reduce post-harvest losses without fuel.

## 15.5 Farm operations calendar

Operations in a natural farm follow the crop calendar but with the natural farming approach at each step.

| Stage | Natural farming operation |
|---|---|
| Land preparation | No ploughing. Clear only what is necessary; retain mulch and residue; repair bunds and water structures. |
| Seed selection and treatment | Select own or indigenous seed; treat with Bijamrit. |
| Sowing | Sow with a no-till drill or dibbler; sow into mulch; follow a mixed or intercropped plan. |
| Input application | Apply Jivamrit to the soil; spray botanical or growth preparations at critical stages. |
| Mulching | Maintain a continuous mulch layer; add residue after each harvest. |

| Stage | Natural farming operation |
|---|---|
| Weeding | Shallow surface weeding only, before weeds set seed. |
| Irrigation | Irrigate on the basis of soil moisture and crop stage; use drip or micro-sprinkler. |
| Pest and disease management | Monitor; act only above threshold; use ecological methods. |
| Harvesting | Harvest at the right maturity; keep residues for mulch and fodder. |
| Post-harvest | Clean, grade, process and store; add value where possible. |

## 15.6 Precautions

- Never use a mouldboard plough or heavy disc harrow, which invert and pulverise the soil.
- Do not work or drive on wet soil, which causes severe compaction.
- Keep machinery weight low and traffic in fixed paths to protect the rest of the field.
- Do not use herbicides; weed by surface tools, mulch and competition.
- Spray botanicals in the evening to avoid harming bees and other pollinators.
- Service and calibrate sprayers to avoid over-application.

**WORKED EXAMPLE - Cost of tools for a one-hectare natural farm**

- Basic hand tools (spade, hoe, sickle, dibbler, cono-weeder): low one-time cost, long life.
- Battery knapsack sprayer: moderate one-time cost, low running cost.
- Drip system for part of the holding: higher one-time cost, but large water and labour savings; often supported by subsidy.
- Solar pump where the water source allows: higher one-time cost, but near-zero running cost and no diesel.
- Jivamrit drums and compost pit: very low cost, built from local materials.
- Rule: the total machinery investment should be recovered within a few seasons from reduced input cost and labour saving.
- Note: exact prices change with the market and with government subsidy schemes; obtain current rates locally.

## EXAM FOCUS

**Key points to remember**

- Mechanisation is welcome in natural farming when it reduces drudgery and cost without disturbing the soil or ecosystem.
- Principles: minimum soil disturbance, minimum compaction, low cost, appropriate scale, energy efficiency, ecosystem protection and timeliness.
- Avoid the mouldboard plough and heavy discing; prefer no-till drills, dibblers and surface weeders.
- Low-cost technologies: seed balls, dibbler, cono-weeder, battery sprayer, solar pump, bullock implements.
- Farm operations follow the crop calendar but with no tillage, permanent mulch and ecological pest management.
- Do not work wet soil; keep machinery light and traffic in fixed paths.
- Spray botanicals in the evening to protect pollinators.

**Short-answer questions with model answers**

Q1. Why is the mouldboard plough avoided in natural farming?

**Model answer:** The mouldboard plough inverts and pulverises the soil, destroys soil structure and aggregates, breaks root channels and fungal networks, oxidises organic matter and buries residue that should stay on the

surface as mulch. This harms soil life and increases erosion, so it is avoided. No-till drills, dibblers and surface tools are used instead.

Q2. Name any four low-cost tools or technologies suitable for natural farming.

**Model answer:** Seed-ball making, the dibbler or jab planter, the cono-weeder and the battery knapsack sprayer. The solar pump, bullock-drawn implements and farm-made Jivamrit units are further examples.

Q3. State two precautions to be observed while mechanising a natural farm.

**Model answer:** Never work or drive on wet soil, because it causes severe compaction; and keep machinery light with traffic confined to fixed paths, so that the rest of the field and its soil life are protected.

**Long-answer questions with model answer outlines**

Q1. Explain the principles of mechanisation in natural farming and describe suitable tools for the main farm operations.

**Outline:**

- Definition and place of mechanisation.
- Principles: minimum disturbance, minimum compaction, low cost, appropriate scale, energy efficiency, ecosystem protection, timeliness.
- Tools by operation (table): sowing, weeding, spraying, irrigation, harvesting, threshing, transport, processing.
- Low-cost and appropriate technologies.
- Precautions.
- Conclusion: mechanisation should serve the soil, not damage it.

Q2. Describe the farm operations calendar of a natural farm and the machinery suited to each stage.

**Outline:**

- Land preparation without tillage; bund and water structure repair.
- Seed selection, Bijamrit treatment and no-till sowing.
- Input application and mulching.
- Surface weeding and irrigation.
- Ecological pest management and monitoring.
- Harvesting, threshing, post-harvest and value addition.
- Conclusion: each operation is matched to a low-disturbance tool.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 The ____ plough is avoided in natural farming.

2 A ____ planter places seed without ploughing.

3 A ____ weeder is a light, hand-pulled weeder.

4 A ____ pump reduces irrigation cost using solar energy.

5 Machinery should not be driven on ____ soil, to avoid compaction.

6 Botanical sprays should be applied in the ____ to protect bees.

### B. State whether true or false

1 Heavy machinery is ideal for natural farming. (False - light, low-disturbance machinery is preferred)

2 No-till drills place seed with minimal soil disturbance. (True)

3 Working wet soil causes compaction. (True)

4 Herbicides are part of natural farming. (False - they are not used)

### C. Match the following

| Column A | Column B |
|---|---|
| No-till drill | Sowing without ploughing |
| Cono-weeder | Surface weeding |
| Dibbler | Hand tool for placing seed |
| Solar pump | Energy-efficient irrigation |
| Battery sprayer | Low-cost spraying |

### D. Answer in one word or one line

1 Why is the mouldboard plough avoided?

2 Name three low-cost tools for natural farming.

3 State two precautions in mechanisation.

4 How is weeding done without herbicides?

**Practical application in Tamil Nadu - mechanisation**

- Custom hiring centres provide small machinery and implements to farmers on hire.
- Solar pumps and drip systems are supported under state and central irrigation schemes.
- Cono-weeders and battery sprayers are widely used on smallholdings.

# Lecture 16. Processing, Labelling and Economic Viability of Natural Farming

**Learning objectives**

- Describe post-harvest handling and value addition for natural farming produce.
- Explain packaging and labelling requirements and their market value.
- Assess the economic viability of natural farming with worked examples.

## 16.1 Why processing and value addition matter

A natural farmer who sells only raw grain at the farm gate captures a small part of the final price the consumer pays. By cleaning, grading, processing, packaging and labelling, the farmer moves up the value chain and keeps a larger share. Value addition is therefore central to the economic viability of natural farming.

## 16.2 Post-harvest handling

- **Timely harvest:** harvest at correct maturity to get the best quality and storability.
- **Cleaning:** remove stones, chaff, damaged grain and foreign matter.
- **Grading:** separate produce by size, colour and quality to get better prices.
- **Drying:** dry to safe moisture to prevent mould and insect damage; solar dryers are ideal.
- **Storage:** store in clean, dry, well-ventilated conditions; use neem leaves, ash or airtight containers to control insects without chemicals.
- **Reducing losses:** good handling can cut post-harvest losses substantially, which is a direct gain to the farmer.

## 16.3 Value addition

| Crop / produce | Value-addition options |
|---|---|
| Paddy | Traditional rice (hand-pounded, unpolished), rice flour, puffed rice, rice bran oil. |
| Millets | Millet flour, ragi malt, millet cookies, noodles, ready mixes. |
| Pulses | Dal, papad, flour, sprouted and roasted snacks. |
| Oilseeds | Cold-pressed oil, oil cake as feed and manure. |
| Fruits (mango, banana) | Pulp, jam, pickle, chips, dried fruit. |
| Vegetables | Dried vegetables, pickles, powders, sauces. |
| Coconut | Copra, virgin coconut oil, coconut milk, coir, shell charcoal. |
| Spices | Cleaned, graded and ground spice powders. |
| Milk | Curd, ghee, paneer, value-added dairy products. |
| Honey and beeswax | Filtered honey, beeswax products. |

## 16.4 Packaging

- Packaging protects the produce, prevents contamination and carries the brand message.
- Use food-grade, moisture-proof and tamper-evident packaging.
- Choose eco-friendly packaging - cloth, paper, jute, glass or recyclable material - to match the natural image.

- Include the product name, net weight, ingredients, farmer or group name, contact, batch number, date of packing and best-before date.
- Attractive, honest labelling raises perceived value and supports a premium price.

## 16.5 Labelling

A label is the identity of a product. For natural and organic produce, the label also carries the certification mark and the traceability information that builds consumer trust.

| Label element | Purpose |
|---|---|
| Product name and description | Tells the buyer what the product is. |
| Net quantity and weight | Mandatory legal information. |
| Ingredients and nutrition | For processed foods; builds trust. |
| Farmer / group / brand name | Creates identity and traceability. |
| Certification mark (e.g. PGS-India / Jaivik Bharat) | Assures the buyer of organic or natural status. |
| Batch number and date of packing | Enables traceability and recall if needed. |
| Best-before or expiry date | Quality and safety information. |
| Contact details and QR code | Links the buyer to the farm for trust and repeat sales. |
| FSSAI licence number (for processed foods) | Mandatory for food business operators. |

## 16.6 Economic viability

Economic viability means the farm can earn enough to cover its costs and give the family a reasonable income over the long term. Natural farming improves viability mainly by reducing cost and by opening premium markets, even where yield is similar.

### 16.6.1 Sources of higher income

- Lower input cost (fertiliser and pesticide replaced by on-farm preparations).
- Price premium for chemical-free and certified produce.
- Value addition through processing and packaging.
- Direct marketing to consumers, reducing middlemen.
- Diversified income from livestock, honey, mushrooms and trees.
- Reduced post-harvest losses through better handling.

**WORKED EXAMPLE - Cost comparison - natural farming vs conventional farming for one acre of paddy**

- Conventional paddy (one acre, one season):

- Seed Rs. 600; fertiliser Rs. 6,000; pesticide Rs. 2,500; land preparation (tractor) Rs. 3,000; labour Rs. 10,000; irrigation Rs. 2,000; harvesting and threshing Rs. 4,000. Total input cost about Rs. 28,100.

- Natural farming paddy (one acre, one season):

- Seed Rs. 400 (own/indigenous); fertiliser Rs. 0 (Jivamrit from farm); pesticide Rs. 300 (botanical materials); land preparation Rs. 500 (no tillage, only bund repair); labour Rs. 11,500 (more labour for mulching and input preparation); irrigation Rs. 1,500; harvesting and threshing Rs. 4,000. Total input cost about Rs. 18,200.

- Difference: natural farming costs about Rs. 9,900 less per acre per season.

- If the natural produce also gets a price premium of 10-20 per cent, the net advantage widens further.

- Note: figures are illustrative for teaching; actual values vary with location, season and prices.

**WORKED EXAMPLE - Partial and complete nutrient and financial budget for a natural farm (one acre)**

- PARTIAL BUDGET - nutrients supplied from on-farm sources:
- Compost 2 tonnes: about 10-30 kg N, plus P and K.
- Jivamrit (2 applications): microbial boost plus a few kg of readily available N and K.
- Legume intercrop / green manure: equivalent of about 16-24 kg N per acre.
- Residue mulch returned: several kg each of N, P and K.
- PARTIAL BUDGET - financial:
- Income: grain (Rs. 40,000) + straw/fodder value (Rs. 3,000) + intercrop (Rs. 4,000) = about Rs. 47,000.
- Variable cost: seed Rs. 400 + labour Rs. 11,500 + botanical materials Rs. 300 + irrigation Rs. 1,500 = about Rs. 13,700 (excluding harvest, which is common to both systems).
- Gross margin: about Rs. 33,300.
- COMPLETE BUDGET - include fixed and family costs:
- Add land rent or opportunity cost, depreciation of tools, and imputed value of family labour.
- Net return after these: still positive, and higher than the conventional system because the input cost is lower.
- Note: a complete budget is essential for a correct comparison; a partial budget alone can overstate the advantage.




Fig. 16.1 The value chain from farm to certified market.

## 16.7 Collective marketing and farmer producer organisations

Individual small farmers cannot usually meet the volume, quality consistency or paperwork that buyers and exporters demand. Farmer Producer Organisations (FPOs) and self-help groups solve this by aggregating produce, arranging certification, processing and marketing collectively, and giving farmers bargaining power.

**WORKED EXAMPLE - Economics of a chemical-free produce value chain**

- Take one tonne of paddy valued at Rs. 22,000 as raw grain at the farm gate.
- If milled and sold as hand-pounded traditional rice at a premium, the same grain can return substantially more, after allowing for milling cost and losses.
- If processed further into rice flour or puffed rice and packed under a brand, the return rises again, though with additional cost and effort.
- The key point: value addition and direct or certified marketing capture a larger share of the consumer's rupee than selling raw grain.
- Note: the exact multiples depend on the product, the market and the cost of processing; the principle is that each step of processing and each step closer to the consumer increases the farmer's share.

## EXAM FOCUS

**Key points to remember**

- Value addition lets the farmer capture a larger share of the consumer price.
- Post-harvest steps: timely harvest, cleaning, grading, drying, safe storage and loss reduction.
- Value-addition options exist for paddy, millets, pulses, oilseeds, fruits, vegetables, coconut, spices and milk.
- Packaging protects and identifies the produce; eco-friendly packaging matches the natural image.
- Labels carry product details, weight, farmer or brand name, certification mark, batch number and dates.
- Economic viability improves through lower cost, price premium, value addition and direct marketing.
- FPOs and self-help groups help small farmers with volume, certification and collective marketing.

**Short-answer questions with model answers**

Q1. Why is value addition important for a natural farmer?

**Model answer:** Because selling raw produce at the farm gate captures only a small part of the final price. By cleaning, grading, processing, packaging and labelling, the farmer moves up the value chain and keeps a larger share of the consumer's rupee. Value addition also reduces dependence on middlemen and creates local employment.

Q2. List any six elements that must appear on the label of a packaged natural produce.

**Model answer:** Product name and description, net quantity or weight, ingredients or nutrition information, farmer or group or brand name, certification mark such as PGS-India, and batch number with date of packing. Best-before date, contact details and the FSSAI licence number for processed foods are further requirements.

Q3. How does natural farming improve the economic viability of a farm?

**Model answer:** Mainly by reducing the cost of cultivation, because fertiliser and pesticide are replaced by on-farm preparations; and by opening premium markets for chemical-free and certified produce. Value addition, direct marketing, diversified income from livestock and trees, and reduced post-harvest losses add to viability.

**Long-answer questions with model answer outlines**

Q1. Explain processing, packaging and labelling practices that enhance the market value of natural farming produce.

**Outline:**

- Importance of value addition and moving up the value chain.
- Post-harvest handling: harvest, cleaning, grading, drying, storage, loss reduction.
- Value-addition options for major crops (table).
- Packaging: protection, food-grade material, eco-friendly choice, required information.
- Labelling: elements and their purpose; certification mark and traceability.

- Marketing channels and the role of FPOs.
- Conclusion: processing and labelling convert a commodity into a premium product.

Q2. Assess the economic viability of natural farming with a cost comparison and a nutrient and financial budget for one crop.

**Outline:**

- Concept of economic viability.
- Cost comparison: natural vs conventional paddy, item by item.
- Nutrient budget from on-farm inputs (compost, Jivamrit, legumes, mulch).
- Partial budget: income and variable cost, gross margin.
- Complete budget: adding fixed cost, land and family labour.
- Value chain and premium; role of FPOs.
- Conclusion: lower cost and premium markets make natural farming viable.

## Quick Revision - Objective Type Questions

### A. Fill in the blanks

1 Cleaning, grading and ____ are important post-harvest operations.

2 A ____ dryer uses solar energy to dry produce.

3 Value addition moves the farmer up the ____ chain.

4 An ____ is a collective of farmers for marketing and certification.

5 The certification mark for Indian organic produce is ____ Bharat.

6 FSSAI licence is required for processed ____.

### B. State whether true or false

1 Selling raw grain at the farm gate earns the farmer the largest share of the consumer price. (False - value addition earns more)

2 Good packaging protects the produce and carries the brand. (True)

3 Labelling includes the certification mark and batch number. (True)

4 Natural farming has no effect on post-harvest losses. (False - good handling reduces them)

### C. Match the following

| Column A | Column B |
|---|---|
| Value addition | Processing to raise market value |
| FPO | Collective marketing and certification |
| Solar dryer | Fuel-free drying |
| Label | Product identity and traceability |
| Premium price | Reward for chemical-free produce |

### D. Answer in one word or one line

1 Name four post-harvest operations.

2 Name three value-added products from millets.

3 Name four elements of a product label.

4 What is an FPO?

**Practical application in Tamil Nadu - value chains**

- Traditional rice varieties are hand-pounded and sold at a premium in the state.
- Millet-based value-added products are produced by women's self-help groups.
- Cold-pressed oil units add value to groundnut and coconut.

# Lecture 17. Certification, Standards, Export Potential, Government Initiatives and Entrepreneurship

**Learning objectives**

- Explain certification systems for organic and natural produce, including PGS and third-party certification.
- Describe quality assurance mechanisms and export potential.
- List government initiatives, NGO programmes and farmer case studies.
- Identify entrepreneurship opportunities in natural and chemical-free agriculture.

## 17.1 Why certification is needed

A consumer cannot tell by looking whether produce was grown without chemicals. Certification is an independent assurance, backed by inspection and record-keeping, that the produce meets a defined standard. It builds consumer trust and gives the farmer access to premium markets and exports. For natural farming, certification is still developing, and several systems are used.

## 17.2 Types of certification

| System | How it works | Suitability |
|---|---|---|
| Participatory Guarantee System (PGS-India) | Peer certification by a group of farmers, with a local group, records, peer inspection and a regional council. Low cost, farmer-driven. | Small and marginal farmers, local and domestic markets, direct sales. |
| Third-party certification (NPOP / NOP / EU) | Inspection and certification by an accredited certification body against a written standard. | Export markets and large buyers who demand accredited certification. |
| Self-declaration / trust-based | Farmer declares chemical-free practice; trust built through direct relationship with consumers. | Direct marketing, community-supported agriculture, farmers' markets. |
| Group / cluster certification | A group of farmers certified together, sharing the cost of inspection and records. | FPOs and clusters seeking scale. |

## 17.3 PGS vs third-party certification

| Basis | PGS-India (participatory) | Third-party certification |
|---|---|---|
| Who certifies | The farmers themselves, through peer review | An independent accredited certification body |
| Cost | Low; shared among the group | High; per-farm and per-inspection fees |
| Approach | Trust, transparency and local participation | Verification against a formal written standard |
| Inspection | Peer inspection by neighbouring farmers | External inspector |
| Documentation | Simple group records and a farm diary | Detailed records and internal control system |
| Market | Domestic, local and direct markets | Export and premium institutional buyers |
| Suitability | Small and marginal farmers, groups | Large farms, exporters, contract buyers |
| Certification mark | PGS-India / Jaivik Bharat mark | NPOP (India Organic) or the buyer-country mark (NOP, EU) |
| Strength | Affordable, inclusive, builds community | Widely recognised, opens export markets |
| Limitation | Less recognised internationally | Expensive and paperwork-heavy for small farms |

## 17.4 Quality assurance and traceability

- **Farm records:** a farm diary of inputs, operations and yields is the base of any certification.
- **Internal control system:** for groups, an internal system to check member compliance before external inspection.

- **Traceability:** batch numbers and records that link a packet back to the field and the farmer.
- **Residue testing:** laboratory testing for pesticide residues where required by buyers.
- **Conversion period:** a defined period (commonly two to three years for perennial crops and shorter for annuals under organic standards) during which the land is managed organically before certification.
- **Certification mark:** the logo or mark that assures the buyer, e.g. PGS-India or the India Organic mark.

Note: the exact length of the conversion period and the detailed standards differ between PGS and third-party systems and are revised from time to time; the general principle of a conversion period should be stated rather than a fixed number.

## 17.5 Export potential

World demand for organic and chemical-free food is growing, and India is a significant exporter of organic produce. Natural farming produce, once certified, can enter this market.

| Aspect | Position |
|---|---|
| Main export products | Organic spices, tea, coffee, basmati rice, pulses, oilseeds, fruits, nuts, honey and processed foods. |
| Key markets | European Union, United States, United Kingdom, Canada, Japan, Australia and the Gulf countries. |
| Requirements | Accredited certification (NPOP and the buyer-country standard), traceability, residue-free testing and consistent quality. |
| Advantages for India | Diverse agro-climatic zones, traditional knowledge, low-input smallholder base and a large domestic market. |
| Challenges | High certification cost, small and scattered holdings, logistics, quality consistency and documentation. |
| Way forward | Cluster and group certification, FPOs, contract farming with exporters and government support for certification. |

## 17.6 Government initiatives

Both the central government and the states support organic and natural farming through programmes, missions and certification support. The names below are the standard national programmes; scheme details, subsidy amounts and coverage change with each budget and should be confirmed from current official documents.

| Programme / initiative | Purpose (general description) |
|---|---|
| Paramparagat Krishi Vikas Yojana (PKVY) | Promotes organic and traditional farming through cluster-based support for inputs, certification and marketing. |
| PGS-India | Provides a low-cost, farmer-participatory organic certification system and the Jaivik Bharat mark. |
| Mission Organic Value Chain Development for North Eastern Region (MOVCDNER) | Develops the organic value chain in the north-eastern states. |
| National Mission on Natural Farming / natural farming initiatives | Promotes natural farming through demonstration, training and support to farmers. |
| National Project on Organic Farming (NPOF) | Supports organic inputs, quality control and certification infrastructure. |
| Rashtriya Krishi Vikas Yojana (RKVY) | State-level support for organic and natural farming projects. |
| Andhra Pradesh Community Managed Natural Farming (APCNF) | A large state programme that promotes natural farming through community institutions; a widely cited model. |

| Programme / initiative | Purpose (general description) |
|---|---|
| Tamil Nadu organic and natural farming initiatives | State agriculture department support for organic and natural farming, certification support through the Tamil Nadu Organic Certification Department, and farmer outreach through the Uzhavan platform. |
| State seed and input support schemes | Support for indigenous seed, bio-inputs and organic input production units. |

Note on the Uzhavan platform: it is the Tamil Nadu government's agriculture information and service platform, used for farmer registration, scheme information, subsidy applications and advisory services. The specific schemes offered on it change over time.

## 17.7 NGO and community initiatives

- Non-governmental organisations that train farmers in ZBNF, organic and natural farming methods.
- Community seed banks and seed savers' groups that conserve indigenous varieties.
- Farmer producer organisations that aggregate produce and arrange certification and marketing.
- Self-help groups, especially of women, involved in input production, processing and marketing.
- Participatory guarantee groups formed under PGS-India for peer certification.
- Farmer field schools and demonstration plots that spread knowledge farmer to farmer.

## 17.8 Case studies

The case studies below are presented in general terms, based on widely reported experience. They are meant to show the pattern of success, not to state precise figures, which vary by farm and year.

| Case | What was done | Reported outcome (general) |
|---|---|---|
| Andhra Pradesh community-managed natural farming | Village-level community institutions supported large numbers of farmers to adopt natural farming methods. | Reduced input cost, improved soil and reported increases in net income and resilience in many villages. |
| Subhash Palekar's ZBNF movement | Farmer-to-farmer training across states in the four pillars. | Widespread adoption, particularly among smallholders and in rain-fed areas. |
| Tamil Nadu traditional paddy farmers | Cultivation of indigenous rice varieties under natural methods. | Premium prices in niche markets and conservation of land races. |
| Organic spice and plantation growers | Certification through PGS or third-party systems and direct or export marketing. | Access to premium markets and higher realisation per unit. |
| Women's self-help group enterprises | Collective production of value-added products such as millet mixes and pickles. | Additional household income and greater participation of women. |

## 17.9 Entrepreneurship opportunities

Natural and chemical-free agriculture creates business opportunities beyond the farm gate. These are the avenues a graduate can pursue or advise farmers on.

| Opportunity | Description |
|---|---|
| Natural input production | Producing and selling Jivamrit, Bijamrit, vermicompost, panchagavya and bio-inputs. |
| Desi cow based enterprises | Dairy, dung and urine products, and a desi-cow-centred input unit. |
| Organic and natural produce trading | Aggregating, grading and selling chemical-free produce. |
| Value-added food processing | Millet mixes, flours, oils, pickles, dried products and ready-to-cook foods. |
| Direct marketing and e-commerce | Farm-to-consumer sales, subscription boxes and online platforms. |
| Certification and advisory services | Helping farmers prepare records and obtain PGS or third-party certification. |

| Opportunity | Description |
|---|---|
| Nursery and seed enterprise | Producing and selling indigenous seed and planting material. |
| Agri-tourism and farm experience | Hosting visitors for natural farming experience and education. |
| Training and knowledge services | Conducting farmer training and demonstration. |
| Beekeeping and honey | Pollination services and honey production. |
| Mushroom and millet enterprises | Using crop residues and millets for high-value products. |

## 17.10 Standards and quality control

- Standards define what may and may not be used and how records must be kept.
- For organic produce, the National Programme for Organic Production (NPOP) sets the Indian standard, and PGS-India provides the participatory route.
- The India Organic (NPOP) mark and the Jaivik Bharat mark identify certified produce.
- Quality control includes residue testing, input verification and inspection.
- For natural farming produce, standards are still developing; many farmers rely on PGS, group certification or direct consumer trust.
- FSSAI regulations apply to processed and packaged foods.

**WORKED EXAMPLE - Economics of a certified chemical-free produce value chain**

- Raw produce sold at the farm gate earns the base price.

- Certified produce, sold to an organic store or buyer, earns a premium over the base price because certification assures quality.

- Value-added, branded and packaged produce, sold directly or online, earns a further premium, after allowing for processing and packaging cost.

- For a farmer group, the cost of PGS certification is shared and low, so the premium is largely retained.

- For an exporter, third-party certification is costly but opens higher-value markets, so the net gain depends on volume.

- Conclusion: certification and value addition together turn a low-margin commodity into a higher-margin branded product.

## EXAM FOCUS

**Key points to remember**

- Certification is independent assurance that produce meets a defined standard; it builds trust and opens premium and export markets.
- PGS-India is participatory, low-cost and peer-certified; third-party certification is external, costly and required for export.
- Quality assurance rests on farm records, an internal control system, traceability, residue testing and a conversion period.
- India exports organic spices, tea, coffee, rice, pulses, oilseeds, fruits and honey, mainly to the EU, USA, UK, Canada, Japan and the Gulf.
- Government programmes include PKVY, PGS-India, MOVCDNER, the National Mission on Natural Farming, NPOF and RKVY, plus state programmes such as APCNF and Tamil Nadu initiatives.
- Entrepreneurship opportunities include input production, value-added processing, direct marketing, certification services and agri-tourism.
- Standards for organic produce are set by NPOP and PGS-India; natural farming standards are still developing.

**Short-answer questions with model answers**

Q1. Distinguish PGS-India from third-party certification.

**Model answer:** PGS-India is a participatory system in which farmers themselves, through peer inspection and group records, certify their produce; it is low-cost and suited to small farmers and domestic markets. Third-party certification is carried out by an independent accredited body against a written standard; it is costly and paperwork-heavy but is widely recognised and required for export markets.

Q2. Name any five government programmes that support organic or natural farming.

**Model answer:** Paramparagat Krishi Vikas Yojana (PKVY), PGS-India, Mission Organic Value Chain Development for North Eastern Region (MOVCDNER), National Project on Organic Farming (NPOF) and Rashtriya Krishi Vikas Yojana (RKVY). Andhra Pradesh Community Managed Natural Farming and Tamil Nadu's organic and natural farming initiatives are state examples.

Q3. List any four entrepreneurship opportunities in natural farming.

**Model answer:** Natural input production (Jivamrit, vermicompost, bio-inputs); value-added food processing (millet mixes, flours, oils); direct marketing and e-commerce of chemical-free produce; and certification and advisory services. Nursery and seed enterprises, agri-tourism and beekeeping are further opportunities.

**Long-answer questions with model answer outlines**

Q1. Explain the certification systems for natural and organic produce and compare PGS with third-party certification.

**Outline:**

- Why certification is needed.
- Types: PGS-India, third-party, self-declaration, group certification.
- Comparison table on certifier, cost, inspection, documentation, market, suitability, mark.
- Quality assurance: records, internal control, traceability, residue testing, conversion period.
- Conclusion: choice of system depends on market and scale.

Q2. Discuss the government initiatives, case studies and entrepreneurship opportunities in natural farming.

**Outline:**

- Government initiatives: PKVY, PGS-India, MOVCDNER, National Mission on Natural Farming, NPOF, RKVY.
- State programmes: APCNF and Tamil Nadu initiatives, Uzhavan platform.
- NGO and community initiatives: training, seed banks, FPOs, SHGs, participatory guarantee groups.
- Case studies: APCNF, ZBNF movement, Tamil Nadu traditional rice, spice growers, women's groups.
- Entrepreneurship: input production, processing, direct marketing, certification services, agri-tourism, beekeeping.
- Conclusion: policy, community action and enterprise together scale natural farming.

Q3. Describe the export potential of natural farming produce and the requirements for export.

**Outline:**

- Growing global demand for organic and chemical-free food.
- Main export products and key markets.
- Requirements: accredited certification, traceability, residue-free testing, quality consistency.
- Advantages for India: agro-climatic diversity, traditional knowledge, smallholder base.
- Challenges: certification cost, scattered holdings, logistics, documentation.
- Way forward: cluster certification, FPOs, contract farming, government support.

- Conclusion: certification is the key to the export market.

# Appendix A. Practical Exercises and Field Assignments

These exercises are designed for the practical and field-work component of the course. Each gives an aim, the materials needed, the procedure and what to record. Students should maintain a practical record book.

### Exercise 1. Preparation of Bijamrit and seed treatment

**Aim:** To prepare Bijamrit and treat seed before sowing.

**Materials:** Desi cow dung (5 kg), desi cow urine (5 L), lime (50-100 g), a handful of farm soil, jaggery (optional), seed, water, a mixing vessel.

**Procedure:** Mix the dung and urine in water to a thin slurry; add lime, farm soil and jaggery; soak the seed briefly; dry in shade and sow the same day.

**Record:** Record the quantity of seed treated, the appearance of the slurry and the germination percentage compared with untreated seed.

### Exercise 2. Preparation of Jivamrit

**Aim:** To prepare Jivamrit and observe fermentation.

**Materials:** Desi cow dung (10 kg), desi cow urine (10 L), jaggery (2 kg), pulse flour (2 kg), a handful of farm soil, 200 L water, a drum and a stirrer.

**Procedure:** Dissolve jaggery in water; add dung, urine, pulse flour and soil; stir twice daily; ferment in shade for two to three days.

**Record:** Record the smell and appearance each day and the volume prepared. Note the changes that show the culture is ready.

### Exercise 3. Preparation of a botanical pest-repellent spray

**Aim:** To prepare a neem-garlic-chilli spray and test it on a crop.

**Materials:** Neem kernel or leaves, garlic, chilli, soap, water, a knapsack sprayer.

**Procedure:** Grind and soak the plant material overnight; strain; add a little soap solution; dilute and spray in the evening.

**Record:** Record the pest observed, the dilution used and the change in pest numbers after a few days.

### Exercise 4. Composting by the Indore method

**Aim:** To prepare compost from farm waste.

**Materials:** Crop residues, cattle dung, urine, water, soil, a pit or heap area.

**Procedure:** Layer residues, dung and soil, moisten, and turn periodically over several weeks until the material is dark and crumbly.

**Record:** Record the weight of materials used, the temperature and the time taken for maturity.

### Exercise 5. Vermicomposting

**Aim:** To prepare vermicompost from organic waste.

**Materials:** Earthworms, cattle dung, crop waste, a vermibed or tank, water.

**Procedure:** Half-fill the bed with dung and waste; introduce earthworms; keep moist and shaded; harvest the compost after it matures and separate the worms.

**Record:** Record the input weight, the compost weight and the time taken.

### Exercise 6. Mulching and soil moisture observation

**Aim:** To study the effect of mulching on soil moisture and temperature.

**Materials:** Straw or crop residue, a small plot, a soil thermometer or moisture meter.

**Procedure:** Mulch one plot and leave a control plot bare; measure soil moisture and temperature at intervals.

**Record:** Record the difference in moisture and temperature between mulched and unmulched plots.

### Exercise 7. Identification of natural enemies and pests

**Aim:** To identify common pests and their natural enemies in the field.

**Materials:** A hand lens, a sweep net, a notebook, the field.

**Procedure:** Collect and observe insects; classify each as pest or natural enemy; note the crop and the stage.

**Record:** Record the names and numbers observed and the predator-prey relationship.

### Exercise 8. Setting up traps and monitoring

**Aim:** To monitor pest populations using traps.

**Materials:** Yellow and blue sticky traps, pheromone traps, pegs.

**Procedure:** Place traps at the recommended height and spacing; count the insects caught twice a week.

**Record:** Record the counts and decide whether the action threshold has been crossed.

### Exercise 9. Seed collection and treatment of indigenous varieties

**Aim:** To collect, clean and store indigenous seed.

**Materials:** Indigenous seed, neem leaves or ash, storage containers.

**Procedure:** Collect seed from healthy plants; clean and dry it; treat it with Bijamrit or cow urine; store with neem leaves or ash.

**Record:** Record the variety name, the source and the germination percentage after storage.

### Exercise 10. Preparing a farm layout plan

**Aim:** To design a natural farm layout for a given holding.

**Materials:** A map of the holding, drawing materials.

**Procedure:** Mark the main field, home garden, cattle shed, fodder plot, farm pond, input unit and boundary trees.

**Record:** Record the area allotted to each component and the reason for the arrangement.

### Exercise 11. Cost of cultivation comparison

**Aim:** To compare the cost of cultivation of a crop under natural and conventional methods.

**Materials:** Records of inputs, labour and yields; current market prices.

**Procedure:** List every input and operation for each method; total the cost; compute the net return.

**Record:** Record the difference in cost and net return per acre.

### Exercise 12. Soil sampling and organic carbon estimation

**Aim:** To collect a representative soil sample and estimate organic carbon.

**Materials:** An auger or spade, a sample bag, a laboratory or field kit.

**Procedure:** Collect samples from several spots, mix, quarter and label; test for organic carbon and other parameters.

**Record:** Record the soil test values and interpret them for natural farming.

### Exercise 13. Water harvesting structure visit

**Aim:** To study a farm pond, percolation pond or recharge structure.

**Materials:** A notebook, measuring tape.

**Procedure:** Visit the structure; measure its size and capacity; note how runoff is directed into it.

**Record:** Record the capacity, the catchment and the use of the stored water.

### Exercise 14. Value addition and labelling exercise

**Aim:** To process a farm product and design a label.

**Materials:** A farm product, processing equipment, packaging material.

**Procedure:** Clean, grade and process the product; pack it; design a label with all required information.

**Record:** Record the cost of processing, the packaging cost and the expected selling price.

### Exercise 15. Survey of a certified or natural farm

**Aim:** To survey a natural or certified farm and document its practices.

**Materials:** A questionnaire, a notebook.

**Procedure:** Visit the farm; record the practices, inputs, crops, certification status, cost and income.

**Record:** Prepare a short report on the farm's practices and its economics.

# Appendix B. Model End-Semester Question Paper

This model paper follows the usual pattern of a final theory examination. Marks and duration are indicative.

**Model end-semester question paper (indicative)**

- Time: 2.5 hours. Maximum marks: 100. Answer all questions in Part A, any four in Part B and any two in Part C.

- PART A - Objective and very short answer (20 x 1 = 20 marks). Covering all units.

- PART B - Short answer (4 x 5 = 20 marks):

- 1. Differentiate natural farming from organic farming. 2. Explain the composition and function of Jivamrit. 3. Write short notes on Korean Natural Farming. 4. Describe the integrated crop-tree-animal nutrient cycle. 5. Explain ecological pest management. 6. Write short notes on PGS-India certification.

- PART C - Long answer (2 x 15 = 30 marks):

- 7. Describe the history and evolution of natural farming from ancient Indian agriculture to the present. 8. Explain the four pillars of natural farming with their composition and function. 9. Discuss the design of a natural farm with a suitable layout and its nutrient cycle. 10. Explain water conservation, renewable energy and nutrient management in natural farming.

- PART D - Application and case-based (2 x 15 = 30 marks):

- 11. Prepare a cost comparison and a nutrient budget for one acre of a crop under natural and conventional farming. 12. Discuss the certification systems, government initiatives and entrepreneurship opportunities in natural farming. 13. Design a plan to convert a two-hectare chemically farmed holding into a certified natural farm.

## Blueprint of the paper

| Unit | Objective | Short | Long | Application |
|---|---|---|---|---|
| Heritage and history (1-2) | 3 | 1 | 1 | - |
| Concept, principles, pillars (3-5) | 5 | 2 | 1 | - |
| Scope, schools, design (6-8) | 4 | 1 | 1 | 1 |
| Ecology, integration, biodiversity (10-12) | 4 | 1 | 1 | 1 |
| Water, energy, pest, mechanisation (13-15) | 2 | 1 | - | - |
| Processing, certification, enterprise (16-17) | 2 | 1 | 1 | 1 |

# Appendix C. Master Revision Tables

These are the comparison tables that appear most often in examinations, gathered in one place for quick revision.

## C.1 Natural farming vs organic farming vs conventional farming

| Basis | Natural | Organic | Conventional |
|---|---|---|---|
| Chemical fertiliser and pesticide | Not used | Not used | Used heavily |
| Tillage | None or minimum | Permitted | Intensive |
| External organic inputs | Minimal | Permitted and often purchased | Not the main input |
| Soil cover | Always mulched | Encouraged | Often bare |
| Source of nutrition | Soil biology and on-farm recycling | Organic amendments | Synthetic NPK |
| Input cost | Very low | Moderate to high | High |
| Certification | Community or PGS where applicable | PGS or NPOP | Not applicable |

## C.2 The four pillars

| Pillar | Composition | Function |
|---|---|---|
| Bijamrit | Desi cow dung, cow urine, lime, farm soil, water | Seed treatment; microbial coat; better germination and protection |
| Jivamrit | Desi cow dung, cow urine, jaggery, pulse flour, farm soil, water | Living microbial culture; decomposition; nutrient mobilisation |
| Mulching | Crop residues, straw, leaves or live cover | Moisture, temperature, weed control, soil life, organic matter |
| Waaphasa | Achieved by mulching and judicious irrigation | Balance of air and moisture in soil pores |

## C.3 Schools of natural farming

| School | Founder / origin | Signature practice |
|---|---|---|
| Fukuoka natural farming | Masanobu Fukuoka, Japan | Seed balls, clover cover, no tillage, minimal intervention |
| ZBNF | Subhash Palekar, India | Four pillars; desi cow based; farmer training |
| Rishi Krishi | Ancient Indian texts | Amritpani, Amritjal, panchagavya, spiritual calendar |
| Korean Natural Farming | Cho Han-Kyu, South Korea | Indigenous Micro-Organisms, FPJ, FFJ, OHN, LAB |

## C.4 Certification systems

| Basis | PGS-India | Third-party certification |
|---|---|---|
| Who certifies | Farmers, by peer review | Independent accredited body |
| Cost | Low, shared | High |
| Market | Domestic and direct | Export and premium institutional |
| Documentation | Simple group records | Detailed records and internal control |
| Mark | PGS-India / Jaivik Bharat | NPOP (India Organic) or buyer-country mark |

## C.5 Key government programmes

| Programme | Purpose |
|---|---|
| PKVY | Cluster support for organic and traditional farming |
| PGS-India | Low-cost participatory organic certification |
| MOVCDNER | Organic value chain in the north-eastern states |
| National Mission on Natural Farming | Promotion of natural farming through demonstration and training |
| NPOF | Organic inputs, quality control and certification infrastructure |
| RKVY | State-level organic and natural farming projects |
| APCNF | State programme on community-managed natural farming |

# Appendix D. Abbreviations

| Abbreviation | Full form |
|---|---|
| APCNF | Andhra Pradesh Community Managed Natural Farming |
| CA | Conservation Agriculture |
| FPO | Farmer Producer Organisation |
| FFJ | Fermented Fruit Juice |
| FPJ | Fermented Plant Juice |
| FSSAI | Food Safety and Standards Authority of India |
| IFS | Integrated Farming System |
| IMO | Indigenous Micro-Organisms |
| IPM | Integrated Pest Management |
| KNF | Korean Natural Farming |
| LAB | Lactic Acid Bacteria |
| MOVCDNER | Mission Organic Value Chain Development for North Eastern Region |
| NPOF | National Project on Organic Farming |
| NPOP | National Programme for Organic Production |
| PGS | Participatory Guarantee System |
| PKVY | Paramparagat Krishi Vikas Yojana |
| RKVY | Rashtriya Krishi Vikas Yojana |
| SDG | Sustainable Development Goal |
| SHG | Self-Help Group |
| SOC | Soil Organic Carbon |
| TNAU | Tamil Nadu Agricultural University |
| ZBNF | Zero Budget Natural Farming |

# Glossary of Key Terms

The terms below are the ones most likely to appear in the examination. Learn the definition of each in one or two lines.

| Term | Meaning |
|---|---|
| Natural farming | A chemical-free, low-input farming system that works with nature, avoids soil disturbance and relies on soil biology for nutrition and on ecological methods for pest control. |
| Zero Budget Natural Farming (ZBNF) | A natural farming system developed by Subhash Palekar, based on four pillars and designed to bring the external cash cost of inputs close to zero. |
| Bijamrit | A seed treatment preparation of desi cow dung, cow urine, lime and farm soil, used to coat seed with beneficial microbes. |
| Jivamrit | A fermented microbial culture of desi cow dung, cow urine, jaggery, pulse flour and farm soil, applied to the soil. |
| Mulching | Covering the soil surface with crop residues, straw, leaves or a live cover crop to conserve moisture and suppress weeds. |
| Waaphasa / Whapasa | The balanced state of air and moisture in soil pores, ideal for roots and aerobic microorganisms. |
| Desi cow | An indigenous cow breed, valued in natural farming for the microbial richness of its dung and urine. |

| Term | Meaning |
|---|---|
| Panchagavya | A preparation of five cow products (dung, urine, milk, curd and ghee) used as a growth promoter. |
| Amritpani / Amritjal | Liquid preparations of cow dung, cow urine, jaggery and gram flour used in Rishi Krishi. |
| Green manure | A crop grown and then incorporated or mulched to add organic matter and nitrogen to the soil. |
| Cover crop | A crop grown mainly to cover and protect the soil and to provide live mulch. |
| Vermicompost | Nutrient-rich compost produced by earthworms from organic waste. |
| Farmyard manure | Decomposed mixture of cattle dung, urine, litter and feed waste used as organic manure. |
| Humus | The stable, dark organic fraction of soil formed by the decomposition of organic matter. |
| Soil organic carbon | The carbon held in soil organic matter; a key indicator of soil health. |
| Rhizobium | A bacterium that fixes atmospheric nitrogen in the root nodules of legumes. |
| Mycorrhiza | A beneficial association between a fungus and plant roots that improves nutrient and water uptake. |
| Azadirachtin | The main active compound in neem, acting as an antifeedant, repellent and growth regulator. |
| Trichoderma | A beneficial fungus used to control soil-borne plant pathogens. |
| Pseudomonas fluorescens | A beneficial bacterium used for biological control of disease and plant growth promotion. |
| Indigenous Micro-Organisms (IMO) | Locally collected and multiplied microorganisms, central to Korean Natural Farming. |
| Ecological engineering | Designing the farm habitat to support natural enemies and pollinators, so that pests are controlled naturally. |
| Trap crop | A crop grown to attract pests away from the main crop, which is then destroyed with the pest. |
| Banker plant | A plant that hosts alternative prey, allowing natural enemies to build up before the pest arrives. |
| Action threshold | The pest population level at which control measures become economically justified. |
| Integrated farming system | A system combining crops, livestock, trees and other enterprises so that the waste of one becomes the input of another. |
| Land race | A traditional crop variety developed and maintained by farmers over generations, adapted to local conditions. |
| Community seed bank | A farmer-managed store of indigenous seed for conservation and exchange. |
| Mulch | The material spread on the soil surface as a cover. |
| PGS-India | Participatory Guarantee System, a low-cost, peer-based organic certification system of India. |
| Third-party certification | Certification by an independent accredited body against a written organic standard, used mainly for export. |
| NPOP | National Programme for Organic Production, which sets the Indian organic standard and the India Organic mark. |
| Jaivik Bharat | The Indian organic certification mark. |
| PKVY | Paramparagat Krishi Vikas Yojana, a central programme promoting organic and traditional farming in clusters. |
| APCNF | Andhra Pradesh Community Managed Natural Farming, a large state programme promoting natural farming. |
| Farmer Producer Organisation (FPO) | A collective of farmers that aggregates produce, arranges certification and markets jointly. |
| Value chain | The sequence of steps from production to the final consumer, at each of which value is added. |
| Conversion period | The period during which land must be managed organically before produce can be certified. |
| Biodiversity | The variety of life at genetic, species and ecosystem levels. |
| Ecological balance | The stable relationship among the living and non-living parts of an ecosystem. |

| Term | Meaning |
|---|---|
| Renewable energy | Energy from sources that are naturally replenished, such as the sun, wind and biomass. |
| Waaphasa-based irrigation | Irrigation managed to maintain the soil's air-moisture balance rather than to saturate it. |



Mounish Raj · MouRa · Agri AI

# Reference List and Suggested Reading

The following standard texts, government documents and institutional materials are the sources for this course. Students should consult the current editions and the latest official documents, since programme details change over time.

## Books on natural and organic farming


- Fukuoka, M. *The One-Straw Revolution: An Introduction to Natural Farming.* (English translation, New York Review Books).
- Fukuoka, M. *The Natural Way of Farming: The Theory and Practice of Green Philosophy.*
- Palekar, S. *The Philosophy of Spiritual Farming* (ZBNF series, Zero Budget Natural Farming).
- Palekar, S. *Zero Budget Natural Farming* - booklets on the four pillars and crop-wise practice.
- Howard, A. *An Agricultural Testament.*
- Howard, A. and Wad, Y.D. *The Waste Products of Agriculture: Their Utilization as Humus.*
- Balfour, E.B. *The Living Soil.*
- Lampkin, N. *Organic Farming.*
- Sharma, A.K. *A Handbook of Organic Farming.*
- Cho Han-Kyu. *Korean Natural Farming* - material on Indigenous Micro-Organisms and natural inputs.


## Ancient and traditional Indian sources


- Krishi-Parashara (attributed to Parashara) - ancient Indian treatise on agriculture.
- Kautilya's Arthashastra - sections on agriculture, irrigation and land administration.
- Surapala's Vrikshayurveda - treatise on trees and plant health.
- Brihat Samhita and Agni Purana - sections on rainfall and agriculture.
- Sangam Tamil literature - references to dryland farming, tanks and cattle.


## ICAR, TNAU and institutional e-course material


- ICAR e-course material on organic farming and natural farming.
- TNAU e-course and lecture material on organic farming, sustainable agriculture and natural farming.
- TNAU Agritech Portal - technology and package-of-practice pages on organic and natural farming.
- National Centre of Organic and Natural Farming (NCONF) - technical and training material.
- ICAR-Indian Institute of Farming Systems Research - integrated farming system material.


## Government programmes and documents


- Paramparagat Krishi Vikas Yojana (PKVY) - operational guidelines.
- PGS-India - programme guidelines and certification manual.
- National Programme for Organic Production (NPOP) - standards document.

- Mission Organic Value Chain Development for North Eastern Region (MOVCDNER) - guidelines.
- National Project on Organic Farming (NPOF) - scheme documents.
- Andhra Pradesh Community Managed Natural Farming (APCNF) - programme reports.
- Tamil Nadu Department of Agriculture - organic and natural farming scheme orders and the Uzhavan platform.
- National Mission on Natural Farming - scheme documents and guidelines.

## Suggested reading for examinations

- Revise the comparison tables of every lecture - they are the most examinable content.
- Learn the four pillars and the ZBNF preparations with their ingredients and quantities.
- Learn the schools of natural farming with their founders and signature practices.
- Learn the certification systems, government programmes and entrepreneurship opportunities.
- Practise the worked examples so that you can reproduce a cost or nutrient budget.
- Draw the text diagrams - pillars, principles wheel, farm layout, nutrient cycle, pest triangle, value chain - from memory.



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# End Note

This study material follows the 2025 syllabus of AGR 202 - Principles and Practices of Natural Farming for the Third Semester of B.Sc. (Hons) Agriculture at Tamil Nadu Agricultural University. It covers all seventeen lectures of the course in the order given in the syllabus. Quantitative figures in the worked examples are illustrative teaching values; where a figure is uncertain, a standard range has been given rather than a false precision. Scheme names, subsidy amounts and certification details change over time and should be confirmed from current official documents.